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Biomarkers predictive of anti-immune checkpoint response

Abstract

The present invention is based on the identification of novel biomarkers predictive of responsiveness to anti-immune checkpoint therapies.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 62/445,105, filed on 11 Jan. 2017; the entire contents of said application are incorporated herein in their entirety by this reference.

BACKGROUND OF THE INVENTION

(1) Immune checkpoint therapies can yield durable responses and long-lasting survival benefit across some cancer types (Topalian et al. (2015) *Cancer Cell* 27:450-461). Indeed, checkpoint therapies have been approved for use in metastatic melanoma, non-small cell lung cancer, bladder cancer, and renal cell carcinoma, including as a first-line therapy for non-small cell lung cancer. However, many subjects among a population of subjects having the same cancer type do not exhibit a therapeutic benefit or relapse despite being treated with the same immune checkpoint therapy. It is presently unclear which factors associated with a cancer or type thereof, such as mutational load, neoantigen presentation, transcriptomic signatures, microbiome features, immune cell infiltration, or other indicators, are predictive of response to immune checkpoint therapies. Accordingly, there remains a great need in the art to identify biomarkers predictive of immune checkpoint therapy in order to better treat cancer of subjects in need thereof.

SUMMARY OF THE INVENTION

(2) The present invention is based, at least in part, on the discovery that alterations in multiple oncogenic signaling pathways, including SWI/SNF pathway but also other chromatin modifiers, such as KDM6A, and EGFR signaling, predict response or resistance to immune checkpoint therapies, including (but not limited to) monoclonal antibodies targeting PD-1, PD-L1, and CTLA-4, across multiple cancer types. The SWI/SNF chromatin remodeling complex, which contains ARID1A, ARID1B, ARID2, SMARCA2, SMARCA4, SMARCB1, and PBRM1 subunits, among other subunits, plays a role in replication, transcription, DNA repair, and control of cell proliferation and differentiation. Although alterations in SWI/SNF subunits are known to play a role in the pathogenesis of ~20% of human cancers, including clear cell renal cell carcinoma, lung cancer, squamous cell carcinomas, hepatocellular carcinoma, small cell lung cancer, colorectal cancer, and pancreatic cancer (Kadoch and Crabtree (2015) *Sci. Adv.* 1:e150047), it was heretofore unknown that a mutation in one or more subunits of the SWI/SNF complex (e.g., mutations in one or more subunits of the PBAF complex, such as PBRM1 and ARID2), is predictive of response to immune checkpoint inhibitors. The same lack of predictive response applies to mutations in certain chromatin modifiers, such as KDM6A, and certain EGFR signaling components described herein. Since mutations in certain SWI/SNF complex subunits, chromatin modifiers, and/or EGFR signaling components described herein are found within a variety of cancers and types thereof, including bladder cancer, renal cell carcinoma, lung cancer, and head and neck squamous cell carcinoma, these biomarkers have wide-ranging implications for patient stratification for immune checkpoint therapy across a wide variety of hyperproliferative disorders.

(3) In one aspect, a method of identifying the likelihood of a cancer in a subject to be responsive to an immune checkpoint therapy, the method comprising a) obtaining or providing a subject sample from a patient having cancer; b) measuring the amount or activity of at least one biomarker listed in Table 1 in the subject sample; and c) comparing said amount or activity of the at least one biomarker listed in Table 1 in a control sample, wherein the absence of or a significantly decreased amount or activity of the at least one biomarker listed in Table 1 in the subject sample and/or the presence of or a significantly increased amount or activity of the at least one biomarker listed in Table 1 having a loss of function mutation in the subject sample, relative to the control sample identifies the cancer as being more likely to be responsive to the immune checkpoint therapy; and wherein the presence of or a significantly increased amount or activity of the at least one biomarker

listed in Table 1 in the subject sample and/or the absence of or a decreased amount or activity of the at least one biomarker listed in Table 1 having a loss of function mutation in the subject sample, relative to the control sample identifies the cancer as being less likely to be responsive to the immune checkpoint therapy, is provided.

(4) In another aspect, a method of identifying the likelihood of a cancer in a subject to be responsive to immune checkpoint therapy, the method comprising a) obtaining or providing a subject sample from a patient having cancer, wherein the sample comprises nucleic acid molecules from the subject; b) determining the copy number of at least one biomarker listed in Table 1 in the subject sample; and c) comparing said copy number to that of a control sample, wherein a decreased copy number of the at least one biomarker listed in Table 1 in the in the subject sample and/or an increased copy number of the at least one biomarker listed in Table 1 having a loss of function mutation in the subject sample, relative to the control sample identifies the cancer as being more likely to be responsive to the immune checkpoint therapy; and wherein a wild type or increased copy number of the biomarker in the subject sample and/or or a decreased copy number of the at least one biomarker listed in Table 1 having a loss of function mutation in the sample relative to the control sample identifies the cancer as being less likely to be responsive to the immune checkpoint therapy, is provided.

(5) Numerous embodiments are further provided that can be applied to any aspect of the present invention and/or combined with any other embodiment described herein. For example, in one embodiment, the method provided herein further comprises recommending, prescribing, or administering the immune checkpoint therapy if the cancer is determined likely to be responsive to the immune checkpoint therapy or administering an anti-cancer therapy other than the immune checkpoint therapy if the cancer is determined be less likely to be responsive to the immune checkpoint therapy. The anti-cancer therapy may be, for example, selected from the group consisting of targeted therapy, chemotherapy, radiation therapy, and/or hormonal therapy. In another embodiment, the control sample described herein is determined from a cancerous or non-cancerous sample from either the patient or a member of the same species to which the patient belongs. In still another embodiment, the control sample is a cancerous or non-cancerous sample from the patient obtained from an earlier point in time than the patient sample. In yet another embodiment, the control sample is obtained before the patient has received immune checkpoint therapy and the patient sample is obtained after the patient has received immune checkpoint therapy. In another embodiment, the control sample described herein comprises cells or does not comprise cells. In still another embodiment, the control sample comprises cancer cells known to be responsive or non-responsive to the immune checkpoint therapy.

(6) In another aspect, a method of assessing the efficacy of an agent for treating a cancer in a subject that is unlikely to be responsive to an immune checkpoint therapy, comprising a) detecting in a first subject sample and maintained in the presence of the agent the amount or activity of at least one biomarker listed in Table 1; b) detecting the amount or activity of the at least one biomarker listed in Table 1 in a second subject sample and maintained in the absence of the test compound; and c) comparing the amount or activity of the at least one biomarker listed in Table 1 from steps a) and b), wherein the presence of or a significantly increased amount or activity of the at least one biomarker listed in Table 1 in the first subject sample and/or the absence of or a decreased amount or activity of the at least one biomarker listed in Table 1 having a loss of function mutation in the first subject sample, relative to at least one subsequent subject sample, indicates that the agent treats the cancer in the subject, is provided.

(7) In another aspect, a method of assessing the efficacy of an agent for treating a cancer in a subject or prognosing progression of a cancer in a subject, comprising a) detecting in a subject sample at a first point in time the amount or activity of at least one biomarker listed in Table 1; b) repeating step a) during at least one subsequent point in time after administration of the agent; and c) comparing the expression and/or activity detected in steps a) and b), wherein the presence of or a

significantly increased amount or activity of the at least one biomarker listed in Table 1 in the first subject sample and/or the absence of or a decreased amount or activity of the at least one biomarker listed in Table 1 having a loss of function mutation in the first subject sample, relative to at least one subsequent subject sample, indicates that the cancer is unlikely to progress or that the agent treats the cancer in the subject, is provided. In one embodiment, between the first point in time and the subsequent point in time, the subject has undergone treatment, completed treatment, and/or is in remission for the cancer. In another embodiment, the first and/or at least one subsequent sample is selected from the group consisting of ex vivo and in vivo samples. In still another embodiment, the first and/or at least one subsequent sample is obtained from an animal model of the cancer. In yet another embodiment, the first and/or at least one subsequent sample is a portion of a single sample or pooled samples obtained from the subject.

(8) In another aspect, a cell-based assay for screening for agents that have a cytotoxic or cytostatic effect on a cancer cell that is unresponsive to an immune checkpoint therapy comprising, contacting the cancer cell with a test agent, and determining the ability of the test agent to decrease the amount or activity of at least one biomarker listed in Table 1 in the subject sample and/or increase the amount or activity of the at least one biomarker listed in Table 1 having a loss of function mutation, is provided. In one embodiment, the step of contacting occurs in vivo, ex vivo, or in vitro. In another embodiment, the subject sample and/or the control sample has not been contacted with any anti-cancer treatment or inhibitor of an immune checkpoint. In still another embodiment, the subject has not been administered any anti-cancer treatment or inhibitor of an immune checkpoint. In yet another embodiment, the method or the cell-based assay provided herein further comprises recommending, prescribing, or administering at least one additional anti-cancer therapeutic agent. In another embodiment, the at least one additional anti-cancer therapeutic agent comprises an anti-PD-1 antibody and/or an anti-CTLA4 antibody.

(9) As described above, numerous embodiments are contemplated for any aspect of the present invention described herein. For example, in one embodiment, the subject sample is selected from the group consisting of serum, whole blood, plasma, urine, cells, cell lines, and biopsies. In another embodiment, the amount of the at least one biomarker listed in Table 1 is detected using a reagent which specifically binds with the protein. For example, the reagent may be selected from the group consisting of an antibody, an antibody derivative, and an antibody fragment. In still another embodiment, the at least one biomarker listed in Table 1 is assessed by detecting the presence in the sample of a transcribed polynucleotide or portion thereof. For example, the transcribed polynucleotide may be an mRNA or a cDNA. The transcribed polynucleotide can be detected by identifying a nucleic acid that anneals with the biomarker nucleic acid, or a portion thereof, under stringent hybridization conditions. In yet another embodiment, the step of detecting further comprises amplifying the transcribed polynucleotide. In another embodiment, the at least one biomarker listed in Table 1 is human PBRM1, ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, or EGFR, or a fragment thereof. In still another embodiment, the immune checkpoint therapy described herein comprises at least one antibody selected from the group consisting of anti-PD-1 antibodies, anti-CTLA-4 antibodies, anti-PD-L1 antibodies, anti-PD-L2 antibodies, and combinations thereof. For example, the immune checkpoint therapy may comprise an anti-PD-1 antibody and/or an anti-CTLA4 antibody. In yet another embodiment, the likelihood of the cancer in the subject to be responsive to immune checkpoint therapy is the likelihood of at least one criteria selected from the group consisting of cellular proliferation, tumor burden, m-stage, metastasis, progressive disease, clinical benefit rate, survival until mortality, pathological complete response, semi-quantitative measures of pathologic response, clinical complete remission, clinical partial remission, clinical stable disease, recurrence-free survival, metastasis free survival, disease free survival, circulating tumor cell decrease, circulating marker response, and RECIST criteria. In another embodiment, the cancer is a solid tumor. In still another embodiment, the cancer is selected from the group consisting of melanoma, lung cancer, head and

neck squamous cell carcinoma (HNSCC), sarcoma, bladder cancer, and renal cell cancer. In another embodiment, the cancer is melanoma. In still another embodiment, the cancer is metastatic. In still another embodiment, the subject described herein is a mammal. In yet another embodiment, the mammal is an animal model of cancer. In another embodiment, the mammal is a human.

Description

BRIEF DESCRIPTION OF FIGURES

- (1) FIG. 1 summarizes the different types of cancer samples and their sources for analysis.
- (2) FIG. 2 depicts two criteria (exclusion and inclusion) for selecting quality controls for analysis.
- (3) FIG. 3 depicts that different patients had different degrees of clinical benefit from immune checkpoint therapy.
- (4) FIG. 4 compares the amount of nonsynonymous mutations in patients having different degrees of clinical benefit from immune checkpoint therapy.
- (5) FIG. 5 shows genes significantly mutated in responders vs. non-responders.
- (6) FIG. 6 shows genes significantly mutated in responders vs. non-responders or intermediate responders (such as those having intermediate clinical benefit).
- (7) FIG. 7 shows genes significantly mutated (such as those having truncating mutations) in responders vs. non-responders.
- (8) FIG. 8 shows genes significantly mutated (such as those having truncating mutations) in responders vs. non-responders or intermediate responders (those having intermediate clinical benefit).
- (9) FIG. 9 depicts protein subunits of the SWI/SNF protein complex.
- (10) FIG. 10 shows SWI/SNF-relevant genes significantly mutated in responders vs. non-responders.
- (11) FIG. 11 shows SWI/SNF-relevant genes significantly mutated (such as those having truncating mutations) in responders vs. non-responders.
- (12) FIG. 12 depicts an enzymatic function scheme of KDM6A.
- (13) FIG. 13 includes 4 panels, identified as panels A, B, C, and D, which show the Kaplan-Meier analysis result for baseline clinical variables as predictors of PFS for SU2C cohort (N=39).
- (14) FIG. 14 shows the quality control processes for analyzing the SU2C cohort.
- (15) FIG. 15 depicts the different responses of 39 SU2C lung cancer patients to anti-PD-1/PD-L1 therapy.
- (16) FIG. 16 shows the mutational burden and response to immune checkpoint therapies of each patient (N=31).
- (17) FIG. 17 shows the relationship between clinical burden and clinical benefit in a cohort in Rizvi et al. (2015) Science 348:124-128. RECIST was not taken into account (such that 2 patients with PR and PFS of ~4 months were considered nonresponders).
- (18) FIG. 18 shows that pre-treatment tumor mutational load was a strong predictor of response to immune checkpoint therapy in anti-PD1/PD-L1-treated lung cancer. All mutations: CB vs. NCB; $p=0.003$. All mutations: CB or SD vs. NCB; $p=0.004$. Nonsyns: CB vs. NCB; $p=0.0047$. Nonsyns: CB or SD vs. NCB; $p=0.0064$. Clonal: CB vs. NCB; $p=0.024$. Clonal: CB or SD vs. NCB; $p=0.007$. If dropping two large outliers (highest mutational load CB and SD), p -values for all mutations go to 0.009 and 0.011.
- (19) FIG. 19 shows commonly mutated genes in lung cancer. NF1 alterations were more frequent in responders (3/6 clinical benefit, 3/13 stable disease, 0/12 NCB). EGFR hotspot alterations were seen more frequently in nonresponders. KRAS hotspot alterations seen more frequently in responders (1/6 clinical benefit, 4/13 SD, 1/12 NCB). SU2C-1006: splice site mutation in MET; missense mutation in LTBP1. SU2C-1066: 3 missense mutations in LEPR. SU2C-1068: 2 missense

mutations in LEPR. SU2C-1067: Missense mutations in STAG2 and SRCAP. EGFR hotspot is L858R. SU2C-1066 may be excluded, since its Purity=0.36.

(20) FIG. 20 shows significantly mutated genes (N=6 clinical benefit vs. 12 no clinical benefit).

(21) FIG. 21 shows that patients with hotspot mutations in EGFR uniformly did not respond to immune checkpoint therapy.

(22) FIG. 22 shows that SAFB2 indels were likely caused by sequencing artifact.

DETAILED DESCRIPTION OF THE INVENTION

(23) It has been determined herein that certain SWI/SNF complex subunits (e.g., PBRM1, ARID2, and other SWI/SNF complex subunits described herein, such as in the Tables and Examples), additional chromatin modifiers (e.g., such as KDM6A), and EGFR signaling components are specific biomarkers for predicted clinical outcome in a wide variety of cancers afflicting patients who have received anti-immune checkpoint-based therapy (e.g., anti-PD1 and/or anti-CTLA4 agents). Accordingly, the present invention relates, in part, to methods for stratifying patients and predicting response of a cancer in a subject to immune checkpoint therapy based upon a determination and analysis of mutations, described herein, of biomarkers, compared to a control. In addition, such analyses can be used in order to provide useful anti-immune checkpoint treatment regimens (e.g., based on predictions of clinical response, subject survival or relapse, timing of adjuvant or neoadjuvant treatment, etc.).

I. Definitions

(24) The articles “a” and “an” are used herein to refer to one or to more than one (i.e. to at least one) of the grammatical object of the article. By way of example, “an element” means one element or more than one element.

(25) The term “altered amount” or “altered level” refers to increased or decreased copy number (e.g., germline and/or somatic) of a biomarker nucleic acid, e.g., increased or decreased expression level in a cancer sample, as compared to the expression level or copy number of the biomarker nucleic acid in a control sample. The term “altered amount” of a biomarker also includes an increased or decreased protein level of a biomarker protein in a sample, e.g., a cancer sample, as compared to the corresponding protein level in a normal, control sample. Furthermore, an altered amount of a biomarker protein may be determined by detecting posttranslational modification such as methylation status of the marker, which may affect the expression or activity of the biomarker protein.

(26) The amount of a biomarker in a subject is “significantly” higher or lower than the normal amount of the biomarker, if the amount of the biomarker is greater or less, respectively, than the normal level by an amount greater than the standard error of the assay employed to assess amount, and preferably at least 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 150%, 200%, 300%, 350%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or than that amount. Such “significance” can be assessed from any desired or known point of comparison, such as a particular post-treatment versus pre-treatment biomarker measurement ratio (e.g., 1-fold, 1.1-fold, 1.2-fold, 1.3-fold, 1.4-fold, 1.5-fold, and the like) or a particular pre-treatment serum biomarker protein measurement (e.g., 2,500 pg/ml, 2,750 pg/ml, 3,000 pg/ml, 3,175 pg/ml, 3,250 pg/ml, 3,500 pg/ml, and the like). Alternately, the amount of the biomarker in the subject can be considered “significantly” higher or lower than the normal amount if the amount is at least about two, and preferably at least about three, four, or five times, higher or lower, respectively, than the normal amount of the biomarker. Such “significance” can also be applied to any other measured parameter described herein, such as for expression, inhibition, cytotoxicity, cell growth, and the like.

(27) The term “altered level of expression” of a biomarker refers to an expression level or copy number of the biomarker in a test sample, e.g., a sample derived from a patient suffering from cancer, that is greater or less than the standard error of the assay employed to assess expression or copy number, and is preferably at least twice, and more preferably three, four, five or ten or more times the expression level or copy number of the biomarker in a control sample (e.g., sample from

a healthy subjects not having the associated disease) and preferably, the average expression level or copy number of the biomarker in several control samples. The altered level of expression is greater or less than the standard error of the assay employed to assess expression or copy number, and is preferably at least 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 150%, 200%, 300%, 350%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more times the expression level or copy number of the biomarker in a control sample (e.g., sample from a healthy subjects not having the associated disease) and preferably, the average expression level or copy number of the biomarker in several control samples.

(28) The term “altered activity” of a biomarker refers to an activity of the biomarker which is increased or decreased in a disease state, e.g., in a cancer sample, as compared to the activity of the biomarker in a normal, control sample. Altered activity of the biomarker may be the result of, for example, altered expression of the biomarker, altered protein level of the biomarker, altered structure of the biomarker, or, e.g., an altered interaction with other proteins involved in the same or different pathway as the biomarker or altered interaction with transcriptional activators or inhibitors.

(29) The term “altered structure” of a biomarker refers to the presence of mutations or allelic variants within a biomarker nucleic acid or protein, e.g., mutations which affect expression or activity of the biomarker nucleic acid or protein, as compared to the normal or wild-type gene or protein. For example, mutations include, but are not limited to substitutions, deletions, or addition mutations. Mutations may be present in the coding or non-coding region of the biomarker nucleic acid.

(30) The term “SWI/SNF complex” refers to SWItch/Sucrose Non-Fermentable, a nucleosome remodeling complex found in both eukaryotes and prokaryotes (Neugeborn Carlson (1984) *Genetics* 108:845-858; Stem et al. (1984) *J Mol. Biol.* 178:853-868). The SWI/SNF complex was first discovered in the yeast, *Saccharomyces cerevisiae*, named after yeast mating types switching (SWI) and sucrose nonfermenting (SNF) pathways (Workman and Kingston (1998) *Annu Rev Biochem.* 67:545-579; Sudarsanam and Winston (2000) *Trends Genet.* 16:345-351). It is a group of proteins comprising, at least, SWI1, SWI2/SNF2, SWI3, SWI5, and SWI6, as well as other polypeptides (Pazin and Kadonaga (1997) *Cell* 88:737-740). A genetic screening for suppressive mutations of the SWI/SNF phenotypes identified different histones and chromatin components, suggesting that these proteins were possibly involved in histone binding and chromatin organization (Winston and Carlson (1992) *Trends Genet.* 8:387-391). Biochemical purification of the SWI/SNF2p in *S. cerevisiae* demonstrated that this protein was part of a complex containing an additional 11 polypeptides, with a combined molecular weight over 1.5 MDa. The SWI/SNF complex contains the ATPase Swi2/Snf2p, two actin-related proteins (Arp7p and Arp9) and other subunits involved in DNA and protein-protein interactions. The purified SWI/SNF complex was able to alter the nucleosome structure in an ATP-dependent manner (Workman and Kingston (1998), supra; Vignali et al. (2000) *Mol Cell Biol.* 20:1899-1910). The structures of the SWI/SNF and RSC complexes are highly conserved but not identical, reflecting an increasing complexity of chromatin (e.g., an increased genome size, the presence of DNA methylation, and more complex genetic organization) through evolution. For this reason, the SWI/SNF complex in higher eukaryotes maintains core components, but also substitute or add on other components with more specialized or tissue-specific domains. Yeast contains two distinct and similar remodeling complexes, SWI/SNF and RSC (Remodeling the Structure of Chromatin). In *Drosophila*, the two complexes are called BAP (Brahma Associated Protein) and PBAP (Polybromo-associated BAP) complexes. The human analogs are BAF (Brg1 Associated Factors, or SWI/SNF-A) and PBAF (Polybromo-associated BAF, or SWI/SNF-B). As shown in FIG. 9, the BAF complex comprises, at least, BAF250A (ARID1A), BAF250B (ARID1B), BAF57 (SMARCE1), BAF190/BRM (SMARCA2), BAF47 (SMARCB1), BAF53A (ACTL6A), BRG1/BAF190 (SMARCA4), BAF155 (SMARCC1), and BAF170 (SMARCC2). The PBAF complex comprises, at last, BAF200

(ARID2), BAF180 (PBRM1), BRD7, BAF45A (PHF10), BRG1/BAF190 (SMARCA4), BAF155 (SMARCC1), and BAF170 (SMARCC2). As in *Drosophila*, human BAF and PBAF share the different core components BAF47, BAF57, BAF60, BAF155, BAF170, BAF45 and the two actins b-Actin and BAF53 (Mohrmann and Verrijzer (2005) *Biochim Biophys Acta*. 1681:59-73). The central core of the BAF and PBAF is the ATPase catalytic subunit BRG1/hBRM, which contains multiple domains to bind to other protein subunits and acetylated histones. For a summary of different complex subunits and their domain structure, see Tang et al. (2010) *Prog Biophys Mol Biol*. 102:122-128 (e.g., FIG. 3), Hohmann and Vakoc (2014) *Trends Genet*. 30:356-363 (e.g., FIG. 1), and Kadoch and Crabtree (2015) *Sci. Adv*. 1:e1500447. For chromatin remodeling, the SWI/SNF complex use the energy of ATP hydrolysis to slide the DNA around the nucleosome. The first step consists in the binding between the remodeler and the nucleosome. This binding occurs with nanomolar affinity and reduces the digestion of nucleosomal DNA by nucleases. The 3-D structure of the yeast RSC complex was first solved and imaged using negative stain electron microscopy (Asturias et al. (2002) *Proc Natl Acad Sci USA* 99:13477-13480). The first Cryo-EM structure of the yeast SWI/SNF complex was published in 2008 (Dechassa et al. 2008). DNA footprinting data showed that the SWI/SNF complex makes close contacts with only one gyre of nucleosomal DNA. Protein crosslinking showed that the ATPase SWI2/SNF2p and Swi5p (the homologue of Inilp in human), Snf6, Swi29, Snf11 and Sw82p (not conserved in human) make close contact with the histones. Several individual SWI/SNF subunits are encoded by gene families, whose protein products are mutually exclusive in the complex (Wu et al. (2009) *Cell* 136:200-206). Thus, only one paralog is incorporated in a given SWI/SNF assembly. The only exceptions are BAF155 and BAF170, which are always present in the complex as homo- or heterodimers. Combinatorial association of SWI/SNF subunits could in principle give rise to hundreds of distinct complexes, although the exact number has yet to be determined (Wu et al. (2009), supra). Genetic evidence suggests that distinct subunit configurations of SWI/SNF are equipped to perform specialized functions. As an example, SWI/SNF contains one of two ATPase subunits, BRG1 or BRM/SMARCA2, which share 75% amino acid sequence identity (Khavari et al. (1993) *Nature* 366:170-174). While in certain cell types BRG1 and BRM can compensate for loss of the other subunit, in other contexts these two ATPases perform divergent functions (Strobeck et al. (2002) *J Biol Chem*. 277:4782-4789; Hoffman et al. (2014) *Proc Natl Acad Sci USA*. 111:3128-3133). In some cell types, BRG1 and BRM can even functionally oppose one another to regulate differentiation (Flowers et al. (2009) *J Biol Chem*. 284:10067-10075). The functional specificity of BRG1 and BRM has been linked to sequence variations near their N-terminus, which have different interaction specificities for transcription factors (Kadam and Emerson (2003) *Mol Cell*. 11:377-389). Another example of paralogous subunits that form mutually exclusive SWI/SNF complexes are ARID1A/BAF250A, ARID1B/BAF250B, and ARID2/BAF200. ARID1A and ARID1B share 60% sequence identity, but yet can perform opposing functions in regulating the cell cycle, with MYC being an important downstream target of each paralog (Nagl et al. (2007) *EMBO J*. 26:752-763). ARID2 has diverged considerably from ARID1A/ARID1B and exists in a unique SWI/SNF assembly known as PBAF (or SWI/SNF-B), which contains several unique subunits not found in ARID1A/B-containing complexes. The composition of SWI/SNF can also be dynamically reconfigured during cell fate transitions through cell type-specific expression patterns of certain subunits. For example, BAF53A/ACTL6A is repressed and replaced by BAF53B/ACTL6B during neuronal differentiation, a switch that is essential for proper neuronal functions in vivo (Lessard et al. (2007) *Neuron* 55:201-215). These studies stress that SWI/SNF in fact represents a collection of multi-subunit complexes whose integrated functions control diverse cellular processes, which is also incorporated in the scope of definitions of the instant disclosure. Two recently published meta-analyses of cancer genome sequencing data estimate that nearly 20% of human cancers harbor mutations in one (or more) of the genes encoding SWI/SNF (Kadoch et al. (2013) *Nat Genet*. 45:592-601; Shain and Pollack (2013) *PLoS One*. 8:e55119). Such mutations

are generally loss-of-function, implicating SWI/SNF as a major tumor suppressor in diverse cancers. Specific SWI/SNF gene mutations are generally linked to a specific subset of cancer lineages: SNF5 is mutated in malignant rhabdoid tumors (MRT), PBRM1/BAF180 is frequently inactivated in renal carcinoma, and BRG1 is mutated in non-small cell lung cancer (NSCLC) and several other cancers. In the instant disclosure, the scope of “SWI/SNF complex” may cover at least one fraction or the whole complex (e.g., some or all subunit proteins/other components), either in the human BAF/PBAF forms or their homologs/orthologs in other species (e.g., the yeast and *drosophila* forms described herein). Preferably, a “SWI/SNF complex” described herein contains at least part of the full complex bio-functionality, such as binding to other subunits/components, binding to DNA/histone, catalyzing ATP, promoting chromatin remodeling, etc.

(31) The term “BAF complex” refers to at least one type of mammalian SWI/SNF complexes. Its nucleosome remodeling activity can be reconstituted with a set of four core subunits (BRG1/SMARCA4, SNF5/SMARCB1, BAF155/SMARCC1, and BAF170/SMARCC2), which have orthologs in the yeast complex (Phelan et al. (1999) *Mol Cell.* 3:247-253). However, mammalian SWI/SNF contains several subunits not found in the yeast counterpart, which can provide interaction surfaces for chromatin (e.g. acetyl-lysine recognition by bromodomains) or transcription factors and thus contribute to the genomic targeting of the complex (Wang et al. (1996) *EMBO J* 15:5370-5382; Wang et al. (1996) *Genes Dev.* 10:2117-2130; Nie et al. (2000)). A key attribute of mammalian SWI/SNF is the heterogeneity of subunit configurations that can exist in different tissues and even in a single cell type (e.g., as BAF, PBAF, neural progenitor BAF (npBAF), neuron BAF (nBAF), embryonic stem cell BAF (esBAF), etc.). In some embodiments, the BAF complex described herein refers to one type of mammalian SWI/SNF complexes, which is different from PBAF complexes.

(32) The term “PBAF complex” refers to one type of mammalian SWI/SNF complexes originally known as SWI/SNF-B. It is highly related to the BAF complex and can be separated with conventional chromatographic approaches. For example, human BAF and PBAF complexes share multiple identical subunits (such as BRG, BAF170, BAF155, BAF60, BAF57, BAF53, BAF45, actin, SS18, and hSNF5/INI1, as illustrated in FIG. 9). However, while BAF contains BAF250 subunit, PBAF contains BAF180 and BAF200, instead (Lemon et al. (2001) *Nature* 414:924-998; Yan et al. (2005) *Genes Dev.* 19:1662-1667). Moreover, they do have selectivity in regulating interferon-responsive genes (Yan et al. (2005), supra, showing that BAF200, but not BAF180, is required for PBAF to mediate expression of IFITM1 gene induced by IFN- α , while the IFITM3 gene expression is dependent on BAF but not PBAF). Due to these differences, PBAF, but not BAF, was able to activate vitamin D receptor-dependent transcription on a chromatinized template in vitro (Lemon et al. (2001), supra). The 3-D structure of human PBAF complex preserved in negative stain was found to be similar to yeast RSC but dramatically different from yeast SWI/SNF (Leschziner et al. (2005) *Structure* 13:267-275).

(33) The term “BRG” or “BRG1/BAF190 (SMARCA4)” refers to a subunit of the SWI/SNF complex, which can be found in either BAF or PBAF complex. It is an ATP-dependent helicase and a transcription activator, encoded by the SMARCA4 gene. BRG1 can also bind BRCA1, as well as regulate the expression of the tumorigenic protein CD44. BRG1 is important for development past the pre-implantation stage. Without having a functional BRG1, exhibited with knockout research, the embryo will not hatch out of the zona pellucida, which will inhibit implantation from occurring on the endometrium (uterine wall). BRG1 is also crucial to the development of sperm. During the first stages of meiosis in spermatogenesis there are high levels of BRG1. When BRG1 is genetically damaged, meiosis is stopped in prophase 1, hindering the development of sperm and would result in infertility. More knockout research has concluded BRG1's aid in the development of smooth muscle. In a BRG1 knockout, smooth muscle in the gastrointestinal tract lacks contractility, and intestines are incomplete in some cases. Another defect occurring in knocking out

BRG1 in smooth muscle development is heart complications such as an open ductus arteriosus after birth (Kim et al. (2012) *Development* 139:1133-1140; Zhang et al. (2011) *Mol. Cell. Biol.* 31:2618-2631). Mutations in SMARCA4 were first recognized in human lung cancer cell lines (Medina et al. (2008) *Hum. Mut.* 29:617-622). Later it was recognized that mutations exist in a significant frequency of medulloblastoma and pancreatic cancers among other tumor subtypes (Jones et al. (2012) *Nature* 488:100-105; Shain et al. (2012) *Proc Natl Acad Sci USA* 109:E252-E259; Shain and Pollack (2013), supra). Mutations in BRG1 (or SMARCA4) appear to be mutually exclusive with the presence of activation at any of the MYC-genes, which indicates that the BRG1 and MYC proteins are functionally related. Another recent study demonstrated a causal role of BRG1 in the control of retinoic acid and glucocorticoid-induced cell differentiation in lung cancer and in other tumor types. This enables the cancer cell to sustain undifferentiated gene expression programs that affect the control of key cellular processes. Furthermore, it explains why lung cancer and other solid tumors are completely refractory to treatments based on these compounds that are effective therapies for some types of leukemia (Romero et al. (2012) *EMBO Mol. Med.* 4:603-616). The role of BRG1 in sensitivity or resistance to anti-cancer drugs had been recently highlighted by the elucidation of the mechanisms of action of darinaparsin, an arsenic-based anti-cancer drugs. Darinaparsin has been shown to induce phosphorylation of BRG1, which leads to its exclusion from the chromatin. When excluded from the chromatin, BRG1 can no longer act as a transcriptional co-regulator. This leads to the inability of cells to express HO-1, a cytoprotective enzyme. BRG1 has been shown to interact with proteins such as ACTL6A, ARID1A, ARID1B, BRCA1, CTNNB1, CBX5, CREBBP, CCNE1, ESR1, FANCA, HSP90B1, ING1, Myc, NR3C1, P53, POLR2A, PHB, SIN3A, SMARCB1, SMARCC1, SMARCC2, SMARCE1, STAT2, STK11, etc.

(34) The term “BRG” or “BRG1/BAF190 (SMARCA4)” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BRG1(SMARCA4) cDNA and human BRG1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, seven different human BRG1 isoforms are known. Human BRG1 isoform A (NP_001122321.1) is encodable by the transcript variant 1 (NM_001128849.1), which is the longest transcript. Human BRG1 isoform B (NP_001122316.1 or NP_003063.2) is encodable by the transcript variant 2 (NM_001128844.1), which differs in the 5' UTR and lacks an alternate exon in the 3' coding region, compared to the variant 1, and also by the transcript variant 3 (NM_003072.3), which lacks an alternate exon in the 3' coding region compared to variant 1. Human BRG1 isoform C (NP_001122317.1) is encodable by the transcript variant 4 (NM_001128845.1), which lacks two alternate in-frame exons and uses an alternate splice site in the 3' coding region, compared to variant 1. Human BRG1 isoform D (NP_001122318.1) is encodable by the transcript variant 5 (NM_001128846.1), which lacks two alternate in-frame exons and uses two alternate splice sites in the 3' coding region, compared to variant 1. Human BRG1 isoform E (NP_001122319.1) is encodable by the transcript variant 6 (NM_001128847.1), which lacks two alternate in-frame exons in the 3' coding region, compared to variant 1. Human BRG1 isoform F (NP_001122320.1) is encodable by the transcript variant 7 (NM_001128848.1), which lacks two alternate in-frame exons and uses an alternate splice site in the 3' coding region, compared to variant 1. Nucleic acid and polypeptide sequences of BRG1 orthologs in organisms other than humans are well known and include, for example, chimpanzee BRG1 (XM_016935029.1 and XP_016790518.1, XM_016935038.1 and XP_016790527.1, XM_016935039.1 and XP_016790528.1, XM_016935036.1 and XP_016790525.1, XM_016935037.1 and XP_016790526.1, XM_016935041.1 and XP_016790530.1, XM_016935040.1 and XP_016790529.1, XM_016935042.1 and XP_016790531.1, XM_016935043.1 and XP_016790532.1, XM_016935035.1 and XP_016790524.1, XM_016935032.1 and XP_016790521.1, XM_016935033.1 and XP_016790522.1, XM_016935030.1 and XP_016790519.1, XM_016935031.1 and XP_016790520.1, and

XM_016935034.1 and XP_016790523.1), Rhesus monkey BRG1 (XM_015122901.1 and XP_014978387.1, XM_015122902.1 and XP_014978388.1, XM_015122903.1 and XP_014978389.1, XM_015122906.1 and XP_014978392.1, XM_015122905.1 and XP_014978391.1, XM_015122904.1 and XP_014978390.1, XM_015122907.1 and XP_014978393.1, XM_015122909.1 and XP_014978395.1, and XM_015122910.1 and XP_014978396.1), dog BRG1 (XM_014122046.1 and XP_013977521.1, XM_014122043.1 and XP_013977518.1, XM_014122042.1 and XP_013977517.1, XM_014122041.1 and XP_013977516.1, XM_014122045.1 and XP_013977520.1, and XM_014122044.1 and XP_013977519.1), cattle BRG1 (NM_001105614.1 and NP_001099084.1), mouse BRG1 (NM_001174078.1 and NP_001167549.1, NM_001174079.1 and NP_001167550.1, and NM_011417.3 and NP_035547.2), rat BRG1 (NM_134368.1 and NP_599195.1), chicken BRG1 (NM_205059.1 and NP_990390.1), and zebrafish BRG1 (NM_181603.1 and NP_853634.1).

(35) Anti-BRG1 antibodies suitable for detecting BRG1 protein are well-known in the art and include, for example, MABE1118, MABE121, MABE60, and 07-478 (poly- and mono-clonal antibodies from EMD Millipore, Billerica, MA), AM26021PU-N, AP23972PU-N, TA322909, TA322910, TA327280, TA347049, TA347050, TA347851, and TA349038 (antibodies from OnGene Technologies, Rockville, MD), NB100-2594, AF5738, NBP2-22234, NBP2-41270, NBP1-51230, and NBP1-40379 (antibodies from Novus Biologicals, Littleton, CO), ab110641, ab4081, ab215998, ab108318, ab70558, ab118558, ab133257, ab92496, ab196535, and ab196315 (antibodies from AbCam, Cambridge, MA), Cat #: 720129, 730011, 730051, MA1-10062, PA5-17003, and PA5-17008 (antibodies from ThermoFisher Scientific, Waltham, MA), GTX633391, GTX32478, GTX31917, GTX16472, and GTX50842 (antibodies from GeneTex, Irvine, CA), antibody 7749 (ProSci, Poway, CA), Brg-1 (N-15), Brg-1 (N-15) X, Brg-1 (H-88), Brg-1 (H-88) X, Brg-1 (P-18), Brg-1 (P-18) X, Brg-1 (G-7), Brg-1 (G-7) X, Brg-1 (H-10), and Brg-1 (H-10) X (antibodies from Santa Cruz Biotechnology, Dallas, TX), antibody of Cat. AF5738 (R&D Systems, Minneapolis, MN), etc. In addition, reagents are well-known for detecting BRG1 expression. Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing BRG1 Expression can be found in the commercial product lists of the above-referenced companies. PFI 3 is a known small molecule inhibitor of polybromo 1 and BRG1 (e.g., Cat. B7744 from APEX BIO, Houston, TX). It is to be noted that the term can further be used to refer to any combination of features described herein regarding BRG1 molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an BRG1 molecule of the present invention.

(36) The term “BRM” or “BRM/BAF190 (SMARCA2)” refers to a subunit of the SWI/SNF complex, which can be found in either BAF or PBAF complexes. It is an ATP-dependent helicase and a transcription activator, encoded by the SMARCA2 gene. The catalytic core of the SWI/SNF complex can be either of two closely related ATPases, BRM or BRG1, with the potential that the choice of alternative subunits is a key determinant of specificity. Instead of impeding differentiation as was seen with BRG1 depletion, depletion of BRM caused accelerated progression to the differentiation phenotype. BRM was found to regulate genes different from those as BRG1 targets and be capable of overriding BRG1-dependent activation of the osteocalcin promoter, due to its interaction with different ARID family members (Flowers et al. (2009), supra). The known binding partners for BRM include, for example, ACTL6A, ARID1B, CEBPB, POLR2A, Prohibitin, SIN3A, SMARCB1, and SMARCC1.

(37) The term “BRM” or “BRM/BAF190 (SMARCA2)” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BRM (SMARCA2) cDNA and human BRM protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, seven different human BRM isoforms are known. Human BRM isoform A (NP_003061.3 or NP_001276325.1) is encodable by the transcript variant 1 (NM_003070.4), which is the longest transcript, or the transcript variant 3

(NM_001289396.1), which differs in the 5' UTR, compared to variant 1. Human BRM isoform B (NP_620614.2) is encodable by the transcript variant 2 (NM_139045.3), which lacks an alternate in-frame exon in the coding region, compared to variant 1. Human BRM isoform C (NP_001276326.1) is encodable by the transcript variant 4 (NM_001289397.1), which uses an alternate in-frame splice site and lacks an alternate in-frame exon in the 3' coding region, compared to variant 1. Human BRM isoform D (NP_001276327.1) is encodable by the transcript variant 5 (NM_001289398.1), which differs in the 5' UTR, lacks a portion of the 5' coding region, and initiates translation at an alternate downstream start codon, compared to variant 1. Human BRM isoform E (NP_001276328.1) is encodable by the transcript variant 6 (NM_001289399.1), which differs in the 5' UTR, lacks a portion of the 5' coding region, and initiates translation at an alternate downstream start codon, compared to variant 1. Human BRM isoform F (NP_001276329.1) is encodable by the transcript variant 7 (NM_001289400.1), which differs in the 5' UTR, lacks a portion of the 5' coding region, and initiates translation at an alternate downstream start codon, compared to variant 1. Nucleic acid and polypeptide sequences of BRM orthologs in organisms other than humans are well known and include, for example, chimpanzee BRM (XM_016960529.1 and XP_016816018.1), dog BRG1 (XM_005615906.2 and XP_005615963.1, XM_845066.4 and XP_850159.1, XM_005615905.2 and XP_005615962.1, XM_005615904.2 and XP_005615961.1, XM_005615903.2 and XP_005615960.1, and XM_005615902.2 and XP_005615959.1), cattle BRM (NM_001099115.2 and NP_001092585.1), mouse BRM (NM_001347439.1 and NP_001334368.1, NM_011416.2 and NP_035546.2, and NM_026003.2 and NP_080279.1), rat BRM (NM_001004446.1 and NP_001004446.1), chicken BRM (NM_205139.1 and NP_990470.1), tropical clawed frog BRM (XM_012952601.1 and XP_012808055.1, XM_012952608.2 and XP_012808062.1, XM_012952597.2 and XP_012808051.1, XM_012952613.2 and XP_012808067.1, and XM_002941009.4 and XP_002941055.2), and zebrafish BRM (NM_001044775.2 and NP_001038240.1).

(38) Anti-BRM antibodies suitable for detecting BRM protein are well-known in the art and include, for example, antibody MABE89 (EMD Millipore, Billerica, MA), antibody TA351725 (OnGene Technologies, Rockville, MD), NBP1-90015, NBP1-80042, NB100-55308, NB100-55309, NB100-55307, and H00006595-M06 (antibodies from Novus Biologicals, Littleton, CO), ab15597, ab12165, ab58188, and ab200480 (antibodies from AbCam, Cambridge, MA), Cat #: 11966 and 6889 (antibodies from Cell Signaling, Danvers, MA), etc. In addition, reagents are well-known for detecting BRM expression. Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing BRM Expression can be found in the commercial product lists of the above-referenced companies. For example, BRM RNAi product H00006595-R02 (Novus Biologicals), CRISPER gRNA products from GenScript, Piscataway, NJ, and other inhibitory RNA products from Origene, ViGene Biosciences (Rockville, MD), and Santa Cruz. It is to be noted that the term can further be used to refer to any combination of features described herein regarding BRM molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an BRM molecule of the present invention.

(39) The term “BAF200” or “ARID2” refers to AT-rich interactive domain-containing protein 2, a subunit of the SWI/SNF complex, which can be found in PBAF but not BAF complexes. It facilitates ligand-dependent transcriptional activation by nuclear receptors. The ARID2 gene, located on chromosome 12q in humans, consists of 21 exons; orthologs are known from mouse, rat, cattle, chicken, and mosquito (Zhao et al. (2011) *Oncotarget* 2:886-891). A conditional knockout mouse line, called Arid2.sup.tm1a(EUCOMM)Wtsi was generated as part of the International Knockout Mouse Consortium program, a high-throughput mutagenesis project to generate and distribute animal models of disease (Skames et al. (2011) *Nature* 474:337-342). Human ARID2 protein has 1835 amino acids and a molecular mass of 197391 Da. The ARID2 protein contains two conserved C-terminal C2H2 zinc fingers motifs, a region rich in the amino acid residues

proline and glutamine, a RFX (regulatory factor X)-type winged-helix DNA-binding domain (e.g., amino acids 521-601 of SEQ ID NO:8), and a conserved N-terminal AT-rich DNA interaction domain (e.g., amino acids 19-101 of SEQ ID NO:8; Zhao et al. (2011), supra). Mutation studies have revealed ARID2 to be a significant tumor suppressor in many cancer subtypes. ARID2 mutations are prevalent in hepatocellular carcinoma (Li et al. (2011) *Nature Genetics*. 43:828-829) and melanoma (Hodis et al. (2012) *Cell* 150:251-263; Krauthammer et al. (2012) *Nature Genetics*. 44:1006-1014). Mutations are present in a smaller but significant fraction in a wide range of other tumors (Shain and Pollack (2013), supra). ARID2 mutations are enriched in hepatitis C virus-associated hepatocellular carcinoma in the U.S. and European patient populations compared with the overall mutation frequency (Zhao et al. (2011), supra). The known binding partners for ARID2 include, e.g., Serum Response Factor (SRF) and SRF cofactors MYOCD, NKX2-5 and SRFBP1.

(40) The term “BAF200” or “ARID2” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. ReRepresentative human ARID2 cDNA and human ARID2 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human ARID2 isoforms are known. Human ARID2 isoform A (NP_689854.2) is encodable by the transcript variant 1 (NM_152641.3), which is the longer transcript. Human ARID2 isoform B (NP_001334768.1) is encodable by the transcript variant 2 (NM_001347839.1), which differs in the 3' UTR and 3' coding region compared to isoform A. The encoded isoform B has a shorter C-terminus compared to isoform A. Nucleic acid and polypeptide sequences of ARID2 orthologs in organisms other than humans are well known and include, for example, chimpanzee ARID2 (XM_016923581.1 and XP_016779070.1, and XM_016923580.1 and XP_016779069.1), Rhesus monkey ARID2 (XM_015151522.1 and XP_015007008.1), dog ARID2 (XM_003433553.2 and XP_003433601.2; and XM_014108583.1 and XP_013964058.1), cattle ARID2 (XM_002687323.5 and XP_002687369.1; and XM_015463314.1 and XP_015318800.1), mouse ARID2 (NM_175251.4 and NP_780460.3), rat ARID2 (XM_345867.8 and XP_345868.4; and XM_008776620.1 and XP_008774842.1), chicken ARID2 (XM_004937552.2 and XP_004937609.1, XM_004937551.2 and XP_004937608.1, XM_004937554.2 and XP_004937611.1, and XM_416046.5 and XP_416046.2), tropical clawed frog ARID2 (XM_002932805.4 and XP_002932851.1, XM_018092278.1 and XP_017947767.1, and XM_018092279.1 and XP_017947768.1), and zebrafish ARID2 (NM_001077763.1 and NP_001071231.1, and XM_005164457.3 and XP_005164514.1). ReRepresentative sequences of ARID2 orthologs are presented below in Table 1.

(41) Anti-ARID2 antibodies suitable for detecting ARID2 protein are well-known in the art and include, for example, antibodies ABE316 and 04-080 (EMD Millipore, Billerica, MA), antibodies NBP1-26615, NBP2-43567, and NBP1-26614 (Novus Biologicals, Littleton, CO), antibodies ab51019, ab166850, ab113283, and ab56082 (AbCam, Cambridge, MA), antibodies Cat #: PA5-35857 and PA5-51258 (ThermoFisher Scientific, Waltham, MA), antibodies GTX129444, GTX129443, and GTX632011 (GeneTex, Irvine, CA), ARID2 (H-182) Antibody, ARID2 (H-182) X Antibody, ARID2 (5-13) Antibody, ARID2 (5-13) X Antibody, ARID2 (E-3) Antibody, and ARID2 (E-3) X Antibody (Santa Cruz Biotechnology), etc. In addition, reagents are well-known for detecting ARID2 expression. Multiple clinical tests of PBRM1 are available in NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000541481.2, offered by Fulgent Clinical Diagnostics Lab (Temple City, CA)). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing ARID2 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA product #SR316272, shRNA products #TR306601, TR505226, TG306601, SR420583, and CRISPR products #KN212320 and KN30154 from Origene Technologies (Rockville, MD), RNAi product H00196528-R01 (Novus Biologicals), CRISPR gRNA products from GenScript (Cat. #KN301549 and KN212320, Piscataway, NJ) and from Santa Cruz (sc-401863), and RNAi products from Santa Cruz (Cat #sc-96225 and sc-77400). It is to be

noted that the term can further be used to refer to any combination of features described herein regarding ARID2 molecules. For example, any combination of sequence composition, percentage identify, sequence length, domain structure, functional activity, etc. can be used to describe an ARID2 molecule of the present invention.

(42) The term “loss-of-function mutation” for BAF200/ARID2 refers to any mutation in a ARID2-related nucleic acid or protein that results in reduced or eliminated ARID2 protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of ARID2. Such mutations reduce or eliminate ARID2 protein amounts and/or function by eliminating proper coding sequences required for proper ARID2 protein translation and/or coding for ARID2 proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a reRepresentative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated ARID2 protein amounts and/or function is described in the Tables and the Examples.

(43) The term “BRD7” refers to Bromodomain-containing protein 7, a subunit of the SWI/SNF complex, which can be found in PBAF but not BAF complexes. BRD7 is a transcriptional corepressor that binds to target promoters (e.g., the ESR1 promoter) and down-regulates the expression of target genes, leading to increased histone H3 acetylation at Lys-9 (H3K9ac). BRD7 can recruit other proteins such as BRCA1 and POU2F1 to, e.g., the ESR1 promoter for its function. BRD7 activates the Wnt signaling pathway in a DVL1-dependent manner by negatively regulating the GSK3B phosphotransferase activity, while BRD7 induces dephosphorylation of GSK3B at Tyr-216. BRD7 is also a coactivator for TP53-mediated activation of gene transcription and is required for TP53-mediated cell-cycle arrest in response to oncogene activation. BRD7 promotes acetylation of TP53 at Lys-382, and thereby promotes efficient recruitment of TP53 to target promoters. BRD7 also inhibits cell cycle progression from G1 to S phase. For studies on BRD7 functions, see Zhou et al. (2006) *J. Cell. Biochem.* 98:920-930; Harte et al. (2010) *Cancer Res.* 70:2538-2547; Drost et al. (2010) *Nat. Cell Biol.* 12:380-389. The known binding partners for BRD7 also include, e.g., Tripartite Motif Containing 24 (TRIM24), Protein Tyrosine Phosphatase, Non-Receptor Type 13 (PTPN13), Disheveled Segment Polarity Protein 1 (DVL1), interferon regulatory factor 2 (IRF2) (Staal et al. (2000) *J. Cell. Physiol.* US 185:269-279) and heterogeneous nuclear ribonucleoprotein U-like protein 1 (HNRPU1) (Kzhyshkowska et al. (2003) *Biochem. J. England.* 371:385-393). Human BRD7 protein has 651 amino acids and a molecular mass of 74139 Da, with a N-terminal nuclear localization signal (e.g., amino acids 65-96 of SEQ ID NO:14), a Bromo-BRD7-like domain (e.g., amino acids 135-232 of SEQ ID NO:14), and a DUF3512 domain (e.g., amino acids 287-533 of SEQ ID NO:14).

(44) The term “BRD7” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. ReRepresentative human BRD7 cDNA and human BRD7 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human BRD7 isoforms are known. Human BRD7 isoform A (NP_001167455.1) is encodable by the transcript variant 1 (NM_001173984.2), which is the longer transcript. Human BRD7 isoform B (NP_037395.2) is encodable by the transcript variant 2 (NM_013263.4), which uses an alternate in-frame splice site in the 3' coding region, compared to variant 1. The resulting isoform B lacks one internal residue, compared to isoform A. Nucleic acid and polypeptide sequences of BRD7 orthologs in organisms other than humans are well known and include, for example, chimpanzee BRD7 (XM_009430766.2 and

XP_009429041.1, XM_016929816.1 and XP_016785305.1, XM_016929815.1 and XP_016785304.1, and XM_003315094.4 and XP_003315142.1), Rhesus monkey BRD7 (XM_015126104.1 and XP_014981590.1, XM_015126103.1 and XP_014981589.1, XM_001083389.3 and XP_001083389.2, and XM_015126105.1 and XP_014981591.1), dog BRD7 (XM_014106954.1 and XP_013962429.1), cattle BRD7 (NM_001103260.2 and NP_001096730.1), mouse BRD7 (NM_012047.2 and NP_036177.1), chicken BRD7 (NM_001005839.1 and NP_001005839.1), tropical clawed frog BRD7 (NM_001008007.1 and NP_001008008.1), and zebrafish BRD7 (NM_213366.2 and NP_998531.2). Representative sequences of BRD7 orthologs are presented below in Table 1.

(45) Anti-BRD7 antibodies suitable for detecting BRD7 protein are well-known in the art and include, for example, antibody TA343710 (Origene), antibody NBP1-28727 (Novus Biologicals, Littleton, CO), antibodies ab56036, ab46553, ab202324, and ab114061 (AbCam, Cambridge, MA), antibodies Cat #: 15125 and 14910 (Cell Signaling), antibody GTX118755 (GeneTex, Irvine, CA), BRD7 (P-13) Antibody, BRD7 (T-12) Antibody, BRD7 (H-77) Antibody, BRD7 (H-2) Antibody, and BRD7 (B-8) Antibody (Santa Cruz Biotechnology), etc. In addition, reagents are well-known for detecting BRD7 expression. A clinical test of BRD7 is available in NIH Genetic Testing Registry (GTR®) with GTR Test ID: GTR000540400.2, offered by Fulgent Clinical Diagnostics Lab (Temple City, CA)). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing BRD7 expression can be found in the commercial product lists of the above-referenced companies, such as shRNA product #TR100001 and CRISPER products #KN302255 and KN208734 from Origene Technologies (Rockville, MD), RNAi product H00029117-R01 (Novus Biologicals), and small molecule inhibitors BI 9564 and TP472 (Tocris Bioscience, UK). It is to be noted that the term can further be used to refer to any combination of features described herein regarding BRD7 molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an BRD7 molecule of the present invention.

(46) The term “loss-of-function mutation” for BRD7 refers to any mutation in a BRD7-related nucleic acid or protein that results in reduced or eliminated BRD7 protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of BRD7. Such mutations reduce or eliminate BRD7 protein amounts and/or function by eliminating proper coding sequences required for proper BRD7 protein translation and/or coding for BRD7 proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a reRepresentative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated BRD7 protein amounts and/or function is described in the Tables and the Examples.

(47) The term “BAF45A” or “PHF10” refers to PHD finger protein 10, a subunit of the PBAF complex having two zinc finger domains at its C-terminus. PHF10 belongs to the neural progenitors-specific chromatin remodeling complex (npBAF complex) and is required for the proliferation of neural progenitors. During neural development a switch from a stem/progenitor to a post-mitotic chromatin remodeling mechanism occurs as neurons exit the cell cycle and become committed to their adult state. The transition from proliferating neural stem/progenitor cells to post-mitotic neurons requires a switch in subunit composition of the npBAF and nBAF complexes. As neural progenitors exit mitosis and differentiate into neurons, npBAF complexes which contain ACTL6A/BAF53A and PHF10/BAF45A, are exchanged for homologous alternative

ACTL6B/BAF53B and DPF1/BAF45B or DPF3/BAF45C subunits in neuron-specific complexes (nBAF). The npBAF complex is essential for the self-renewal/proliferative capacity of the multipotent neural stem cells. The nBAF complex along with CREST plays a role regulating the activity of genes essential for dendrite growth. PHF10 gene encodes at least two types of evolutionarily conserved, ubiquitously expressed isoforms that are incorporated into the PBAF complex in a mutually exclusive manner. One isoform contains C-terminal tandem PHD fingers, which in the other isoform are replaced by the consensus sequence for phosphorylation-dependent SUMO 1 conjugation (PDSM) (Brechalov et al. (2014) *Cell Cycle* 13:1970-1979). PBAF complexes containing different PHF10 isoforms can bind to the promoters of the same genes but produce different effects on the recruitment of Pol II to the promoter and on the level of gene transcription. PHF10 is a transcriptional repressor of caspase 3 and impares the programmed cell death pathway in human gastric cancer at the transcriptional level (Wei et al. (2010) *Mol Cancer Ther.* 9:1764-1774). Knockdown of PHF10 expression in gastric cancer cells led to significant induction of caspase-3 expression at both the RNA and protein levels and thus induced alteration of caspase-3 substrates in a time-dependent manner (Wei et al. (2010), supra). Results from luciferase assays by the same group indicated that PHF10 acted as a transcriptional repressor when the two PHD domains contained in PHF10 were intact. Human PHF10 protein has 498 amino acids and a molecular mass of 56051 Da, with two domains essential to induce neural progenitor proliferation (e.g., amino acids 89-185 and 292-334 of SEQ ID NO:20) and two PHD finger domains (e.g., amino acids 379-433 and 435-478 of SEQ ID NO:20). By similarity, PHF 10 binds to ACTL6A/BAF53A, SMARCA2/BRM/BAF190B, SMARCA4/BRG1/BAF190A and PBRM1/BAF180.

(48) The term “BAF45A” or “PHF10” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. ReRepresentative human PHF10 cDNA and human PHF10 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human PHF10 isoforms are known. Human PHF10 isoform A (NP_060758.2) is encodable by the transcript variant 1 (NM_018288.3), which is the longer transcript. Human PHF10 isoform B (NP_579866.2) is encodable by the transcript variant 2 (NM_133325.2), which uses an alternate splice junction which results in six fewer nt when compared to variant 1. The isoform B lacks 2 internal amino acids compared to isoform A. Nucleic acid and polypeptide sequences of PHF10 orthologs in organisms other than humans are well known and include, for example, chimpanzee PHF10 (XM_016956680.1 and XP_016812169.1, XM_016956679.1 and XP_016812168.1, and XM_016956681.1 and XP_016812170.1), Rhesus monkey PHF10 (XM_015137735.1 and XP_014993221.1, and XM_015137734.1 and XP_014993220.1), dog PHF10 (XM_005627727.2 and XP_005627784.1, XM_005627726.2 and XP_005627783.1, XM_532272.5 and XP_532272.4, XM_014118230.1 and XP_013973705.1, and XM_014118231.1 and XP_013973706.1), cattle PHF10 (NM_001038052.1 and NP_001033141.1), mouse PHF10 (NM_024250.4 and NP_077212.3), rat PHF10 (NM_001024747.2 and NP_001019918.2), chicken PHF10 (XM_015284374.1 and XP_015139860.1), tropical clawed frog PHF10 (NM_001030472.1 and NP_001025643.1), zebrafish PHF10 (NM_200655.3 and NP_956949.3), and *C. elegans* PHF10 (NM_001047648.2 and NP_001041113.1, NM_001047647.2 and NP_001041112.1, and NM_001313168.1 and NP_001300097.1). Representative sequences of PHF10 orthologs are presented below in Table 1.

(49) Anti-PHF10 antibodies suitable for detecting PHF10 protein are well-known in the art and include, for example, antibody TA346797 (Origene), antibodies NBP1-52879, NBP2-19795, NBP2-33759, and H00055274-B01P (Novus Biologicals, Littleton, CO), antibodies ab154637, ab80939, and ab68114 (AbCam, Cambridge, MA), antibody Cat #PA5-30678 (ThermoFisher Scientific), antibody Cat #26-352 (ProSci, Poway, CA), etc. In addition, reagents are well-known for detecting PHF10 expression. A clinical test of PHF10 for hereditary disease is available with the test ID no. GTR000536577 in NIH Genetic Testing Registry (GTR*), offered by Fulgent

Clinical Diagnostics Lab (Temple City, CA). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing PHF10 expression can be found in the commercial product lists of the above-referenced companies, such as siRNA product #sc-95343 and sc-152206 and CRISPR products #sc-410593 from Santa Cruz Biotechnology, RNAi products H00055274-R01 and H00055274-R02 (Novus Biologicals), and multiple CRISPR products from GenScript (Piscataway, NJ). Human PHF10 knockout cell (from HAP1 cell line) is also available from Horizon Discovery (Cat #HZGHC002778c011, UK). It is to be noted that the term can further be used to refer to any combination of features described herein regarding PHF10 molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an PHF10 molecule of the present invention.

(50) The term “loss-of-function mutation” for BAF45A/PHF10 refers to any mutation in a PHF10-related nucleic acid or protein that results in reduced or eliminated PHF10 protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of PHF10. Such mutations reduce or eliminate PHF10 protein amounts and/or function by eliminating proper coding sequences required for proper PHF10 protein translation and/or coding for PHF10 proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated PHF10 protein amounts and/or function is described in the Tables and the Examples.

(51) The term “PBRM1” or “BAF180” refers to protein Polybromo-1, which is a subunit of ATP-dependent chromatin-remodeling complexes. PBRM1 functions in the regulation of gene expression as a constituent of the evolutionary-conserved SWI/SNF chromatin remodeling complexes (Euskirchen et al. (2012) *J Biol. Chem.* 287:30897-30905). Beside BRD7 and BAF200, PBRM1 is one of the unique components of the SWI/SNF-B complex, also known as polybromo/BRG1-associated factors (or PBAF), absent in the SWI/SNF-A (BAF) complex (Xue et al. (2000) *Proc Natl Acad Sci USA.* 97:13015-13020; Brownlee et al. (2012) *Biochem Soc Trans.* 40:364-369). On that account, and because it contains bromodomains known to mediate binding to acetylated histones, PBRM1 has been postulated to target PBAF complex to specific chromatin sites, therefore providing the functional selectivity for the complex (Xue et al. (2000), *supra*; Lemon et al. (2001) *Nature* 414:924-928; Brownlee et al. (2012), *supra*). Although direct evidence for PBRM1 involvement is lacking, SWI/SNF complexes have also been shown to play a role in DNA damage response (Park et al. (2006) *EMBO J.* 25:3986-3997). In vivo studies have shown that PBRM1 deletion leads to embryonic lethality in mice, where PBRM1 is required for mammalian cardiac chamber maturation and coronary vessel formation (Wang et al. (2004) *Genes Dev.* 18:3106-3116; Huang et al. (2008) *Dev Biol.* 319:258-266). PBRM1 mutations are most predominant in renal cell carcinomas (RCCs) and have been detected in over 40% of cases, placing PBRM1 second (after VHL) on the list of most frequently mutated genes in this cancer (Varela et al. (2011) *Nature* 469:539-542; Hakimi et al. (2013) *Eur Urol.* 63:848-854; Pena-Llopis et al. (2012) *Nat Genet.* 44:751-759; Pawlowski et al. (2013) *Int J Cancer.* 132:E11-E17). PBRM1 mutations have also been found in a smaller group of breast and pancreatic cancers (Xia et al. (2008) *Cancer Res.* 68:1667-1674; Shain et al. (2012) *Proc Natl Acad Sci USA.* 109:E252-E259; Numata et al. (2013) *Int J Oncol.* 42:403-410). PBRM1 mutations are more common in patients with advanced stages (Hakimi et al. (2013), *supra*) and loss of PBRM1 protein expression has been

associated with advanced tumour stage, low differentiation grade and worse patient outcome (Pawlowski et al. (2013), supra). In another study, no correlation between PBRM1 status and tumour grade was found (Pena-Llopis et al. (2012), supra). Although PBRM1-mutant tumours are associated with better prognosis than BAP1-mutant tumours, tumours mutated for both PBRM1 and BAP1 exhibit the greatest aggressiveness (Kapur et al. (2013) *Lancet Oncol.* 14:159-167). PBRM1 is ubiquitously expressed during mouse embryonic development (Wang et al. (2004), supra) and has been detected in various human tissues including pancreas, kidney, skeletal muscle, liver, lung, placenta, brain, heart, intestine, ovaries, testis, prostate, thymus and spleen (Xue et al. (2000), supra; Horikawa and Barrett (2002) *DNA Seq.* 13:211-215).

(52) PBRM1 protein localises to the nucleus of cells (Nicolas and Goodwin (1996) *Gene* 175:233-240). As a component of the PBAF chromatin-remodelling complex, it associates with chromatin (Thompson (2009) *Biochimie.* 91:309-319), and has been reported to confer the localisation of PBAF complex to the kinetochores of mitotic chromosomes (Xue et al. (2000), supra). Human PBRM1 gene encodes a 1582 amino acid protein, also referred to as BAF180. Six bromodomains (BD1-6), known to recognize acetylated lysine residues and frequently found in chromatin-associated proteins, constitute the N-terminal half of PBRM1 (e.g., six BD domains at amino acid residue no. 44-156, 182-284, 383-484, 519-622, 658-762, and 775-882 of SEQ ID NO:2). The C-terminal half of PBRM1 contains two bromo-adjacent homology (BAH) domains (BAH1 and BAH2, e.g., at amino acid residue no. 957-1049 and 1130-1248 of SE ID NO:2), present in some proteins involved in transcription regulation. High mobility group (HMG) domain is located close to the C-terminus of PBRM1 (e.g., amino acid residue no. 1328-1377 of SEQ ID NO:2). HMG domains are found in a number of factors regulating DNA-dependent processes where HMG domains often mediate interactions with DNA.

(53) The term “PBRM1” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. ReRepresentative human PBRM1 cDNA and human PBRM1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human PBRM1 isoforms are known. Human PBRM1 transcript variant 2 (NM_181042.4) represents the longest transcript. Human PBRM1 transcript variant 1 (NM_018313.4, having a CDS from the 115-4863 nucleotide residue of SEQ ID NO:1) differs in the 5' UTR and uses an alternate exon and splice site in the 3' coding region, thus encoding a distinct protein sequence (NP_060783.3, as SEQ ID NO:2) of the same length as the isoform (NP_851385.1) encoded by variant 2. Nucleic acid and polypeptide sequences of PBRM1 orthologs in organisms other than humans are well known and include, for example, chimpanzee PBRM1 (XM_009445611.2 and XP_009443886.1, XM_009445608.2 and XP_009443883.1, XM_009445602.2 and XP_009443877.1, XM_016941258.1 and XP_016796747.1, XM_016941256.1 and XP_016796745.1, XM_016941249.1 and XP_016796738.1, XM_016941260.1 and XP_016796749.1, XM_016941253.1 and XP_016796742.1, XM_016941250.1 and XP_016796739.1, XM_016941261.1 and XP_016796750.1, XM_009445605.2 and XP_009443880.1, XM_016941252.1 and XP_016796741.1, XM_009445603.2 and XP_009443878.1, XM_016941263.1 and XP_016796752.1, XM_016941262.1 and XP_016796751.1, XM_009445604.2 and XP_009443879.1, XM_016941251.1 and XP_016796740.1, XM_016941257.1 and XP_016796746.1, XM_016941255.1 and XP_016796744.1, XM_016941254.1 and XP_016796743.1, XM_016941265.1 and XP_016796754.1, XM_016941264.1 and XP_016796753.1, XM_016941248.1 and XP_016796737.1, XM_009445617.2 and XP_009443892.1, XM_009445616.2 and XP_009443891.1, XM_009445619.2 and XP_009443894.1, XM_009445615.2 and XP_009443890.1, XM_009445618.2 and XP_009443893.1, and XM_016941266.1 and XP_016796755.1), rhesus monkey PBRM1 (XM_015130736.1 and XP_014986222.1, XM_015130739.1 and XP_014986225.1, XM_015130737.1 and XP_014986223.1, XM_015130740.1 and XP_014986226.1, XM_015130727.1 and XP_014986213.1,

XM_015130726.1 and XP_014986212.1, XM_015130728.1 and XP_014986214.1, XM_015130743.1 and XP_014986229.1, XM_015130731.1 and XP_014986217.1, XM_015130745.1 and XP_014986231.1, XM_015130741.1 and XP_014986227.1, XM_015130734.1 and XP_014986220.1, XM_015130744.1 and XP_014986230.1, XM_015130748.1 and XP_014986234.1, XM_015130746.1 and XP_014986232.1, XM_015130742.1 and XP_014986228.1, XM_015130747.1 and XP_014986233.1, XM_015130730.1 and XP_014986216.1, XM_015130732.1 and XP_014986218.1, XM_015130733.1 and XP_014986219.1, XM_015130735.1 and XP_014986221.1, XM_015130738.1 and XP_014986224.1, and XM_015130725.1 and XP_014986211.1), dog PBRM1 (XM_005632441.2 and XP_005632498.1, XM_014121868.1 and XP_013977343.1, XM_005632451.2 and XP_005632508.1, XM_014121867.1 and XP_013977342.1, XM_005632440.2 and XP_005632497.1, XM_005632446.2 and XP_005632503.1, XM_533797.5 and XP_533797.4, XM_005632442.2 and XP_005632499.1, XM_005632439.2 and XP_005632496.1, XM_014121869.1 and XP_013977344.1, XM_005632448.1 and XP_005632505.1, XM_005632449.1 and XP_005632506.1, XM_005632452.1 and XP_005632509.1, XM_005632445.1 and XP_005632502.1, XM_005632450.1 and XP_005632507.1, XM_005632453.1 and XP_005632510.1, XM_014121870.1 and XP_013977345.1, XM_005632443.1 and XP_005632500.1, XM_005632444.1 and XP_005632501.1, and XM_005632447.2 and XP_005632504.1), cow PBRM1 (XM_005222983.3 and XP_005223040.1, XM_005222979.3 and XP_005223036.1, XM_015459550.1 and XP_015315036.1, XM_015459551.1 and XP_015315037.1, XM_015459548.1 and XP_015315034.1, XM_010817826.1 and XP_010816128.1, XM_010817829.1 and XP_010816131.1, XM_010817830.1 and XP_010816132.1, XM_010817823.1 and XP_010816125.1, XM_010817824.2 and XP_010816126.1, XM_010817819.2 and XP_010816121.1, XM_010817827.2 and XP_010816129.1, XM_010817828.2 and XP_010816130.1, XM_010817817.2 and XP_010816119.1, and XM_010817818.2 and XP_010816120.1), mouse PBRM1 (NM_001081251.1 and NP_001074720.1), chicken PBRM1 (NM_205165.1 and NP_990496.1), tropical clawed frog PBRM1 (XM_018090224.1 and XP_017945713.1), zebrafish PBRM1 (XM_009305786.2 and XP_009304061.1, XM_009305785.2 and XP_009304060.1, and XM_009305787.2 and XP_009304062.1), fruit fly PBRM1 (NM_143031.2 and NP_651288.1), and worm PBRM1 (NM_001025837.3 and NP_001021008.1 and .NM_001025838.2 and NP_001021009.1). ReRepresentative sequences of PBRM1 orthologs are presented below in Table 1.

(54) Anti-PBRM1 antibodies suitable for detecting PBRM1 protein are well-known in the art and include, for example, ABE70 (rabbit polyclonal antibody, EMD Millipore, Billerica, MA), TA345237 and TA345238 (rabbit polyclonal antibodies, OriGene Technologies, Rockville, MD), NBP2-30673 (mouse monoclonal) and other polyclonal antibodies (Novus Biologicals, Littleton, CO), ab196022 (rabbit mAb, AbCam, Cambridge, MA), PAH437Hu01 and PAH437Hu02 (rabbit polyclonal antibodies, Cloud-Clone Corp., Houston, TX), GTX100781 (GeneTex, Irvine, CA), 25-498 (ProSci, Poway, CA), sc-367222 (Santa Cruz Biotechnology, Dallas, TX), etc. In addition, reagents are well-known for detecting PBRM1 expression (see, for example, PBRM1 Hu-Cy3 or Hu-Cy5 SmartFlare™ RNA Detection Probe (EMD Millipore)). Multiple clinical tests of PBRM1 are available in NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000537378.2 which is offered by Fulgent Clinical Diagnostics Lab (Temple City, CA)). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing PBRM1 expression can be found in the commercial product lists of the above-referenced companies. Ribavirin and PFI 3 are known PBRM1 inhibitors. It is to be noted that the term can further be used to refer to any combination of features described herein regarding PBRM1 molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe a PBRM1 molecule of the present invention.

(55) The term “PBRM1 loss-of-function mutation” refers to any mutation in a PBRM1-related nucleic acid or protein that results in reduced or eliminated PBRM1 protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of PBRM1. Such mutations reduce or eliminate PBRM1 protein amounts and/or function by eliminating proper coding sequences required for proper PBRM1 protein translation and/or coding for PBRM1 proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated PBRM1 protein amounts and/or function is described in the Tables and the Examples.

(56) The term “BAF250A” or “ARID1A” refers to AT-rich interactive domain-containing protein 1A, a subunit of the SWI/SNF complex, which can be found in BAF but not PBAF complex. In humans there are two BAF250 isoforms, BAF250A/ARID1A and BAF250B/ARID1B. They are thought to be E3 ubiquitin ligases that target histone H2B (Li et al. (2010) *Mol. Cell. Biol.* 30:1673-1688). ARID1A is highly expressed in the spleen, thymus, prostate, testes, ovaries, small intestine, colon and peripheral leukocytes. ARID1A is involved in transcriptional activation and repression of select genes by chromatin remodeling. It is also involved in vitamin D-coupled transcription regulation by associating with the WINAC complex, a chromatin-remodeling complex recruited by vitamin D receptor. ARID1A belongs to the neural progenitors-specific chromatin remodeling (npBAF) and the neuron-specific chromatin remodeling (nBAF) complexes, which are involved in switching developing neurons from stem/progenitors to post-mitotic chromatin remodeling as they exit the cell cycle and become committed to their adult state. ARID1A also plays key roles in maintaining embryonic stem cell pluripotency and in cardiac development and function (Lei et al. (2012) *J Biol. Chem.* 287:24255-24262; Gao et al. (2008) *Proc. Natl. Acad. Sci. U.S.A.* 105:6656-6661). Loss of BAF250a expression was seen in 42% of the ovarian clear cell carcinoma samples and 21% of the endometrioid carcinoma samples, compared with just 1% of the high-grade serous carcinoma samples. ARID1A deficiency also impairs the DNA damage checkpoint and sensitizes cells to PARP inhibitors (Shen et al. (2015) *Cancer Discov.* 5:752-767). Human ARID1A protein has 2285 amino acids and a molecular mass of 242045 Da, with at least a DNA-binding domain that can specifically bind an AT-rich DNA sequence, recognized by a SWI/SNF complex at the beta-globin locus, and a C-terminus domain for glucocorticoid receptor-dependent transcriptional activation. ARID1A has been shown to interact with proteins such as SMARCB1/BAF47 (Kato et al. (2002) *J. Biol. Chem.* 277:5498-505; Wang et al. (1996) *EMBO J.* 15:5370-5382) and SMARCA4/BRG1 (Wang et al. (1996), supra; Zhao et al. (1998) *Cell* 95:625-636), etc.

(57) The term “BAF250A” or “ARID1A” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BAF250A (ARID1A) cDNA and human BAF250A (ARID1A) protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, two different human ARID1A isoforms are known. Human ARID1A isoform A (NP_006006.3) is encodable by the transcript variant 1 (NM_006015.4), which is the longer transcript. Human ARID1A isoform B (NP_624361.1) is encodable by the transcript variant 2 (NM_139135.2), which lacks a segment in the coding region compared to variant 1. Isoform B thus lacks an internal segment, compared to isoform A. Nucleic acid and polypeptide sequences of ARID1A orthologs in organisms other than humans are well known and include, for example, chimpanzee ARID1A (XM_016956953.1 and XP_016812442.1, XM_016956958.1 and XP_016812447.1, and XM_009451423.2 and

XP_009449698.2), Rhesus monkey ARID1A (XM_015132119.1 and XP_014987605.1, and XM_015132127.1 and XP_014987613.1), dog ARID1A (XM_847453.5 and XP_852546.3, XM_005617743.2 and XP_005617800.1, XM_005617742.2 and XP_005617799.1, XM_005617744.2 and XP_005617801.1, XM_005617746.2 and XP_005617803.1, and XM_005617745.2 and XP_005617802.1), cattle ARID1A (NM_001205785.1 and NP_001192714.1), mouse ARID1A (NM_001080819.1 and NP_001074288.1), rat ARID1A (NM_001106635.1 and NP_001100105.1), chicken ARID1A (XM_015297557.1 and XP_015153043.1, XM_015297556.1 and XP_015153042.1, and XM_417693.5 and XP_417693.5), tropical clawed frog ARID1A (XM_002934639.4 and XP_002934685.2), and zebrafish ARID1A (XM_009294131.2 and XP_009292406.1, and XM_009294132.2 and XP_009292407.1).

(58) Anti-ARID1A antibodies suitable for detecting ARID1A protein are well-known in the art and include, for example, antibody Cat #04-080 (EMD Millipore, Billerica, MA), antibodies TA349170, TA350870, and TA350871 (OriGene Technologies, Rockville, MD), antibodies NBP1-88932, NB100-55334, NBP2-43566, NB100-55333, and H00008289-Q01 (Novus Biologicals, Littleton, CO), antibodies ab182560, ab182561, ab176395, and ab97995 (AbCam, Cambridge, MA), antibodies Cat #: 12354 and 12854 (Cell Signaling Technology, Danvers, MA), antibodies GTX129433, GTX129432, GTX632013, GTX12388, and GTX31619 (GeneTex, Irvine, CA), etc. In addition, reagents are well-known for detecting ARID1A expression. For example, multiple clinical tests for ARID1A are available at NIH Genetic Testing Registry (GTR©) (e.g., GTR Test ID: GTR000520952.1 for mental retardation, offered by Centogene AG, Germany). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing ARID1A Expression can be found in the commercial product lists of the above-referenced companies, such as RNAi products H00008289-R01, H00008289-R02, and H00008289-R03 (Novus Biologicals) and CRISPR products KN301547G1 and KN301547G2 (Origene). Other CRISPR products include sc-400469 (Santa Cruz Biotechnology) and those from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding ARID1A molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an ARID1A molecule of the present invention.

(59) The term “loss-of-function mutation” for BAF250A/ARID1A refers to any mutation in an ARID1A-related nucleic acid or protein that results in reduced or eliminated ARID1A protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of ARID1A. Such mutations reduce or eliminate ARID1A protein amounts and/or function by eliminating proper coding sequences required for proper ARID1A protein translation and/or coding for ARID1A proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated ARID1A protein amounts and/or function is described in the Tables and the Examples.

(60) The term “BAF250B” or “ARID1B” refers to AT-rich interactive domain-containing protein 1B, a subunit of the SWI/SNF complex, which can be found in BAF but not PBAF complex. ARID1B and ARID1A are alternative and mutually exclusive ARID-subunits of the SWI/SNF complex. Germline mutations in ARID1B are associated with Coffin-Siris syndrome (Tsurusaki et al. (2012) *Nat. Genet.* 44:376-378; Santen et al. (2012) *Nat. Genet.* 44:379-380). Somatic

mutations in ARID1B are associated with several cancer subtypes, suggesting that it is a tumor suppressor gene (Shai and Pollack (2013) *PLoS ONE* 8:e55119; Sausen et al. (2013) *Nat. Genet.* 45:12-17; Shain et al. (2012) *Proc. Natl. Acad. Sci. U.S.A.* 109:E252-E259; Fujimoto et al. (2012) *Nat. Genet.* 44:760-764). Human ARID1A protein has 2236 amino acids and a molecular mass of 236123 Da, with at least a DNA-binding domain that can specifically bind an AT-rich DNA sequence, recognized by a SWI/SNF complex at the beta-globin locus, and a C-terminus domain for glucocorticoid receptor-dependent transcriptional activation. ARID1B has been shown to interact with SMARCA4/BRG1 (Hurlstone et al. (2002) *Biochem. J.* 364:255-264; Inoue et al. (2002) *J Biol. Chem.* 277:41674-41685 and SMARCA2/BRM (Inoue et al. (2002), supra).

(61) The term “BAF250B” or “ARID1B” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human BAF250B (ARID1B) cDNA and human BAF250B (ARID1B) protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, three different human ARID1B isoforms are known. Human ARID1B isoform A (NP_059989.2) is encodable by the transcript variant 1 (NM_017519.2). Human ARID1B isoform B (NP_065783.3) is encodable by the transcript variant 2 (NM_020732.3). Human ARID1B isoform C (NP_001333742.1) is encodable by the transcript variant 3 (NM_001346813.1). Nucleic acid and polypeptide sequences of ARID1B orthologs in organisms other than humans are well known and include, for example, Rhesus monkey ARID1B (XM_015137088.1 and XP_014992574.1), dog ARID1B (XM_014112912.1 and XP_013968387.1), cattle ARID1B (XM_010808714.2 and XP_010807016.1, and XM_015464874.1 and XP_015320360.1), mouse ARID1B (NM_001085355.1 and NP_001078824.1), rat ARID1B (XM_017604567.1 and XP_017460056.1), chicken ARID1B (XM_015284235.1 and XP_015139721.1, XM_015284233.1 and XP_015139719.1, XM_015284238.1 and XP_015139724.1, XM_015284230.1 and XP_015139716.1, XM_015284234.1 and XP_015139720.1, XM_015284231.1 and XP_015139717.1, XM_015284232.1 and XP_015139718.1, XM_015284236.1 and XP_015139722.1, and XM_015284237.1 and XP_015139723.1), tropical clawed frog ARID1B (XM_004914629.3 and XP_004914686.1, XM_004914631.3 and XP_004914688.1, XM_004914630.3 and XP_004914687.1, XM_004914634.3 and XP_004914691.1, XM_002931507.4 and XP_002931553.2, XM_004914632.3 and XP_004914689.1, XM_004914635.3 and XP_004914692.1, XM_004914633.3 and XP_004914690.1, XM_004914636.3 and XP_004914693.1, and XM_004914637.3 and XP_004914694.1), and zebrafish ARID1B (XM_009294544.2 and XP_009292819.1, XM_009294545.2 and XP_009292820.1, XM_005160356.3 and XP_005160413.1, XM_005160355.3 and XP_005160412.1, XM_005160354.3 and XP_005160411.1, and XM_692987.8 and XP_698079.4).

(62) Anti-ARID1B antibodies suitable for detecting ARID1B protein are well-known in the art and include, for example, antibody Cat #ABE316 (EMD Millipore, Billerica, MA), antibody TA315663 (OriGene Technologies, Rockville, MD), antibodies H00057492-M02, H00057492-MO1, NB100-57485, NBP1-89358, and NB100-57484 (Novus Biologicals, Littleton, CO), antibodies ab57461, ab69571, ab84461, and ab163568 (AbCam, Cambridge, MA), antibodies Cat #: PA5-38739, PA5-49852, and PA5-50918 (ThermoFisher Scientific, Danvers, MA), antibodies GTX130708, GTX60275, and GTX56037 (GeneTex, Irvine, CA), ARID1B (KMN1) Antibody and other antibodies (Santa Cruz Biotechnology), etc. In addition, reagents are well-known for detecting ARID1B expression. For example, multiple clinical tests for ARID1B are available at NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000520953.1 for mental retardation, offered by Centogene AG, Germany). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing ARID1B Expression can be found in the commercial product lists of the above-referenced companies, such as RNAi products H00057492-R03, H00057492-R01, and H00057492-R02 (Novus Biologicals) and CRISPR products KN301548 and KN214830 (Origene). Other CRISPR products include sc-402365 (Santa Cruz Biotechnology) and those from GenScript (Piscataway,

NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding ARID1B molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an ARID1B molecule of the present invention.

(63) The term “loss-of-function mutation” for BAF250B/ARID1B refers to any mutation in an ARID1B-related nucleic acid or protein that results in reduced or eliminated ARID1B protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of ARID1B. Such mutations reduce or eliminate ARID1B protein amounts and/or function by eliminating proper coding sequences required for proper ARID1B protein translation and/or coding for ARID1B proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated ARID1B protein amounts and/or function is described in the Tables and the Examples.

(64) The term “CRB1” refers to Crumbs homolog 1, a protein similar to the *Drosophila* crumbs protein and localizes to the inner segment of mammalian photoreceptors. In *Drosophila* crumbs localizes to the stalk of the fly photoreceptor and may be a component of the molecular scaffold that controls proper development of polarity in the eye. CRB1 gene is involved in the Hippo signaling pathway. Mutations in this gene are associated with a severe form of retinitis pigmentosa, RP12, and with Leber congenital amaurosis. One study suggests that mutations in this gene are associated with keratoconus in patients that already have Leber's congenital amaurosis (McMahon et al. (2009) *Invest. Ophthalmol. Vis. Sci.* 50:3185-3187). CRB1 mutation is also related to lung squamous cell carcinoma (SQCC) (Li et al. (2015) *Sci. Rep.* 5:Article 14237) and retinal dystrophy (Li et al. (2014) *Int J Mol Med* 33:913-918). The human CRB1 protein has 1406 amino acids and a molecular mass of 154183 Da.

(65) The term “CRB1” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human CRB1 cDNA and human CRB1 protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, four different human CRB1 isoforms are known. Human CRB1 isoform A (NP_957705.1) is the longest isoform and is encodable by the transcript variant 1 (NM_201253.2). Human CRB1 isoform B (NP_001180569.1) is encodable by the transcript variant 2 (NM_001193640.1), which lacks two in-frame exons compared to variant 1. The resulting isoform B has the same N- and C-termini but is shorter compared to isoform A. Human CRB1 isoform C (NP_001244894.1) is encodable by the transcript variant 3 (NM_001257965.1), which contains three noncoding exons in place of the first exon and contains an alternate in-frame exon compared to variant 1. The resulting isoform C is shorter at the N-terminus and contains an alternate internal segment compared to isoform A. Human CRB1 isoform D (NP_001244895.1) is encodable by the transcript variant 4 (NM_001257966.1), which lacks an alternate in-frame segment of two coding exons and most of a third compared to variant 1. The resulting isoform D has the same N- and C-termini but lacks an alternate internal segment compared to isoform A. Nucleic acid and polypeptide sequences of CRB1 orthologs in organisms other than humans are well known and include, for example, chimpanzee CRB1 (XM_009440300.2 and XP_009438575.1, XM_009440289.2 and XP_009438564.1, XM_009440291.2 and XP_009438566.1, XM_016934908.1 and XP_016790397.1, XM_016934919.1 and XP_016790408.1, XM_016934927.1 and XP_016790416.1, XM_525009.5 and XP_525009.2, and

XM_016934898.1 and XP_016790387.1), Rhesus monkey CRB1 (XM_015120817.1 and XP_014976303.1, XM_001110878.3 and XP_001110878.2, XM_001110912.3 and XP_001110912.2, XM_015120808.1 and XP_014976294.1, and XM_015120812.1 and XP_014976298.1), dog CRB1 (XM_014115056.1 and XP_013970531.1, XM_014115058.1 and XP_013970533.1, XM_005622293.2 and XP_005622350.1, and XM_014115057.1 and XP_013970532.1), cattle CRB1 (XM_010813559.2 and XP_010811861.1), mouse CRB1 (NM_133239.2 and NP_573502.2), rat CRB1 (NM_001107182.1 and NP_001100652.1), chicken CRB1 (XM_015290380.1 and XP_015145866.1, and XM_003641670.3 and XP_003641718.2), tropical clawed frog ARID1B (XM_018093205.1 and XP_017948694.1), and zebrafish CRB1 (NM_001044943.1 and NP_001038408.1).

(66) Anti-CRB1 antibodies suitable for detecting CRB1 protein are well-known in the art and include, for example, antibody Cat #MABN1572 and ABE553 (EMD Millipore, Billerica, MA), antibody TA319859 (OnGene Technologies, Rockville, MD), antibody NBP2-41201 (Novus Biologicals, Littleton, CO), antibody ab156282 (AbCam, Cambridge, MA), antibody GTX32103 (GeneTex, Irvine, CA), CRB1 (H-14) Antibody (Santa Cruz Biotechnology), etc. In addition, reagents are well-known for detecting CRB1 expression. For example, multiple clinical tests for CRB1 are available at NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000515886.2 for retinitis pigmentosa type 12, offered by Centogene AG, Germany).

Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing CRB1 Expression can be found in the commercial product lists of the above-referenced companies, such as RNAi products H00023418-R01 and H00023418-R02 (Novus Biologicals) and CRISPR products KN303799 and KN212347 (Origene). Other CRISPR products include sc-418097 (Santa Cruz Biotechnology) and those from GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding CRB1 molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an CRB1 molecule of the present invention.

(67) The term “loss-of-function mutation” for CRB1 refers to any mutation in a CRB1-related nucleic acid or protein that results in reduced or eliminated CRB1 protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of CRB1. Such mutations reduce or eliminate CRB1 protein amounts and/or function by eliminating proper coding sequences required for proper CRB1 protein translation and/or coding for CRB1 proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated CRB1 protein amounts and/or function is described in the Tables and the Examples.

(68) The term “EGFR” refers to the epidermal growth factor receptor, a transmembrane glycoprotein that is a member of the ErbB family of receptors, a subfamily of four closely related receptor tyrosine kinases: EGFR (ErbB-1), HER2/c-neu (ErbB-2), Her 3 (ErbB-3) and Her 4 (ErbB-4). This protein is a receptor for members of the epidermal growth factor family. Binding of the protein to a ligand induces receptor homo- and/or heterodimerization and tyrosine autophosphorylation on key cytoplasmic residues. The activated EGFR then recruits adapter proteins like GRB2 which in turn activates complex downstream signaling cascades, leading to cell proliferation. Known ligands of EGFR include EGF, TGFA/TGF-alpha, amphiregulin, epigen/EPGN, BTC/betacellulin, epiregulin/EREG and HBEGF/heparin-binding EGF. While being

activated, autophosphorylation of several tyrosine (Y) residues in the C-terminal domain of EGFR occurs. These include Y992, Y1045, Y1068, Y1148 and Y1173, as shown in the adjacent diagram (Downward et al. (1984) *Nature* 311:483-485). This autophosphorylation elicits downstream activation and signaling by several other proteins that associate with the phosphorylated tyrosines through their own phosphotyrosine-binding SH2 domains. These downstream signaling proteins initiate several signal transduction cascades, principally the MAPK, Akt and JNK pathways, leading to DNA synthesis and cell proliferation (Oda et al. (2005) *Mol. Sys. Biol.* 1:2005.0010). Such proteins modulate phenotypes such as cell migration, adhesion, and proliferation. Activation of the receptor is important for the innate immune response in human skin. The kinase domain of EGFR can also cross-phosphorylate tyrosine residues of other receptors it is aggregated with, and can itself be activated in that manner. EGFR activates at least 4 major downstream signaling cascades including the RAS-RAF-MEK-ERK, PI3 kinase-AKT, PLCgamma-PKC and STATs modules. EGFR may also activate the NF-kappa-B signaling cascade and other proteins like RGS16, by activating its GTPase activity, and probably coupling the EGF receptor signaling to the G protein-coupled receptor signaling. EGFR also phosphorylates MUC1 and increases its interaction with SRC and CTNNB1/beta-catenin. Mutations that lead to EGFR overexpression (i.e., upregulation) or overactivity have been associated with a number of cancers, including squamous-cell carcinoma of the lung (80% of cases), anal cancers (Walker et al. (2009) *Hum. Pathol.* 40:1517-1527), glioblastoma (50%) and epithelial tumors of the head and neck (80-100%) (Kumar et al. (2013) *Robbins basic pathology*. Philadelphia: Elsevier/Saunders. p. 179). These somatic mutations involving EGFR lead to its constant activation, which produces uncontrolled cell division. In glioblastoma a more or less specific mutation of EGFR, called EGFRvIII is often observed (Kuan et al. (2001) *Endocr. Relat. Cancer.* 8:83-96). Aberrant EGFR signaling has been implicated in psoriasis, eczema and atherosclerosis (Jost et al. (2000) *Eur. J Dermatol.* 10:505-510; Dreux et al. (2006) *Atherosclerosis* 186:38-53). However, its exact roles in these conditions are ill-defined. Human EGFR protein has 1210 amino acids and a molecular mass of 134277 Da, with at least a receptor L domain (amino acid no. 57-168 of SEQ ID NO:92), a Furin-like domain (amino acid no. 185-335 of SDEQ ID NO:92), another receptor L domain (amino acid no. 361-481 of SEQ ID NO:92), a growth factor receptor domain IV (amino acid no. 505-637 of SEQ ID NO: 92), a transmembrane region (amino acid no. 646-668 of SEQ ID NO:92), and a catalytic domain of the protein tyrosine kinase family (amino acid no. 704-1016 of SEQ ID NO:92). The structure and domains of human EGFR may be found at the World Wide Web address of www.uniprot.org/uniprot/P00533#structure and www.ebi.ac.uk/interpro/protein/P00533. EGFR has been shown to interact with proteins such as AR, ARF4, CAV1, CAV3, CBL, CBLB, CBLC, CD44, CDC25A, CRK, CTNNB1, DCN, EGF, GRB14, Grb2, JAK2, MUC1, NCK1, NCK2, PKC alpha, PLCG1, PLSCR1, PTPN1, PTPN11, PTPN6, PTPRK, SH2D3A, SH3KBP1, SHC1, SOS1, Src, STAT1, STAT3, STAT5A, UBC, and WAS.

(69) The term “EGFR” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human EGFR cDNA and human EGFR protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, nine different human EGFR isoforms are known. Human EGFR isoform A (NP_005219.2), the longest isoform, is encodable by the transcript variant 1 (NM_005228.4). Human EGFR isoform B (NP_958439.1) is encodable by the transcript variant 2 (NM_201282.1), which uses a different 3' terminal exon when compared to variant 1. The resulting isoform B has a shorter and distinct C-terminus. Human EGFR isoform C (also known as ErbB1-S, NP_958440.1) is encodable by the transcript variant 3 (NM_201283.1), which uses a different 3' terminal exon when compared to variant 1. The resulting isoform C has a shorter and distinct C-terminus. Only the extracellular domain is present in isoform C. Human EGFR isoform D (NP_958441.1) is encodable by the transcript variant 4 (NM_201284.1), which uses a different 3' terminal exon when compared to variant 1. The resulting isoform D has a shorter and distinct C-

terminus. Only the extracellular domain is present in isoform D. Human EGFR isoform E (NP_001333826.1) is encodable by the transcript variant 5 (NM_001346897.1), which lacks an in-frame exon in the 5' coding region and its 3' terminal exon extends past a splice site that is used in variant 1. The encoded isoform E is shorter and has a distinct C-terminus compared to isoform A. Human EGFR isoform F (NP_001333827.1) is encodable by the transcript variant 6 (NM_001346898.1), which has a 3' terminal exon that extends past a splice site that is used in variant 1. The encoded isoform F has a shorter and distinct C-terminus compared to isoform A. Human EGFR isoform G (NP_001333828.1) is encodable by the transcript variant 7 (NM_001346899.1), which lacks an in-frame exon in the 5' coding region, compared to variant 1. Human EGFR isoform H (NP_001333829.1) is encodable by the transcript variant 8 (NM_001346900.1), which uses a novel 5' terminal exon compared to variant 1. The encoded isoform H has a shorter and distinct N-terminus compared to isoform A. Human EGFR isoform I (a.k.a. EGFRvIII, delta-EGFR, and de2-7EGFR; NP_001333870.1) is encodable by the transcript variant 9 (NM_001346941.1), which has an in-frame deletion of six exons in the 5' coding region, compared to variant 1. The encoded isoform I has a shorter extracellular domain compared to isoform A. This variant is considered to be tumorigenic and the encoded protein lacks normal ligand binding ability and is constitutively active. Nucleic acid and polypeptide sequences of EGFR orthologs in organisms other than humans are well known and include, for example, chimpanzee EGFR (XM_519102.6 and XP_519102.3, and XM_001156264.5 and XP_001156264.1), Rhesus monkey EGFR (XM_015133436.1 and XP_014988922.1, and XM_015133437.1 and XP_014988923.1), dog EGFR (XM_014120756.1 and XP_013976231.1), cattle EGFR (XM_002696890.4 and XP_002696936.2, and XM_592211.8 and XP_592211.4), mouse EGFR (NM_007912.4 and NP_031938.1, and NM_207655.2 and NP_997538.1), rat EGFR (NM_031507.1 and NP_113695.1, XM_008770416.2 and XP_008768638.1, XM_008770418.2 and XP_008768640.1, and XM_017599073.1 and XP_017454562.1), chicken EGFR (NM_205497.2 and NP_990828.2), tropical clawed frog EGFR (XM_002939914.4 and XP_002939960.2), and zebrafish EGFR (NM_194424.1 and NP_919405.1).

(70) Anti-EGFR antibodies suitable for detecting EGFR protein are well-known in the art and include, for example, antibody Cat #06-847 (EMD Millipore, Billerica, MA), antibodies AM00029BT-N, AM00029PU-N, and others (OnGene Technologies, Rockville, MD), antibodies Cat #MAB8967, AF231, AF1095, and others (R&D Systems, Minneapolis, MN), antibodies NB120-10414, NBP1-84814, and others (Novus Biologicals, Littleton, CO), antibodies ab52894, ab40815, and others (AbCam, Cambridge, MA), antibodies Cat #: 4267, 2244, 48685, and others (Cell Signaling Technology, Danvers, MA), antibodies GTX121919, GTX628887, and others (GeneTex, Irvine, CA), etc. In addition, reagents are well-known for detecting EGFR expression. For example, multiple clinical tests for EGFR are available at NIH Genetic Testing Registry (GTR®) (e.g., GTR Test ID: GTR000514557.2 for EGFR mutation by Sanger Sequencing, offered by Cancer Genetics, Inc. (Rutherford, NJ), GTR Test ID: GTR000510455.1 for lung cancer, offered by Centogene AG, Germany, and other tests). Commercial ELISA kits for detecting EGFR are available, at least, from R&D Systems (Cat #DYC1095B-2, DYC1854-2, DEGFRO, DYC3570-2, etc.). Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing EGFR Expression can be found in the commercial product lists of the above-referenced companies, such as RNAi products SC-29301, SC-44340, and others and CRISPR products sc-400015 (Santa Cruz Biotechnology). Other similar products include TG320326, TR320326, TG509941, and others shRNA products, as well as KN214877, KN204201, and others CRISPR products (Origene). Small molecule compounds are known to regulate EGFR expression, such as Cat. #A8197 and other inhibitors (ApexBio, Houston, TX), CAS 879127-07-8 and other inhibitors or activators (EMD Millipore). Known EGFR inhibitory drugs include, at least, Iressa™ (gefitinib), Tarceva™ (erlotinib), Tykerb™ (lapatinib), Erbitux™ (cetuximab), Vectibix™ (panitumumab), Caprelsa™ (vandetanib), Tagrisso™ (osimertinib), Portrazza™ (necitumumab), etc. It is to be noted that the

term can further be used to refer to any combination of features described herein regarding EGFR molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an EGFR molecule of the present invention.

(71) The term “loss-of-function mutation” for EGFR refers to any mutation in an EGFR-related nucleic acid or protein that results in reduced or eliminated EGFR protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of EGFR. Such mutations reduce or eliminate EGFR protein amounts and/or function by eliminating proper coding sequences required for proper EGFR protein translation and/or coding for EGFR proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated EGFR protein amounts and/or function is described in the Tables and the Examples. In some embodiments, the term “hotspot mutation” for EGFR refers to a mutation that is commonly known to be mutated in EGFR associated with cancer. In some instances, such “hotspot mutations” can be those known to cause resistance to anti-EGFR therapies such as those described in Example 4.

(72) Unless otherwise specified here within, the terms “antibody” and “antibodies” broadly encompass naturally-occurring forms of antibodies (e.g. IgG, IgA, IgM, IgE) and recombinant antibodies such as single-chain antibodies, chimeric and humanized antibodies and multi-specific antibodies, as well as fragments and derivatives of all of the foregoing, which fragments and derivatives have at least an antigenic binding site. Antibody derivatives may comprise a protein or chemical moiety conjugated to an antibody.

(73) The term “antibody” as used herein also includes an “antigen-binding portion” of an antibody (or simply “antibody portion”). The term “antigen-binding portion”, as used herein, refers to one or more fragments of an antibody that retain the ability to specifically bind to an antigen (e.g., a biomarker polypeptide or fragment thereof). It has been shown that the antigen-binding function of an antibody can be performed by fragments of a full-length antibody. Examples of binding fragments encompassed within the term “antigen-binding portion” of an antibody include (i) a Fab fragment, a monovalent fragment consisting of the VL, VH, CL and CHI domains; (ii) a F(ab').sub.2 fragment, a bivalent fragment comprising two Fab fragments linked by a disulfide bridge at the hinge region; (iii) a Fd fragment consisting of the VH and CHI domains; (iv) a Fv fragment consisting of the VL and VH domains of a single arm of an antibody, (v) a dAb fragment (Ward et al. (1989) *Nature* 341:544-546), which consists of a VH domain; and (vi) an isolated complementarity determining region (CDR). Furthermore, although the two domains of the Fv fragment, VL and VH, are coded for by separate genes, they can be joined, using recombinant methods, by a synthetic linker that enables them to be made as a single protein chain in which the VL and VH regions pair to form monovalent polypeptides (known as single chain Fv (scFv); see e.g., Bird et al. (1988) *Science* 242:423-426; and Huston et al. (1988) *Proc. Natl. Acad. Sci. USA* 85:5879-5883; and Osbourn et al. 1998, *Nature Biotechnology* 16: 778). Such single chain antibodies are also intended to be encompassed within the term “antigen-binding portion” of an antibody. Any VH and VL sequences of specific scFv can be linked to human immunoglobulin constant region cDNA or genomic sequences, in order to generate expression vectors encoding complete IgG polypeptides or other isotypes. VH and VL can also be used in the generation of Fab, Fv or other fragments of immunoglobulins using either protein chemistry or recombinant DNA

technology. Other forms of single chain antibodies, such as diabodies are also encompassed. Diabodies are bivalent, bispecific antibodies in which VH and VL domains are expressed on a single polypeptide chain, but using a linker that is too short to allow for pairing between the two domains on the same chain, thereby forcing the domains to pair with complementary domains of another chain and creating two antigen binding sites (see e.g., Holliger, P., et al. (1993) *Proc. Natl. Acad. Sci. USA* 90:6444-6448; Poljak, R. J., et al. (1994) *Structure* 2:1121-1123).

(74) Still further, an antibody or antigen-binding portion thereof may be part of larger immunoadhesion polypeptides, formed by covalent or noncovalent association of the antibody or antibody portion with one or more other proteins or peptides. Examples of such immunoadhesion polypeptides include use of the streptavidin core region to make a tetrameric scFv polypeptide (Kipriyanov, S. M., et al. (1995) *Human Antibodies and Hybridomas* 6:93-101) and use of a cysteine residue, biomarker peptide and a C-terminal polyhistidine tag to make bivalent and biotinylated scFv polypeptides (Kipriyanov, S. M., et al. (1994) *Mol. Immunol.* 31:1047-1058). Antibody portions, such as Fab and F(ab').sub.2 fragments, can be prepared from whole antibodies using conventional techniques, such as papain or pepsin digestion, respectively, of whole antibodies. Moreover, antibodies, antibody portions and immunoadhesion polypeptides can be obtained using standard recombinant DNA techniques, as described herein.

(75) Antibodies may be polyclonal or monoclonal; xenogeneic, allogeneic, or syngeneic; or modified forms thereof (e.g. humanized, chimeric, etc.). Antibodies may also be fully human. Preferably, antibodies of the present invention bind specifically or substantially specifically to a biomarker polypeptide or fragment thereof. The terms “monoclonal antibodies” and “monoclonal antibody composition”, as used herein, refer to a population of antibody polypeptides that contain only one species of an antigen binding site capable of immunoreacting with a particular epitope of an antigen, whereas the term “polyclonal antibodies” and “polyclonal antibody composition” refer to a population of antibody polypeptides that contain multiple species of antigen binding sites capable of interacting with a particular antigen. A monoclonal antibody composition typically displays a single binding affinity for a particular antigen with which it immunoreacts.

(76) Antibodies may also be “humanized”, which is intended to include antibodies made by a non-human cell having variable and constant regions which have been altered to more closely resemble antibodies that would be made by a human cell. For example, by altering the non-human antibody amino acid sequence to incorporate amino acids found in human germline immunoglobulin sequences. The humanized antibodies of the present invention may include amino acid residues not encoded by human germline immunoglobulin sequences (e.g., mutations introduced by random or site-specific mutagenesis in vitro or by somatic mutation in vivo), for example in the CDRs. The term “humanized antibody”, as used herein, also includes antibodies in which CDR sequences derived from the germline of another mammalian species, such as a mouse, have been grafted onto human framework sequences.

(77) The term “assigned score” refers to the numerical value designated for each of the biomarkers after being measured in a patient sample. The assigned score correlates to the absence, presence or inferred amount of the biomarker in the sample. The assigned score can be generated manually (e.g., by visual inspection) or with the aid of instrumentation for image acquisition and analysis. In certain embodiments, the assigned score is determined by a qualitative assessment, for example, detection of a fluorescent readout on a graded scale, or quantitative assessment. In one embodiment, an “aggregate score,” which refers to the combination of assigned scores from a plurality of measured biomarkers, is determined. In one embodiment the aggregate score is a summation of assigned scores. In another embodiment, combination of assigned scores involves performing mathematical operations on the assigned scores before combining them into an aggregate score. In certain, embodiments, the aggregate score is also referred to herein as the “predictive score.”

(78) The term “biomarker” refers to a measurable entity of the present invention that has been

determined to be predictive of immune checkpoint therapy effects on a cancer. Biomarkers can include, without limitation, nucleic acids and proteins, including those shown in Table 1, the Examples, and the Figures.

(79) A “blocking” antibody or an antibody “antagonist” is one which inhibits or reduces at least one biological activity of the antigen(s) it binds. In certain embodiments, the blocking antibodies or antagonist antibodies or fragments thereof described herein substantially or completely inhibit a given biological activity of the antigen(s).

(80) The term “body fluid” refers to fluids that are excreted or secreted from the body as well as fluids that are normally not (e.g. amniotic fluid, aqueous humor, bile, blood and blood plasma, cerebrospinal fluid, cerumen and earwax, cowper's fluid or pre-ejaculatory fluid, chyle, chyme, stool, female ejaculate, interstitial fluid, intracellular fluid, lymph, menses, breast milk, mucus, pleural fluid, pus, saliva, sebum, semen, serum, sweat, synovial fluid, tears, urine, vaginal lubrication, vitreous humor, vomit).

(81) The terms “cancer” or “tumor” or “hyperproliferative” refer to the presence of cells possessing characteristics typical of cancer-causing cells, such as uncontrolled proliferation, immortality, metastatic potential, rapid growth and proliferation rate, and certain characteristic morphological features. In some embodiments, such cells exhibit such characteristics in part or in full due to the expression and activity of immune checkpoint proteins, such as PD-1, PD-L1, and/or CTLA-4. Cancer cells are often in the form of a tumor, but such cells may exist alone within an animal, or may be a non-tumorigenic cancer cell, such as a leukemia cell. As used herein, the term “cancer” includes premalignant as well as malignant cancers. Cancers include, but are not limited to, B cell cancer, e.g., multiple myeloma, Waldenstrom's macroglobulinemia, the heavy chain diseases, such as, for example, alpha chain disease, gamma chain disease, and mu chain disease, benign monoclonal gammopathy, and immunocytic amyloidosis, melanomas, breast cancer, lung cancer, bronchus cancer, colorectal cancer, prostate cancer, pancreatic cancer, stomach cancer, ovarian cancer, urinary bladder cancer, brain or central nervous system cancer, peripheral nervous system cancer, esophageal cancer, cervical cancer, uterine or endometrial cancer, cancer of the oral cavity or pharynx, liver cancer, kidney cancer, testicular cancer, biliary tract cancer, small bowel or appendix cancer, salivary gland cancer, thyroid gland cancer, adrenal gland cancer, osteosarcoma, chondrosarcoma, cancer of hematologic tissues, and the like. Other non-limiting examples of types of cancers applicable to the methods encompassed by the present invention include human sarcomas and carcinomas, e.g., fibrosarcoma, myxosarcoma, liposarcoma, chondrosarcoma, osteogenic sarcoma, chordoma, angiosarcoma, endotheliosarcoma, lymphangiosarcoma, lymphangioendotheliosarcoma, synovioma, mesothelioma, Ewing's tumor, leiomyosarcoma, rhabdomyosarcoma, colon carcinoma, colorectal cancer, pancreatic cancer, breast cancer, ovarian cancer, prostate cancer, squamous cell carcinoma, basal cell carcinoma, adenocarcinoma, sweat gland carcinoma, sebaceous gland carcinoma, papillary carcinoma, papillary adenocarcinomas, cystadenocarcinoma, medullary carcinoma, bronchogenic carcinoma, renal cell carcinoma, hepatoma, bile duct carcinoma, liver cancer, choriocarcinoma, seminoma, embryonal carcinoma, Wilms' tumor, cervical cancer, bone cancer, brain tumor, testicular cancer, lung carcinoma, small cell lung carcinoma, bladder carcinoma, epithelial carcinoma, glioma, astrocytoma, medulloblastoma, craniopharyngioma, ependymoma, pinealoma, hemangioblastoma, acoustic neuroma, oligodendroglioma, meningioma, melanoma, neuroblastoma, retinoblastoma; leukemias, e.g., acute lymphocytic leukemia and acute myelocytic leukemia (myeloblastic, promyelocytic, myelomonocytic, monocytic and erythroleukemia); chronic leukemia (chronic myelocytic (granulocytic) leukemia and chronic lymphocytic leukemia); and polycythemia vera, lymphoma (Hodgkin's disease and non-Hodgkin's disease), multiple myeloma, Waldenstrom's macroglobulinemia, and heavy chain disease. In some embodiments, cancers are epithelial in nature and include but are not limited to, bladder cancer, breast cancer, cervical cancer, colon cancer, gynecologic cancers, renal cancer, laryngeal cancer, lung cancer, oral cancer, head and neck

cancer, ovarian cancer, pancreatic cancer, prostate cancer, or skin cancer. In other embodiments, the cancer is breast cancer, prostate cancer, lung cancer, or colon cancer. In still other embodiments, the epithelial cancer is non-small-cell lung cancer, nonpapillary renal cell carcinoma, cervical carcinoma, ovarian carcinoma (e.g., serous ovarian carcinoma), or breast carcinoma. The epithelial cancers may be characterized in various other ways including, but not limited to, serous, endometrioid, mucinous, clear cell, Brenner, or undifferentiated.

(82) The term “coding region” refers to regions of a nucleotide sequence comprising codons which are translated into amino acid residues, whereas the term “noncoding region” refers to regions of a nucleotide sequence that are not translated into amino acids (e.g., 5' and 3' untranslated regions).

(83) The term “complementary” refers to the broad concept of sequence complementarity between regions of two nucleic acid strands or between two regions of the same nucleic acid strand. It is known that an adenine residue of a first nucleic acid region is capable of forming specific hydrogen bonds (“base pairing”) with a residue of a second nucleic acid region which is antiparallel to the first region if the residue is thymine or uracil. Similarly, it is known that a cytosine residue of a first nucleic acid strand is capable of base pairing with a residue of a second nucleic acid strand which is antiparallel to the first strand if the residue is guanine. A first region of a nucleic acid is complementary to a second region of the same or a different nucleic acid if, when the two regions are arranged in an antiparallel fashion, at least one nucleotide residue of the first region is capable of base pairing with a residue of the second region. Preferably, the first region comprises a first portion and the second region comprises a second portion, whereby, when the first and second portions are arranged in an antiparallel fashion, at least about 50%, and preferably at least about 75%, at least about 90%, or at least about 95% of the nucleotide residues of the first portion are capable of base pairing with nucleotide residues in the second portion. More preferably, all nucleotide residues of the first portion are capable of base pairing with nucleotide residues in the second portion.

(84) The terms “conjoint therapy” and “combination therapy,” as used herein, refer to the administration of two or more therapeutic substances, e.g., combinations of anti-immune checkpoint therapies, multiple inhibitors of an immune checkpoint of interest, combinations of immune checkpoint therapy with an inhibitor of PBRM1 (ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, EGFR, and the like), and combinations thereof. The different agents comprising the combination therapy may be administered concomitant with, prior to, or following the administration of one or more therapeutic agents.

(85) The term “control” refers to any reference standard suitable to provide a comparison to the expression products in the test sample. In one embodiment, the control comprises obtaining a “control sample” from which expression product levels are detected and compared to the expression product levels from the test sample. Such a control sample may comprise any suitable sample, including but not limited to a sample from a control cancer patient (can be stored sample or previous sample measurement) with a known outcome; normal tissue or cells isolated from a subject, such as a normal patient or the cancer patient, cultured primary cells/tissues isolated from a subject such as a normal subject or the cancer patient, adjacent normal cells/tissues obtained from the same organ or body location of the cancer patient, a tissue or cell sample isolated from a normal subject, or a primary cells/tissues obtained from a depository. In another preferred embodiment, the control may comprise a reference standard expression product level from any suitable source, including but not limited to housekeeping genes, an expression product level range from normal tissue (or other previously analyzed control sample), a previously determined expression product level range within a test sample from a group of patients, or a set of patients with a certain outcome (for example, survival for one, two, three, four years, etc.) or receiving a certain treatment (for example, standard of care cancer therapy). It will be understood by those of skill in the art that such control samples and reference standard expression product levels can be used in combination as controls in the methods of the present invention. In one embodiment, the control may comprise

normal or non-cancerous cell/tissue sample. In another preferred embodiment, the control may comprise an expression level for a set of patients, such as a set of cancer patients, or for a set of cancer patients receiving a certain treatment, or for a set of patients with one outcome versus another outcome. In the former case, the specific expression product level of each patient can be assigned to a percentile level of expression, or expressed as either higher or lower than the mean or average of the reference standard expression level. In another preferred embodiment, the control may comprise normal cells, cells from patients treated with combination chemotherapy, and cells from patients having benign cancer. In another embodiment, the control may also comprise a measured value for example, average level of expression of a particular gene in a population compared to the level of expression of a housekeeping gene in the same population. Such a population may comprise normal subjects, cancer patients who have not undergone any treatment (i.e., treatment naive), cancer patients undergoing standard of care therapy, or patients having benign cancer. In another preferred embodiment, the control comprises a ratio transformation of expression product levels, including but not limited to determining a ratio of expression product levels of two genes in the test sample and comparing it to any suitable ratio of the same two genes in a reference standard; determining expression product levels of the two or more genes in the test sample and determining a difference in expression product levels in any suitable control; and determining expression product levels of the two or more genes in the test sample, normalizing their expression to expression of housekeeping genes in the test sample, and comparing to any suitable control. In particularly preferred embodiments, the control comprises a control sample which is of the same lineage and/or type as the test sample. In another embodiment, the control may comprise expression product levels grouped as percentiles within or based on a set of patient samples, such as all patients with cancer. In one embodiment a control expression product level is established wherein higher or lower levels of expression product relative to, for instance, a particular percentile, are used as the basis for predicting outcome. In another preferred embodiment, a control expression product level is established using expression product levels from cancer control patients with a known outcome, and the expression product levels from the test sample are compared to the control expression product level as the basis for predicting outcome. As demonstrated by the data below, the methods of the present invention are not limited to use of a specific cut-point in comparing the level of expression product in the test sample to the control.

(86) The “copy number” of a biomarker nucleic acid refers to the number of DNA sequences in a cell (e.g., germline and/or somatic) encoding a particular gene product. Generally, for a given gene, a mammal has two copies of each gene. The copy number can be increased, however, by gene amplification or duplication, or reduced by deletion. For example, germline copy number changes include changes at one or more genomic loci, wherein said one or more genomic loci are not accounted for by the number of copies in the normal complement of germline copies in a control (e.g., the normal copy number in germline DNA for the same species as that from which the specific germline DNA and corresponding copy number were determined). Somatic copy number changes include changes at one or more genomic loci, wherein said one or more genomic loci are not accounted for by the number of copies in germline DNA of a control (e.g., copy number in germline DNA for the same subject as that from which the somatic DNA and corresponding copy number were determined).

(87) The “normal” copy number (e.g., germline and/or somatic) of a biomarker nucleic acid or “normal” level of expression of a biomarker nucleic acid or protein is the activity/level of expression or copy number in a biological sample, e.g., a sample containing tissue, whole blood, serum, plasma, buccal scrape, saliva, cerebrospinal fluid, urine, stool, and bone marrow, from a subject, e.g., a human, not afflicted with cancer, or from a corresponding non-cancerous tissue in the same subject who has cancer.

(88) As used herein, the term “costimulate” with reference to activated immune cells includes the ability of a costimulatory molecule to provide a second, non-activating receptor mediated signal (a

“costimulatory signal”) that induces proliferation or effector function. For example, a costimulatory signal can result in cytokine secretion, e.g., in a T cell that has received a T cell-receptor-mediated signal. Immune cells that have received a cell-receptor mediated signal, e.g., via an activating receptor are referred to herein as “activated immune cells.”

(89) The term “determining a suitable treatment regimen for the subject” is taken to mean the determination of a treatment regimen (i.e., a single therapy or a combination of different therapies that are used for the prevention and/or treatment of the cancer in the subject) for a subject that is started, modified and/or ended based or essentially based or at least partially based on the results of the analysis according to the present invention. One example is determining whether to provide targeted therapy against a cancer to provide immunotherapy that generally increases immune responses against the cancer (e.g., immune checkpoint therapy). Another example is starting an adjuvant therapy after surgery whose purpose is to decrease the risk of recurrence, another would be to modify the dosage of a particular chemotherapy. The determination can, in addition to the results of the analysis according to the present invention, be based on personal characteristics of the subject to be treated. In most cases, the actual determination of the suitable treatment regimen for the subject will be performed by the attending physician or doctor.

(90) The term “diagnosing cancer” includes the use of the methods, systems, and code of the present invention to determine the presence or absence of a cancer or subtype thereof in an individual. The term also includes methods, systems, and code for assessing the level of disease activity in an individual.

(91) A molecule is “fixed” or “affixed” to a substrate if it is covalently or non-covalently associated with the substrate such that the substrate can be rinsed with a fluid (e.g. standard saline citrate, pH 7.4) without a substantial fraction of the molecule dissociating from the substrate.

(92) The term “expression signature” or “signature” refers to a group of two or more coordinately expressed biomarkers. For example, the genes, proteins, metabolites, and the like making up this signature may be expressed in a specific cell lineage, stage of differentiation, or during a particular biological response. The biomarkers can reflect biological aspects of the tumors in which they are expressed, such as the cell of origin of the cancer, the nature of the non-malignant cells in the biopsy, and the oncogenic mechanisms responsible for the cancer. Expression data and gene expression levels can be stored on computer readable media, e.g., the computer readable medium used in conjunction with a microarray or chip reading device. Such expression data can be manipulated to generate expression signatures.

(93) “Homologous” as used herein, refers to nucleotide sequence similarity between two regions of the same nucleic acid strand or between regions of two different nucleic acid strands. When a nucleotide residue position in both regions is occupied by the same nucleotide residue, then the regions are homologous at that position. A first region is homologous to a second region if at least one nucleotide residue position of each region is occupied by the same residue. Homology between two regions is expressed in terms of the proportion of nucleotide residue positions of the two regions that are occupied by the same nucleotide residue. By way of example, a region having the nucleotide sequence 5'-ATTGCC-3' and a region having the nucleotide sequence 5'-TATGGC-3' share 50% homology. Preferably, the first region comprises a first portion and the second region comprises a second portion, whereby, at least about 50%, and preferably at least about 75%, at least about 90%, or at least about 95% of the nucleotide residue positions of each of the portions are occupied by the same nucleotide residue. More preferably, all nucleotide residue positions of each of the portions are occupied by the same nucleotide residue.

(94) The term “immune cell” refers to cells that play a role in the immune response. Immune cells are of hematopoietic origin, and include lymphocytes, such as B cells and T cells; natural killer cells; myeloid cells, such as monocytes, macrophages, eosinophils, mast cells, basophils, and granulocytes.

(95) The term “immune checkpoint” refers to a group of molecules on the cell surface of CD4+

and/or CD8+ T cells that fine-tune immune responses by down-modulating or inhibiting an anti-tumor immune response. Immune checkpoint proteins are well known in the art and include, without limitation, CTLA-4, PD-1, VISTA, B7-H2, B7-H3, PD-L1, B7-H4, B7-H6, 2B4, ICOS, HVEM, PD-L2, CD160, gp49B, PIR-B, KIR family receptors, TIM-1, TIM-3, TIM-4, LAG-3, BTLA, SIRPalpha (CD47), CD48, 2B4 (CD244), B7.1, B7.2, ILT-2, ILT-4, TIGIT, and A2aR (see, for example, WO 2012/177624). The term further encompasses biologically active protein fragment, as well as nucleic acids encoding full-length immune checkpoint proteins and biologically active protein fragments thereof. In some embodiment, the term further encompasses any fragment according to homology descriptions provided herein.

(96) “Immune checkpoint therapy” refers to the use of agents that inhibit immune checkpoint nucleic acids and/or proteins. Inhibition of one or more immune checkpoints can block or otherwise neutralize inhibitory signaling to thereby upregulate an immune response in order to more efficaciously treat cancer. Exemplary agents useful for inhibiting immune checkpoints include antibodies, small molecules, peptides, peptidomimetics, natural ligands, and derivatives of natural ligands, that can either bind and/or inactivate or inhibit immune checkpoint proteins, or fragments thereof; as well as RNA interference, antisense, nucleic acid aptamers, etc. that can downregulate the expression and/or activity of immune checkpoint nucleic acids, or fragments thereof. Exemplary agents for upregulating an immune response include antibodies against one or more immune checkpoint proteins block the interaction between the proteins and its natural receptor(s); a non-activating form of one or more immune checkpoint proteins (e.g., a dominant negative polypeptide); small molecules or peptides that block the interaction between one or more immune checkpoint proteins and its natural receptor(s); fusion proteins (e.g. the extracellular portion of an immune checkpoint inhibition protein fused to the Fc portion of an antibody or immunoglobulin) that bind to its natural receptor(s); nucleic acid molecules that block immune checkpoint nucleic acid transcription or translation; and the like. Such agents can directly block the interaction between the one or more immune checkpoints and its natural receptor(s) (e.g., antibodies) to prevent inhibitory signaling and upregulate an immune response. Alternatively, agents can indirectly block the interaction between one or more immune checkpoint proteins and its natural receptor(s) to prevent inhibitory signaling and upregulate an immune response. For example, a soluble version of an immune checkpoint protein ligand such as a stabilized extracellular domain can binding to its receptor to indirectly reduce the effective concentration of the receptor to bind to an appropriate ligand. In one embodiment, anti-PD-1 antibodies, anti-PD-L1 antibodies, and anti-CTLA-4 antibodies, either alone or in combination, are used to inhibit immune checkpoints.

(97) “Ipilimumab” is a reRepresentative example of an immune checkpoint therapy. Ipilimumab (previously MDX-010; Medarex Inc., marketed by Bristol-Myers Squibb as YERVOY™) is a fully human anti-human CTLA-4 monoclonal antibody that blocks the binding of CTLA-4 to CD80 and CD86 expressed on antigen presenting cells, thereby, blocking the negative down-regulation of the immune responses elicited by the interaction of these molecules (see, for example, WO 2013/169971, U.S. Pat. Publ. 2002/0086014, and U.S. Pat. Publ. 2003/0086930).

(98) The term “immune response” includes T cell mediated and/or B cell mediated immune responses. Exemplary immune responses include T cell responses, e.g., cytokine production and cellular cytotoxicity. In addition, the term immune response includes immune responses that are indirectly effected by T cell activation, e.g., antibody production (humoral responses) and activation of cytokine responsive cells, e.g., macrophages.

(99) The term “immunotherapeutic agent” can include any molecule, peptide, antibody or other agent which can stimulate a host immune system to generate an immune response to a tumor or cancer in the subject. Various immunotherapeutic agents are useful in the compositions and methods described herein.

(100) The term “inhibit” includes the decrease, limitation, or blockage, of, for example a particular

action, function, or interaction. In some embodiments, cancer is “inhibited” if at least one symptom of the cancer is alleviated, terminated, slowed, or prevented. As used herein, cancer is also “inhibited” if recurrence or metastasis of the cancer is reduced, slowed, delayed, or prevented.

(101) The term “interaction”, when referring to an interaction between two molecules, refers to the physical contact (e.g., binding) of the molecules with one another. Generally, such an interaction results in an activity (which produces a biological effect) of one or both of said molecules.

(102) An “isolated protein” refers to a protein that is substantially free of other proteins, cellular material, separation medium, and culture medium when isolated from cells or produced by recombinant DNA techniques, or chemical precursors or other chemicals when chemically synthesized. An “isolated” or “purified” protein or biologically active portion thereof is substantially free of cellular material or other contaminating proteins from the cell or tissue source from which the antibody, polypeptide, peptide or fusion protein is derived, or substantially free from chemical precursors or other chemicals when chemically synthesized. The language “substantially free of cellular material” includes preparations of a biomarker polypeptide or fragment thereof, in which the protein is separated from cellular components of the cells from which it is isolated or recombinantly produced. In one embodiment, the language “substantially free of cellular material” includes preparations of a biomarker protein or fragment thereof, having less than about 30% (by dry weight) of non-biomarker protein (also referred to herein as a “contaminating protein”), more preferably less than about 20% of non-biomarker protein, still more preferably less than about 10% of non-biomarker protein, and most preferably less than about 5% non-biomarker protein. When antibody, polypeptide, peptide or fusion protein or fragment thereof, e.g., a biologically active fragment thereof, is recombinantly produced, it is also preferably substantially free of culture medium, i.e., culture medium represents less than about 20%, more preferably less than about 10%, and most preferably less than about 5% of the volume of the protein preparation.

(103) The term “KDM6A” refers to a particular lysine demethylase containing a JmjC-domain that catalyzes the demethylation of tri-/di-methylated histone H3. The term “KDM6A” is intended to include fragments, variants (e.g., allelic variants), and derivatives thereof. Representative human KDM6A cDNA and human KDM6A protein sequences are well-known in the art and are publicly available from the National Center for Biotechnology Information (NCBI). For example, the nucleic acid and amino acid sequences of a representative human KDM6A biomarker (also known as UTX or MGC141941 or bA386N14.2 or DKFZp686A03225) is available to the public at the GenBank database under NM_021140.2 and NP_006696.2. Nucleic acid and polypeptide sequences of KDM6A orthologs in organisms other than humans are well known and include, for example, mouse KDM6A (NM_009483.1 and NP_033509.1), rat KDM6A (XM_002730185.2 and XP_002730231.1), chimpanzee KDM6A (XM_002806207.1 and XP_002806253.1), chicken KDM6A (XM_416762.3 and XP_416762.3), fruit fly KDM6A (NM_001201844.1 and NP_001188773.1), and worm KDM6A (NM_077049.3 and NP_509450.1). Representative sequences of KDM6A orthologs are presented below in Table 1.

(104) Anti-KDM6A antibodies suitable for detecting KDM6A protein are well-known in the art and include, for example, antibody ab36938 (Abcam), 16F9.1 (EMD Millipore), PA5-31828 (ThermoFisher), NBP1-80628 and H00007403-M05 (Novus Biologicals), etc. Moreover, multiple siRNA, shRNA, CRISPR constructs for reducing KDM6A expression can be found in the commercial product lists of the above-referenced companies, such as siRNA product #sc-76881 and sc-76882 and CRISPR products #sc-514859 from Santa Cruz Biotechnology, as well as multiple RNAi products and CRISPR products from Origene and GenScript (Piscataway, NJ). It is to be noted that the term can further be used to refer to any combination of features described herein regarding KDM6A molecules. For example, any combination of sequence composition, percentage identity, sequence length, domain structure, functional activity, etc. can be used to describe an KDM6A molecule of the present invention.

(105) The term “loss-of-function mutation” for KDM6A refers to any mutation in a KDM6A-related nucleic acid or protein that results in reduced or eliminated KDM6A protein amounts and/or function. For example, nucleic acid mutations include single-base substitutions, multi-base substitutions, insertion mutations, deletion mutations, frameshift mutations, missense mutations, nonsense mutations, splice-site mutations, epigenetic modifications (e.g., methylation, phosphorylation, acetylation, ubiquitylation, sumoylation, histone acetylation, histone deacetylation, and the like), and combinations thereof. In some embodiments, the mutation is a “nonsynonymous mutation,” meaning that the mutation alters the amino acid sequence of KDM6A. Such mutations reduce or eliminate KDM6A protein amounts and/or function by eliminating proper coding sequences required for proper KDM6A protein translation and/or coding for KDM6A proteins that are non-functional or have reduced function (e.g., deletion of enzymatic and/or structural domains, reduction in protein stability, alteration of sub-cellular localization, and the like). Such mutations are well-known in the art. In addition, a representative list describing a wide variety of structural mutations correlated with the functional result of reduced or eliminated KDM6A protein amounts and/or function is described in the Tables and the Examples.

(106) A “kit” is any manufacture (e.g. a package or container) comprising at least one reagent, e.g. a probe or small molecule, for specifically detecting and/or affecting the expression of a marker of the present invention. The kit may be promoted, distributed, or sold as a unit for performing the methods of the present invention. The kit may comprise one or more reagents necessary to express a composition useful in the methods of the present invention. In certain embodiments, the kit may further comprise a reference standard, e.g., a nucleic acid encoding a protein that does not affect or regulate signaling pathways controlling cell growth, division, migration, survival or apoptosis. One skilled in the art can envision many such control proteins, including, but not limited to, common molecular tags (e.g., green fluorescent protein and beta-galactosidase), proteins not classified in any of pathway encompassing cell growth, division, migration, survival or apoptosis by GeneOntology reference, or ubiquitous housekeeping proteins. Reagents in the kit may be provided in individual containers or as mixtures of two or more reagents in a single container. In addition, instructional materials which describe the use of the compositions within the kit can be included.

(107) The term “neoadjuvant therapy” refers to a treatment given before the primary treatment. Examples of neoadjuvant therapy can include chemotherapy, radiation therapy, and hormone therapy. For example, in treating breast cancer, neoadjuvant therapy can allow patients with large breast cancer to undergo breast-conserving surgery.

(108) The “normal” level of expression of a biomarker is the level of expression of the biomarker in cells of a subject, e.g., a human patient, not afflicted with a cancer. An “over-expression” or “significantly higher level of expression” of a biomarker refers to an expression level in a test sample that is greater than the standard error of the assay employed to assess expression, and is preferably at least 10%, and more preferably 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10, 10.5, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 times or more higher than the expression activity or level of the biomarker in a control sample (e.g., sample from a healthy subject not having the biomarker associated disease) and preferably, the average expression level of the biomarker in several control samples. A “significantly lower level of expression” of a biomarker refers to an expression level in a test sample that is at least 10%, and more preferably 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10, 10.5, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 times or more lower than the expression level of the biomarker in a control sample (e.g., sample from a healthy subject not having the biomarker associated disease) and preferably, the average expression level of the biomarker in several control samples.

(109) An “over-expression” or “significantly higher level of expression” of a biomarker refers to an expression level in a test sample that is greater than the standard error of the assay employed to assess expression, and is preferably at least 10%, and more preferably 1.2, 1.3, 1.4, 1.5, 1.6, 1.7,

1.8, 1.9, 2.0, 2.1, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10, 10.5, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 times or more higher than the expression activity or level of the biomarker in a control sample (e.g., sample from a healthy subject not having the biomarker associated disease) and preferably, the average expression level of the biomarker in several control samples. A “significantly lower level of expression” of a biomarker refers to an expression level in a test sample that is at least 10%, and more preferably 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2.0, 2.1, 2.1, 2.2, 2.3, 2.4, 2.5, 2.6, 2.7, 2.8, 2.9, 3, 3.5, 4, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8, 8.5, 9, 9.5, 10, 10.5, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 times or more lower than the expression level of the biomarker in a control sample (e.g., sample from a healthy subject not having the biomarker associated disease) and preferably, the average expression level of the biomarker in several control samples.

(110) The term “pre-determined” biomarker amount and/or activity measurement(s) may be a biomarker amount and/or activity measurement(s) used to, by way of example only, evaluate a subject that may be selected for a particular treatment, evaluate a response to a treatment such as an anti-immune checkpoint inhibitor therapy, and/or evaluate the disease state. A pre-determined biomarker amount and/or activity measurement(s) may be determined in populations of patients with or without cancer. The pre-determined biomarker amount and/or activity measurement(s) can be a single number, equally applicable to every patient, or the pre-determined biomarker amount and/or activity measurement(s) can vary according to specific subpopulations of patients. Age, weight, height, and other factors of a subject may affect the pre-determined biomarker amount and/or activity measurement(s) of the individual. Furthermore, the pre-determined biomarker amount and/or activity can be determined for each subject individually. In one embodiment, the amounts determined and/or compared in a method described herein are based on absolute measurements. In another embodiment, the amounts determined and/or compared in a method described herein are based on relative measurements, such as ratios (e.g., serum biomarker normalized to the expression of a housekeeping or otherwise generally constant biomarker). The pre-determined biomarker amount and/or activity measurement(s) can be any suitable standard. For example, the pre-determined biomarker amount and/or activity measurement(s) can be obtained from the same or a different human for whom a patient selection is being assessed. In one embodiment, the pre-determined biomarker amount and/or activity measurement(s) can be obtained from a previous assessment of the same patient. In such a manner, the progress of the selection of the patient can be monitored over time. In addition, the control can be obtained from an assessment of another human or multiple humans, e.g., selected groups of humans, if the subject is a human. In such a manner, the extent of the selection of the human for whom selection is being assessed can be compared to suitable other humans, e.g., other humans who are in a similar situation to the human of interest, such as those suffering from similar or the same condition(s) and/or of the same ethnic group.

(111) The term “predictive” includes the use of a biomarker nucleic acid and/or protein status, e.g., over- or under-activity, emergence, expression, growth, remission, recurrence or resistance of tumors before, during or after therapy, for determining the likelihood of response of a cancer to anti-immune checkpoint treatment (e.g., therapeutic antibodies against CTLA-4, PD-1, PD-L1, and the like). Such predictive use of the biomarker may be confirmed by, e.g., (1) increased or decreased copy number (e.g., by FISH, FISH plus SKY, single-molecule sequencing, e.g., as described in the art at least at *J. Biotechnol.*, 86:289-301, or qPCR), overexpression or underexpression of a biomarker nucleic acid (e.g., by ISH, Northern Blot, or qPCR), increased or decreased biomarker protein (e.g., by IHC), or increased or decreased activity, e.g., in more than about 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95%, 100%, or more of assayed human cancers types or cancer samples; (2) its absolute or relatively modulated presence or absence in a biological sample, e.g., a sample containing tissue, whole blood, serum, plasma, buccal scrape, saliva, cerebrospinal fluid, urine,

stool, or bone marrow, from a subject, e.g. a human, afflicted with cancer; (3) its absolute or relatively modulated presence or absence in clinical subset of patients with cancer (e.g., those responding to a particular immune checkpoint therapy or those developing resistance thereto).

(112) The term “pre-malignant lesions” as described herein refers to a lesion that, while not cancerous, has potential for becoming cancerous. It also includes the term “pre-malignant disorders” or “potentially malignant disorders.” In particular this refers to a benign, morphologically and/or histologically altered tissue that has a greater than normal risk of malignant transformation, and a disease or a patient's habit that does not necessarily alter the clinical appearance of local tissue but is associated with a greater than normal risk of precancerous lesion or cancer development in that tissue (leukoplakia, erythroplakia, erytroleukoplakia lichen planus (lichenoid reaction) and any lesion or an area which histological examination showed atypia of cells or dysplasia.

(113) The terms “prevent,” “preventing,” “prevention,” “prophylactic treatment,” and the like refer to reducing the probability of developing a disease, disorder, or condition in a subject, who does not have, but is at risk of or susceptible to developing a disease, disorder, or condition.

(114) The term “probe” refers to any molecule which is capable of selectively binding to a specifically intended target molecule, for example, a nucleotide transcript or protein encoded by or corresponding to a biomarker nucleic acid. Probes can be either synthesized by one skilled in the art, or derived from appropriate biological preparations. For purposes of detection of the target molecule, probes may be specifically designed to be labeled, as described herein. Examples of molecules that can be utilized as probes include, but are not limited to, RNA, DNA, proteins, antibodies, and organic molecules.

(115) The term “prognosis” includes a prediction of the probable course and outcome of cancer or the likelihood of recovery from the disease. In some embodiments, the use of statistical algorithms provides a prognosis of cancer in an individual. For example, the prognosis can be surgery, development of a clinical subtype of cancer (e.g., solid tumors, such as lung cancer, melanoma, and renal cell carcinoma), development of one or more clinical factors, development of intestinal cancer, or recovery from the disease.

(116) The term “response to immune checkpoint therapy” relates to any response of the hyperproliferative disorder (e.g., cancer) to an immune checkpoint therapy, such as immune checkpoint therapy, preferably to a change in tumor mass and/or volume after initiation of neoadjuvant or adjuvant chemotherapy. Hyperproliferative disorder response may be assessed, for example for efficacy or in a neoadjuvant or adjuvant situation, where the size of a tumor after systemic intervention can be compared to the initial size and dimensions as measured by CT, PET, mammogram, ultrasound or palpation. Responses may also be assessed by caliper measurement or pathological examination of the tumor after biopsy or surgical resection. Response may be recorded in a quantitative fashion like percentage change in tumor volume or in a qualitative fashion like “pathological complete response” (pCR), “clinical complete remission” (cCR), “clinical partial remission” (cPR), “clinical stable disease” (cSD), “clinical progressive disease” (cPD) or other qualitative criteria. Assessment of hyperproliferative disorder response may be done early after the onset of neoadjuvant or adjuvant therapy, e.g., after a few hours, days, weeks or preferably after a few months. A typical endpoint for response assessment is upon termination of neoadjuvant chemotherapy or upon surgical removal of residual tumor cells and/or the tumor bed. This is typically three months after initiation of neoadjuvant therapy. In some embodiments, clinical efficacy of the therapeutic treatments described herein may be determined by measuring the clinical benefit rate (CBR). The clinical benefit rate is measured by determining the sum of the percentage of patients who are in complete remission (CR), the number of patients who are in partial remission (PR) and the number of patients having stable disease (SD) at a time point at least 6 months out from the end of therapy. The shorthand for this formula is $CBR = CR + PR + SD$ over 6 months. In some embodiments, the CBR for a particular cancer therapeutic regimen is at least 25%,

30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, or more. Additional criteria for evaluating the response to cancer therapies are related to “survival,” which includes all of the following: survival until mortality, also known as overall survival (wherein said mortality may be either irrespective of cause or tumor related); “recurrence-free survival” (wherein the term recurrence shall include both localized and distant recurrence); metastasis free survival; disease free survival (wherein the term disease shall include cancer and diseases associated therewith). The length of said survival may be calculated by reference to a defined start point (e.g., time of diagnosis or start of treatment) and end point (e.g., death, recurrence or metastasis). In addition, criteria for efficacy of treatment can be expanded to include response to chemotherapy, probability of survival, probability of metastasis within a given time period, and probability of tumor recurrence. For example, in order to determine appropriate threshold values, a particular cancer therapeutic regimen can be administered to a population of subjects and the outcome can be correlated to biomarker measurements that were determined prior to administration of any cancer therapy. The outcome measurement may be pathologic response to therapy given in the neoadjuvant setting. Alternatively, outcome measures, such as overall survival and disease-free survival can be monitored over a period of time for subjects following cancer therapy for whom biomarker measurement values are known. In certain embodiments, the doses administered are standard doses known in the art for cancer therapeutic agents. The period of time for which subjects are monitored can vary. For example, subjects may be monitored for at least 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 25, 30, 35, 40, 45, 50, 55, or 60 months. Biomarker measurement threshold values that correlate to outcome of a cancer therapy can be determined using well-known methods in the art, such as those described in the Examples section.

(117) The term “resistance” refers to an acquired or natural resistance of a cancer sample or a mammal to a cancer therapy (i.e., being nonresponsive to or having reduced or limited response to the therapeutic treatment), such as having a reduced response to a therapeutic treatment by 25% or more, for example, 30%, 40%, 50%, 60%, 70%, 80%, or more, to 2-fold, 3-fold, 4-fold, 5-fold, 10-fold, 15-fold, 20-fold or more. The reduction in response can be measured by comparing with the same cancer sample or mammal before the resistance is acquired, or by comparing with a different cancer sample or a mammal who is known to have no resistance to the therapeutic treatment. A typical acquired resistance to chemotherapy is called “multidrug resistance.” The multidrug resistance can be mediated by P-glycoprotein or can be mediated by other mechanisms, or it can occur when a mammal is infected with a multi-drug-resistant microorganism or a combination of microorganisms. The determination of resistance to a therapeutic treatment is routine in the art and within the skill of an ordinarily skilled clinician, for example, can be measured by cell proliferative assays and cell death assays as described herein as “sensitizing.” In some embodiments, the term “reverses resistance” means that the use of a second agent in combination with a primary cancer therapy (e.g., chemotherapeutic or radiation therapy) is able to produce a significant decrease in tumor volume at a level of statistical significance (e.g., $p < 0.05$) when compared to tumor volume of untreated tumor in the circumstance where the primary cancer therapy (e.g., chemotherapeutic or radiation therapy) alone is unable to produce a statistically significant decrease in tumor volume compared to tumor volume of untreated tumor. This generally applies to tumor volume measurements made at a time when the untreated tumor is growing log rhythmically.

(118) The terms “response” or “responsiveness” refers to an anti-cancer response, e.g. in the sense of reduction of tumor size or inhibiting tumor growth. The terms can also refer to an improved prognosis, for example, as reflected by an increased time to recurrence, which is the period to first recurrence censoring for second primary cancer as a first event or death without evidence of recurrence, or an increased overall survival, which is the period from treatment to death from any cause. To respond or to have a response means there is a beneficial endpoint attained when exposed to a stimulus. Alternatively, a negative or detrimental symptom is minimized, mitigated or attenuated on exposure to a stimulus. It will be appreciated that evaluating the likelihood that a

tumor or subject will exhibit a favorable response is equivalent to evaluating the likelihood that the tumor or subject will not exhibit favorable response (i.e., will exhibit a lack of response or be non-responsive).

(119) An “RNA interfering agent” as used herein, is defined as any agent which interferes with or inhibits expression of a target biomarker gene by RNA interference (RNAi). Such RNA interfering agents include, but are not limited to, nucleic acid molecules including RNA molecules which are homologous to the target biomarker gene of the present invention, or a fragment thereof, short interfering RNA (siRNA), and small molecules which interfere with or inhibit expression of a target biomarker nucleic acid by RNA interference (RNAi).

(120) “RNA interference (RNAi)” is an evolutionally conserved process whereby the expression or introduction of RNA of a sequence that is identical or highly similar to a target biomarker nucleic acid results in the sequence specific degradation or specific post-transcriptional gene silencing (PTGS) of messenger RNA (mRNA) transcribed from that targeted gene (see Coburn, G. and Cullen, B. (2002) *J. of Virology* 76(18):9225), thereby inhibiting expression of the target biomarker nucleic acid. In one embodiment, the RNA is double stranded RNA (dsRNA). This process has been described in plants, invertebrates, and mammalian cells. In nature, RNAi is initiated by the dsRNA-specific endonuclease Dicer, which promotes processive cleavage of long dsRNA into double-stranded fragments termed siRNAs. siRNAs are incorporated into a protein complex that recognizes and cleaves target mRNAs. RNAi can also be initiated by introducing nucleic acid molecules, e.g., synthetic siRNAs or RNA interfering agents, to inhibit or silence the expression of target biomarker nucleic acids. As used herein, “inhibition of target biomarker nucleic acid expression” or “inhibition of marker gene expression” includes any decrease in expression or protein activity or level of the target biomarker nucleic acid or protein encoded by the target biomarker nucleic acid. The decrease may be of at least 30%, 40%, 50%, 60%, 70%, 80%, 90%, 95% or 99% or more as compared to the expression of a target biomarker nucleic acid or the activity or level of the protein encoded by a target biomarker nucleic acid which has not been targeted by an RNA interfering agent.

(121) The term “sample” used for detecting or determining the presence or level of at least one biomarker is typically whole blood, plasma, serum, saliva, urine, stool (e.g., feces), tears, and any other bodily fluid (e.g., as described above under the definition of “body fluids”), or a tissue sample (e.g., biopsy) such as a small intestine, colon sample, or surgical resection tissue. In certain instances, the method of the present invention further comprises obtaining the sample from the individual prior to detecting or determining the presence or level of at least one marker in the sample.

(122) The term “sensitize” means to alter cancer cells or tumor cells in a way that allows for more effective treatment of the associated cancer with a cancer therapy (e.g., anti-immune checkpoint, chemotherapeutic, and/or radiation therapy). In some embodiments, normal cells are not affected to an extent that causes the normal cells to be unduly injured by the immune checkpoint therapy. An increased sensitivity or a reduced sensitivity to a therapeutic treatment is measured according to a known method in the art for the particular treatment and methods described herein below, including, but not limited to, cell proliferative assays (Tanigawa N, Kern D H, Kikasa Y, Morton D L, *Cancer Res* 1982; 42: 2159-2164), cell death assays (Weisenthal L M, Shoemaker R H, Marsden J A, Dill P L, Baker J A, Moran E M, *Cancer Res* 1984; 94: 161-173; Weisenthal L M, Lippman M E, *Cancer Treat Rep* 1985; 69: 615-632; Weisenthal L M, In: Kaspers G J L, Pieters R, Twentyman P R, Weisenthal L M, Veerman A J P, eds. *Drug Resistance in Leukemia and Lymphoma*. Langhorne, P A: Harwood Academic Publishers, 1993: 415-432; Weisenthal L M, *Contrib Gynecol Obstet* 1994; 19: 82-90). The sensitivity or resistance may also be measured in animal by measuring the tumor size reduction over a period of time, for example, 6 month for human and 4-6 weeks for mouse. A composition or a method sensitizes response to a therapeutic treatment if the increase in treatment sensitivity or the reduction in resistance is 25% or more, for example, 30%,

40%, 50%, 60%, 70%, 80%, or more, to 2-fold, 3-fold, 4-fold, 5-fold, 10-fold, 15-fold, 20-fold or more, compared to treatment sensitivity or resistance in the absence of such composition or method. The determination of sensitivity or resistance to a therapeutic treatment is routine in the art and within the skill of an ordinarily skilled clinician. It is to be understood that any method described herein for enhancing the efficacy of a cancer therapy can be equally applied to methods for sensitizing hyperproliferative or otherwise cancerous cells (e.g., resistant cells) to the cancer therapy.

(123) The term “synergistic effect” refers to the combined effect of two or more anti-immune checkpoint agents can be greater than the sum of the separate effects of the anticancer agents alone.

(124) “Short interfering RNA” (siRNA), also referred to herein as “small interfering RNA” is defined as an agent which functions to inhibit expression of a target biomarker nucleic acid, e.g., by RNAi. An siRNA may be chemically synthesized, may be produced by in vitro transcription, or may be produced within a host cell. In one embodiment, siRNA is a double stranded RNA (dsRNA) molecule of about 15 to about 40 nucleotides in length, preferably about 15 to about 28 nucleotides, more preferably about 19 to about 25 nucleotides in length, and more preferably about 19, 20, 21, or 22 nucleotides in length, and may contain a 3' and/or 5' overhang on each strand having a length of about 0, 1, 2, 3, 4, or 5 nucleotides. The length of the overhang is independent between the two strands, i.e., the length of the overhang on one strand is not dependent on the length of the overhang on the second strand. Preferably the siRNA is capable of promoting RNA interference through degradation or specific post-transcriptional gene silencing (PTGS) of the target messenger RNA (mRNA).

(125) In another embodiment, an siRNA is a small hairpin (also called stem loop) RNA (shRNA). In one embodiment, these shRNAs are composed of a short (e.g., 19-25 nucleotide) antisense strand, followed by a 5-9 nucleotide loop, and the analogous sense strand. Alternatively, the sense strand may precede the nucleotide loop structure and the antisense strand may follow. These shRNAs may be contained in plasmids, retroviruses, and lentiviruses and expressed from, for example, the pol III U6 promoter, or another promoter (see, e.g., Stewart, et al. (2003) *RNA* 9:493-501 incorporated by reference herein).

(126) RNA interfering agents, e.g., siRNA molecules, may be administered to a patient having or at risk for having cancer, to inhibit expression of a biomarker gene which is overexpressed in cancer and thereby treat, prevent, or inhibit cancer in the subject.

(127) The term “subject” refers to any healthy animal, mammal or human, or any animal, mammal or human afflicted with a cancer, e.g., lung, ovarian, pancreatic, liver, breast, prostate, and colon carcinomas, as well as melanoma and multiple myeloma. The term “subject” is interchangeable with “patient.”

(128) The term “survival” includes all of the following: survival until mortality, also known as overall survival (wherein said mortality may be either irrespective of cause or tumor related); “recurrence-free survival” (wherein the term recurrence shall include both localized and distant recurrence); metastasis free survival; disease free survival (wherein the term disease shall include cancer and diseases associated therewith). The length of said survival may be calculated by reference to a defined start point (e.g. time of diagnosis or start of treatment) and end point (e.g. death, recurrence or metastasis). In addition, criteria for efficacy of treatment can be expanded to include response to chemotherapy, probability of survival, probability of metastasis within a given time period, and probability of tumor recurrence.

(129) The term “therapeutic effect” refers to a local or systemic effect in animals, particularly mammals, and more particularly humans, caused by a pharmacologically active substance. The term thus means any substance intended for use in the diagnosis, cure, mitigation, treatment or prevention of disease or in the enhancement of desirable physical or mental development and conditions in an animal or human. The phrase “therapeutically-effective amount” means that amount of such a substance that produces some desired local or systemic effect at a reasonable

benefit/risk ratio applicable to any treatment. In certain embodiments, a therapeutically effective amount of a compound will depend on its therapeutic index, solubility, and the like. For example, certain compounds discovered by the methods of the present invention may be administered in a sufficient amount to produce a reasonable benefit/risk ratio applicable to such treatment.

(130) The terms “therapeutically-effective amount” and “effective amount” as used herein means that amount of a compound, material, or composition comprising a compound of the present invention which is effective for producing some desired therapeutic effect in at least a sub-population of cells in an animal at a reasonable benefit/risk ratio applicable to any medical treatment. Toxicity and therapeutic efficacy of subject compounds may be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD.sub.50 and the ED.sub.50. Compositions that exhibit large therapeutic indices are preferred. In some embodiments, the LD.sub.50 (lethal dosage) can be measured and can be, for example, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more reduced for the agent relative to no administration of the agent. Similarly, the ED.sub.50 (i.e., the concentration which achieves a half-maximal inhibition of symptoms) can be measured and can be, for example, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more increased for the agent relative to no administration of the agent. Also, Similarly, the IC.sub.50 (i.e., the concentration which achieves half-maximal cytotoxic or cytostatic effect on cancer cells) can be measured and can be, for example, at least 10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, 100%, 200%, 300%, 400%, 500%, 600%, 700%, 800%, 900%, 1000% or more increased for the agent relative to no administration of the agent. In some embodiments, cancer cell growth in an assay can be inhibited by at least about 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or even 100%. In another embodiment, at least about a 10%, 15%, 20%, 25%, 30%, 3%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or even 100% decrease in a solid malignancy can be achieved.

(131) A “transcribed polynucleotide” or “nucleotide transcript” is a polynucleotide (e.g. an mRNA, hnRNA, a cDNA, or an analog of such RNA or cDNA) which is complementary to or homologous with all or a portion of a mature mRNA made by transcription of a biomarker nucleic acid and normal post-transcriptional processing (e.g. splicing), if any, of the RNA transcript, and reverse transcription of the RNA transcript.

(132) As used herein, the term “unresponsiveness” includes refractivity of immune cells to stimulation, e.g., stimulation via an activating receptor or a cytokine. Unresponsiveness can occur, e.g., because of exposure to immunosuppressants or exposure to high doses of antigen. As used herein, the term “anergy” or “tolerance” includes refractivity to activating receptor-mediated stimulation. Such refractivity is generally antigen-specific and persists after exposure to the tolerizing antigen has ceased. For example, anergy in T cells (as opposed to unresponsiveness) is characterized by lack of cytokine production, e.g., IL-2. T cell anergy occurs when T cells are exposed to antigen and receive a first signal (a T cell receptor or CD-3 mediated signal) in the absence of a second signal (a costimulatory signal). Under these conditions, reexposure of the cells to the same antigen (even if reexposure occurs in the presence of a costimulatory polypeptide) results in failure to produce cytokines and, thus, failure to proliferate. Anergic T cells can, however, proliferate if cultured with cytokines (e.g., IL-2). For example, T cell anergy can also be observed by the lack of IL-2 production by T lymphocytes as measured by ELISA or by a proliferation assay using an indicator cell line. Alternatively, a reporter gene construct can be used. For example, anergic T cells fail to initiate IL-2 gene transcription induced by a heterologous promoter under the control of the 5' IL-2 gene enhancer or by a multimer of the AP1 sequence that can be found within the enhancer (Kang et al. (1992) *Science* 257:1134).

(133) There is a known and definite correspondence between the amino acid sequence of a particular protein and the nucleotide sequences that can code for the protein, as defined by the

genetic code (shown below). Likewise, there is a known and definite correspondence between the nucleotide sequence of a particular nucleic acid and the amino acid sequence encoded by that nucleic acid, as defined by the genetic code.

(134) TABLE-US-00001 GENETIC CODE Alanine (Ala, A) GCA, GCC, GCG, GCT Arginine (Arg, R) AGA, ACG, CGA, CGC, CGG, CGT Asparagine (Asn, N) AAC, AAT Aspartic acid (Asp, D) GAC, GAT Cysteine (Cys, C) TGC, TGT Glutamic acid (Glu, E) GAA, GAG Glutamine (Gln, Q) CAA, CAG Glycine (Gly, G) GGA, GGC, GGG, GGT Histidine (His, H) CAC, CAT Isoleucine (Ile, I) ATA, ATC, ATT Leucine (Leu, L) CTA, CTC, CTG, CTT, TTA, TTG Lysine (Lys, K) AAA, AAG Methionine (Met, M) ATG Phenylalanine (Phe, F) TTC, TTT Proline (Pro, P) CCA, CCC, CCG, CCT Serine (Ser, S) AGC, AGT, TCA, TCC, TCG, TCT Threonine (Thr, T) ACA, ACC, ACG, ACT Tryptophan (Trp, W) TGG Tyrosine (Tyr, Y) TAC, TAT Valine (Val, V) GTA, GTC, GTG, GTT Termination signal (end) TAA, TAG, TGA

(135) An important and well known feature of the genetic code is its redundancy, whereby, for most of the amino acids used to make proteins, more than one coding nucleotide triplet may be employed (illustrated above). Therefore, a number of different nucleotide sequences may code for a given amino acid sequence. Such nucleotide sequences are considered functionally equivalent since they result in the production of the same amino acid sequence in all organisms (although certain organisms may translate some sequences more efficiently than they do others). Moreover, occasionally, a methylated variant of a purine or pyrimidine may be found in a given nucleotide sequence. Such methylations do not affect the coding relationship between the trinucleotide codon and the corresponding amino acid.

(136) In view of the foregoing, the nucleotide sequence of a DNA or RNA encoding a biomarker nucleic acid (or any portion thereof) can be used to derive the polypeptide amino acid sequence, using the genetic code to translate the DNA or RNA into an amino acid sequence. Likewise, for polypeptide amino acid sequence, corresponding nucleotide sequences that can encode the polypeptide can be deduced from the genetic code (which, because of its redundancy, will produce multiple nucleic acid sequences for any given amino acid sequence). Thus, description and/or disclosure herein of a nucleotide sequence which encodes a polypeptide should be considered to also include description and/or disclosure of the amino acid sequence encoded by the nucleotide sequence. Similarly, description and/or disclosure of a polypeptide amino acid sequence herein should be considered to also include description and/or disclosure of all possible nucleotide sequences that can encode the amino acid sequence.

(137) Finally, nucleic acid and amino acid sequence information for the loci and biomarkers of the present invention (e.g., biomarkers listed in Table 1) are well known in the art and readily available on publicly available databases, such as the National Center for Biotechnology Information (NCBI). For example, exemplary nucleic acid and amino acid sequences derived from publicly available sequence databases are provided below.

(138) TABLE-US-00002 TABLE 1 SEQ ID NO: 1 Human PBRM1 Transcript Variant 1 cDNA Sequence (NM_018313.4) 1 gcggccgcgg ccgaggagc aatagcagca gccgtggcgg ccacggggcg gggcgcggcg 61 gtcggtgacc gcggccgggg ctgcaggcgg cggagcggct ggaagtggga ttccatgggt 121 tccaagagaa gaagagctac ctccccttc agcagtgtca gcgggggactt tgatgatggg 181 caccattctg tgtaacacc aggcccaagc aggaaaagga ggagactttc caatctcca 241 actgtagatc ctattgccgt gtgccatgaa ctctataata ccatccgaga ctataaggat 301 gaacagggca gacttctctg tgagctcttc attagggcac caaagcgaag aatcaacca 361 gactattatg aagtggttc tcagcccatt gacttgatga aaatccaaca gaaactaaaa 421 atggaagagt atgatgatgt taatttgctg actgctgact tccagcttct tttaacaat 481 gcaaagtcct attataagcc agattctct gaatataaag ccgcttgcaa actctgggat 541 ttgtaccttc gaacaagaaa tgagttgtt cagaaaggag aagcagatga cgaagatgat 601 gatgaagatg ggcaagacaa tcagggcaca gtgactgaag gatcttctcc agcttacttg 661 aaggagatcc tggagcagct tcttgaagcc atagttgtag ctacaaatcc atcaggacgt 721 ctcattagcg aacttttca gaaactgcct tctaaagtgc aatatccaga ttattatgca 781 ataattaagg agcctataga

ttccaagacc attgcccaga ggatacagaa tggaagctac 841 aaaagtattc atgcaatggc caaagatata
gatctcctcg caaaaaatgc caaaacttat 901 aatgagcctg gctctcaagt attcaaggat gcaaattcaa
ttaaaaaaat attttatatg 961 aaaaaggctg aaattgaaca tcatgaaatg gctaagtcaa gtcttcgaat
gaggactcca 1021 tccaacttgg ctgcagccag actgacaggt ccttcacaca gtaaaggcag ccttggtgaa 1081
gagagaaatc ccactagcaa gtattaccgt aataaaagag cagtacaagg aggtcgttta 1141 tcagcaatta
caatggcact tcaatatggc tcagaaagt aagaagatgc tgcttagct 1201 gctgcacgct atgaagaggg
agagtcagaa gcagaaagca tcacttcctt tatggatgtt 1261 tcaaatcctt tttatcagct ttatgacaca
gttaggagtt gtcggaataa ccaagggcag 1321 ctaatagctg aacctttta ccattgcct tcaaagaaaa
aataccctga ttattaccag 1381 caaattaa tgcacatc actacaacag atccgaacaa aactgaagaa
tcaagaatat 1441 gaaactttag atcatttga gtgtgatctg aatttaagt ttgaaatgc caaacgctat 1501
aatgtgccc attcagccat ctacaagcga gttctaaaat tgcagcaagt tatgcaggca 1561 aagaagaaag
agcttgccag gagagacgat atcgaggacg gagacagcat gatctcttca 1621 gccacctctg atactggtag
tgccaaaaga aaaagtaaaa agaacataag aaagcagcga 1681 atgaaaatct tattcaatgt tgttcttgaa
gctcgagagc cagggttcagg cagaagactt 1741 tgtgacctat ttatggttaa accatccaaa aaggactatc
ctgattatta taaaatcatc 1801 ttggagccaa tggacttgaa aataattgag cataacatcc gcaatgacaa
atatgctggt 1861 gaagagggaa tgatagaaga catgaagctg atgttccgga atgccaggca ctataatgag 1921
gagggctccc aggtttataa tgatgcat atcctggaga agttactcaa ggagaaaagg 1981 aaagagctgg
gccactgcc tgatgatgat gacatggctt ctccaaact caagctgagt 2041 aggaagagtg gcatttctc
taaaaaatca aaatacatga ctccaatgca gcagaaacta 2101 aatgaggtct atgaagctgt aaagaactat
actgataaga ggggtcgccg cctcagtgcc 2161 atatttctga ggcttcctc tagatctgag ttgcctgact
actatctgac tattaaaaag 2221 cccatggaca tggaaaaaat tcgaagtcac atgatggcca acaagtacca
agatattgac 2281 tctatggttg aggactttgt catgatgtt aataatgcct gtacatacaa tgagccggag 2341
tctttgatct acaaagatgc tcttgttcta cacaagtc tgcttgaaac acgcagagac 2401 ctggagggag
atgaggactc tcatgtccca aatgtgactt tgctgattca agagcttacc 2461 cacaatctt ttgtgtcagt
catgagtcac caggatgatg aggggaagatg ctacagcgat 2521 tctttagcag aaattcctgc tgtggatccc
aactttcta acaaaccacc cttacattt 2581 gacataatta ggaagaatgt tgaaaataat cgctaccgtc
ggcttgattt attcaagag 2641 catatgttg aagtattgga acgagcaaga aggatgaatc ggacagattc
agaaatatat 2701 gaagatgcag tagaacttca gcagttttt attaaaattc gtgatgaact ctgaaaaat 2761
ggagagattc ttcttcacc ggcactcagc tataccacaa aacatttgca taatgatgtg 2821 gagaaagaga
gaaaggaaaa attgcaaaaa gaaatagagg aagataaact aaaacgagaa 2881 gaagaaaaaa gagaagctga
aaagagtga gattcctctg gtgctgcagg cctctcaggc 2941 ttacatcgca catacagcca ggactgtagc
ttaaaaaca gcatgtacca tgttgagat 3001 tacgtctatg tggaacctgc agaggccaac ctacaaccac
atatcgctg tattgaaaga 3061 ctgtgggagg attcagctga aaaagaagtt ttaagagt actattacaa
caaagttcca 3121 gttagtaaaa ttctaggcaa gtgtgtggc atgtttgtca aggaatactt taagtatgc 3181
ccagaaaact tccgatga ggatgtttt gtctgtgaat cacggattc tgccaaaacc 3241 aaatcttta
agaaaattaa actgtggacc atgcccata gctcagtcag gttgtcct 3301 cgggatgtgc ctctgcctgt
ggttcgcgtg gcctctgtat ttgcaaatgc agataaagg 3361 gatgatgaga agaatacaga caactcagag
gacagtcgag ctgaagacaa ttttaactg 3421 gaaaaggaaa aagaagatgt ccctgtggaa atgtccatg
gtgaaccagg ttgccactac 3481 ttgagcagc tccattacaa tgacatgtg ctgaagggtg gcgactgtg
ctcatcaag 3541 tccatggcc tgggtgcgtc tcgtgtgggc agaattgaaa aagtatgggt tcgagatgga 3601
gctgcatatt ttatggccc catcttcatt caccagaag aaacagagca tgagcccaca 3661 aaaatgttct
acaaaaaaga agtatttctg agtaatctgg aagaaacctg cccatgaca 3721 tgtattctcg gaaagtgtgc
tgtgtgtca ttcaaggact tctctctg caggccaact 3781 gaaataccag aaaatgacat tctgcttgt
gagagccgct acaatgagag cgacaagcag 3841 atgaagaaat tcaaaggatt gaagaggtt tcaactctg
ctaaagtgg agatgatga 3901 atttactact tcagaaaacc aattgttct cagaaggagc catcacctt
gctggaaaag 3961 aagatccagt tgctagaagc taaatttgc gagttagaag gtggagatga tgatattgaa 4021
gagatgggag aagaagatag tgagtctacc ccaaagtctg ccaaaggcag tgcaaagaag 4081 gaaggctcca
aacggaaaa caacatgagt ggctacatcc tgttcagcag tgagatgagg 4141 gctgtgatta aggcccaaca
cccagactac tcttcgggg agctcagccg cctggtgggg 4201 acagaatgga gaaatcttga gacagccaag
aaagcagaat atgaaggcat gatgggtggc 4261 tatccgccag gccttcacc ttgcagggc ccagttgatg

gccttggtag catgggcagc 4321 atgcagccac ttcacctgg ggggcctcca cccaccatc ttcggccagg
tgtgctggc 4381 ctcccgggca tcccaccacc ggggtgtgat aaccaaggag tggcccctat ggtagggact
4441 ccagcaccag gtggaagtcc atatggacaa caggtgggag ttttggggcc tccagggcag 4501
caggcaccac ctccatatcc cggcccacat ccagctggac cccctgtcat acagcagcca 4561 acaacacca
tgtttgtagc tccccacca aagaccagc ggcttttca ctgagaggcc 4621 tacctgaaat acattgaagg
actcagtgcg gagtccaaca gcattagcaa gtgggatcag 4681 aactggcag ctggaagacg cgacgtccat
ttgtcgaaag aacaggagag ccgcctaccc 4741 tctcactggc tgaagacaa aggggcccac accaccatgg
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BRD7 Amino Acid Sequence Isoform B (NP_037395.2) 1 mgkhhkhkhs dkhlyeeyve
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SEQ ID NO: 17 Mouse BRD7 cDNA Sequence (NM_012047.2. CDS: from
238 to 2193) 1 ggtttgccgg cctctgccc tctcgccact ggtgtcgcgc ttcggtcgcg tcccgcgcgt 61
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spivgnsfvd ltgeceepke tstaecgpda s SEQ ID NO: 19 Human PHF10 cDNA

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SEQ ID NO: 21 Human PHF10 cDNA Sequence Variant 2 (NM_133325.2, CDS: From 80 to 1570) 1 ggcgggcgcg
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 aaaaaa SEQ ID NO: 22 Human PHF10 Amino Acid Sequence Isoform B
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 481 ptpkvgrrg knskeg SEQ ID NO: 23 Mouse PHF10 cDNA Sequence
 (NM_024250.4. CDS: from 67 to 1560) 1 gcggcggcgg ccgctgggac taggcgaagg
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dvcdrgyhtf cvglgaipsg rwidccqra 481 pptprkvgrr gknskeg SEQ ID NO: 25 Human
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Variant 7 (NM_001289400.1, CDS: from 521 to 1357) 1 acttcattaa atctagaggc
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Acid Sequence Isoform B (NP_080279.1) 1 meeevrlkk krrnvdldp vkedvekakk
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rinhhkyrs gldkdvml 121 chnaqtnle gsqiyedsiv lqsvfksarq kiakeeese esneeeedd
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ID NO: 89 Mouse BRM cDNA Sequence Variant 3 (NM_001347439.1, CDS:
from 180 to 1010) 1 acacacacac acacacacac acgcaggctg aagtatgctt aactcttta
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1741 agtattacag tgtctagaat accctggatg gcttccttg tccaccgat ctcccggtt 1801 accaatgtgt
atggtctct tctccgaaa gtgtactta tcttgctt cttgcacaa 1861 tgtcttgggt tgcaagtc
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lqsvfksarq kiakeesees esneeeeedd eeseseaks vkviklnkk eekgrdtgkg 241 kkrpnrgkak
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Variant 1 (NM_005228.4, CDS: from 258 to 3890) 1 gtccgggcag cccccggcgc
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ttctaaccg 5641 ttatccatcc ctgactctca tctctgaaa tagatgtcat ccatgaaatt aaggagttag 5701
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tcaggagcgg taatccgtga gagagaaccg ttgcaagcca 5881 ctctaactgt agcaatgaaa ccctagtatt
ttgtacttt gaaatacttt cttataacaa 5941 aataaagtag caaaaaaact gttcaaaaaa aaaaaaaaaa aaa
SEQ ID NO: 110 Mouse EGFR Amino Acid Sequence Isoform A
(NP_997538.1) 1 mrpsgtartt llvltalca aggaleekkv cqgtsnrltq lgtfedhfls lqrmynncev 61
vlgneityv qrnyslflk tiqevagyvl ialntverip lenlqiirgn alyentyala 121 ilsnygtprt
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gggeencqkl tkiicaqqcs hrcgrspds cchnqcaagc 241 tgpresdclv cqkfdeatc kdtcpplmly
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gpcrkvcngi gigeftdls inatnikhfk 361 yctaisgdlh ilpvafkgs ftrtpldpr eleilktvke
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kcwmidadsr 961 pkfrelilef skmardpqry lviqgdermh lpsptdsnfy ralmedeeme dvvdadeyli
1021 pqqgffnsp tsrtplssl satsnnstva cinrngscrv kedaflqrys sdptgavted 1081 niddaflpvp
eyvnqsvpkr pagsvqnpvy hnqplhpapg rdlyhqnphs navgnpeyln 1141 taqptclssg
fnspalwiqk gshqmsldnp dyqqdffpke tkpngifkqp taenaeylr 1201 appssefiga
SEQ ID NO: 111 Mouse EGFR cDNA Sequence Variant 2 (NM_007912.4, CDS: from
281 to 22478) 1 ctccccagt ccgacccga gctaactaga cgtctgggca gcccagcgc aacgcgcagc
61 agccccctc ctcttctcc cgcactgtgc gctccccctg ggctagggcg tctggatcga 121 gtcccggagg
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 ggcatTTTT tctttctt ttttctgaa acacattgcc cagcaaagcc aataaattc 2641 tctcatcatt ttgttctga
 taaattctta ctattgat SEQ ID NO: 112 Mouse EGFR Amino Acid Sequence
 Isoform B (NP_031938.1) 1 mrpsgtartt llvlltalca aggaleekkv cqgtsnrltq lgtfedhfls
 lqrmynncev 61 vlgneityv qrnydlsflk tiqevagyvl ialntverip lenlqiirgn alyentyala 121
 ilsnygtprt glrelpmrnl qeiligavrf snnpilcnmd tiqwrdivqn vfmsnmsmdl 181 qshpsscpkc
 dpscpngscw gggeencqkl tkiicaqqcs hrccgrpsd cchnqcaagc 241 tgpresdclv cqkfqdeatc
 kdtcpplmly npttyqmdvn pegkysfgat cvkkcprnyv 301 vtdhgscvra cgpdyeyevee
 dgirkckkcd gpcrkvcngi gigeftdts inatnikhfk 361 yctaisgdh ilpvafkgds ftrtppldpr
 eleilktvke itgflliqaw pdnwtldhaf 421 enleiirgt kqhggqfslav vglntslgl rslkeisdgd
 viisgnrnlc yantinwkl 481 fgtpnqtki mnraekdck avnhvcnplc ssegcwgppep rdcvscqnvs
 rgrecvekc 541 ilegeprefv enseciqchp eclpqamnit ctgrgpdnci qcahydghph cvktcpagim 601
 genntlvwky adannvchlc hanctygcag pglqgcevwp sgyvqwqwil ktfwi * Included in Table 1
 are RNA nucleic acid molecules (e.g., thymines replaced with uracines), nucleic acid molecules
 encoding orthologs of the encoded proteins, as well as DNA or RNA nucleic acid sequences
 comprising a nucleic acid sequence having at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%,
 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5% or more identity
 across their full length with the nucleic acid sequence of any SEQ ID NO or biomarker described in
 Table 1 (see below for example), or a portion thereof. Such nucleic acid molecules can have a
 function of the full-length nucleic acid as described further herein. * Included in Table 1 are
 orthologs of the proteins, as well as polypeptide molecules comprising an amino acid sequence
 having at least 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%,
 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or more identity across their full length with an amino
 acid sequence of any SEQ ID NO or biomarker described in Table 1 (see below for example), or a
 portion thereof. Such polypeptides can have a function of the full-length polypeptide as described
 further herein. * Included in Table 1 are one or more subunits of a SWI/SNF complex like BAF or
 PBAF, and mutations within the one or more subunits. In some embodiments, the biomarkers
 are a class of mutations encompassing the one or more subunits of a SWI/SNF complex, such as
 the class of synonymous and/or non-synonymous mutations of ARID2 and/or PBRM1, or the class
 of loss-of-function mutations for biomarkers shown in Tables 4-5. In other embodiment, the
 biomarkers are particular mutations of one or more subunits of a SWI/SNF complex, such as
 particular mutations described in the Tables and Examples (e.g., Tables 4-5). Thus, included in
 Table 1 is, for example, PBRM1, ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1,

BRM, CRB1, and or EGFR, including any cDNA or polypeptide of PBRM1, ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, and EGFR. Similarly, included in Table 1 is, for example, PBRM1, ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, and EGFR nucleic acid and/or amino acid sequences encoding or representing PBRM1, ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, and EGFR having reduced or eliminated function (e.g., truncating mutations causing encoding of incomplete protein of PBRM1, ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, and EGFR). Many of these mutations were found in subjects having cancer and who were insensitive to immune checkpoint therapies. It is further determined that EGFR as a biomarker of immune checkpoint efficacy acts in opposite fashion to the other biomarkers described in Table 1 such that EGFR is mutated more frequently (e.g., hotspot mutations) in non-responders or less efficacious responders to immune checkpoint therapy rather than more frequently in subjects who respond to immune checkpoint therapy.

II. Subjects

(139) In one embodiment, the subject for whom predicted likelihood of efficacy of an immune checkpoint therapy is determined, is a mammal (e.g., mouse, rat, primate, non-human mammal, domestic animal, such as a dog, cat, cow, horse, and the like), and is preferably a human.

(140) In another embodiment of the methods of the present invention, the subject has not undergone treatment, such as chemotherapy, radiation therapy, targeted therapy, and/or immune checkpoint therapy. In still another embodiment, the subject has undergone treatment, such as chemotherapy, radiation therapy, targeted therapy, and/or immune checkpoint therapy.

(141) In certain embodiments, the subject has had surgery to remove cancerous or precancerous tissue. In other embodiments, the cancerous tissue has not been removed, e.g., the cancerous tissue may be located in an inoperable region of the body, such as in a tissue that is essential for life, or in a region where a surgical procedure would cause considerable risk of harm to the patient.

(142) The methods of the present invention can be used to determine the responsiveness to anti-immune checkpoint therapies of a cancer. In one embodiment, the cancer is one for which an immune checkpoint therapy (e.g., anti-PD-1 blocking antibody, anti-PD-L1 blocking antibody, CTLA-4 blocking antibody, and the like) is FDA-approved for treatment, such as those described in the Examples. In one embodiment, the cancers are solid tumors, such as lung cancer such as non-small cell lung cancer, bladder cancer, melanoma such as metastatic melanoma, and/or renal cell carcinoma. In another embodiment, the cancer is an epithelial cancer such as, but not limited to, brain cancer (e.g., glioblastomas) bladder cancer, breast cancer, cervical cancer, colon cancer, gynecologic cancers, renal cancer, laryngeal cancer, lung cancer, oral cancer, head and neck cancer, ovarian cancer, pancreatic cancer, prostate cancer, or skin cancer. In still other embodiments, the cancer is breast cancer, prostate cancer, lung cancer, or colon cancer. In still other embodiments, the epithelial cancer is non-small-cell lung cancer, nonpapillary renal cell carcinoma, cervical carcinoma, ovarian carcinoma (e.g., serous ovarian carcinoma), or breast carcinoma. The epithelial cancers may be characterized in various other ways including, but not limited to, serous, endometrioid, mucinous, clear cell, brenner, or undifferentiated. In yet other embodiments, the cancer is a mesenchymal cancer, such as sarcoma.

III. Sample Collection, Preparation and Separation

(143) In some embodiments, biomarker amount and/or activity measurement(s) in a sample from a subject is compared to a predetermined control (standard) sample. The sample from the subject is typically from a diseased tissue, such as cancer cells or tissues. The control sample can be from the same subject or from a different subject. The control sample is typically a normal, non-diseased sample. However, in some embodiments, such as for staging of disease or for evaluating the efficacy of treatment, the control sample can be from a diseased tissue. The control sample can be a combination of samples from several different subjects. In some embodiments, the biomarker amount and/or activity measurement(s) from a subject is compared to a pre-determined level. This

pre-determined level is typically obtained from normal samples. As described herein, a “pre-determined” biomarker amount and/or activity measurement(s) may be a biomarker amount and/or activity measurement(s) used to, by way of example only, evaluate a subject that may be selected for treatment, evaluate a response to an immune checkpoint therapy, and/or evaluate a response to a combination immune checkpoint therapy. A pre-determined biomarker amount and/or activity measurement(s) may be determined in populations of patients with or without cancer. The pre-determined biomarker amount and/or activity measurement(s) can be a single number, equally applicable to every patient, or the pre-determined biomarker amount and/or activity measurement(s) can vary according to specific subpopulations of patients. Age, weight, height, and other factors of a subject may affect the pre-determined biomarker amount and/or activity measurement(s) of the individual. Furthermore, the pre-determined biomarker amount and/or activity can be determined for each subject individually. In one embodiment, the amounts determined and/or compared in a method described herein are based on absolute measurements.

(144) In another embodiment, the amounts determined and/or compared in a method described herein are based on relative measurements, such as ratios (e.g., biomarker copy numbers, level, and/or activity before a treatment vs. after a treatment, such biomarker measurements relative to a spiked or man-made control, such biomarker measurements relative to the expression of a housekeeping gene, and the like). For example, the relative analysis can be based on the ratio of pre-treatment biomarker measurement as compared to post-treatment biomarker measurement. Pre-treatment biomarker measurement can be made at any time prior to initiation of anti-cancer therapy. Post-treatment biomarker measurement can be made at any time after initiation of anti-cancer therapy. In some embodiments, post-treatment biomarker measurements are made 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 weeks or more after initiation of anti-cancer therapy, and even longer toward indefinitely for continued monitoring. Treatment can comprise anti-cancer therapy, such as a therapeutic regimen comprising an anti-PD1 monoclonal antibody (e.g., nivolumab) alone or in combination with other anti-cancer agents, such as anti-PD-L1/PD-L2 antibodies, anti-VEGF agents (e.g., bevacizumab), agents described in the Examples, Figures, and Tables, or anti-PBRM1 (or anti-ARID2, anti-BRD7, anti-PHF10, or anti-KDM6A) agents.

(145) The pre-determined biomarker amount and/or activity measurement(s) can be any suitable standard. For example, the pre-determined biomarker amount and/or activity measurement(s) can be obtained from the same or a different human for whom a patient selection is being assessed. In one embodiment, the pre-determined biomarker amount and/or activity measurement(s) can be obtained from a previous assessment of the same patient. In such a manner, the progress of the selection of the patient can be monitored over time. In addition, the control can be obtained from an assessment of another human or multiple humans, e.g., selected groups of humans, if the subject is a human. In such a manner, the extent of the selection of the human for whom selection is being assessed can be compared to suitable other humans, e.g., other humans who are in a similar situation to the human of interest, such as those suffering from similar or the same condition(s) and/or of the same ethnic group.

(146) In some embodiments of the present invention the change of biomarker amount and/or activity measurement(s) from the pre-determined level is about 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, or 5.0 fold or greater, or any range in between, inclusive. Such cutoff values apply equally when the measurement is based on relative changes, such as based on the ratio of pre-treatment biomarker measurement as compared to post-treatment biomarker measurement.

(147) Biological samples can be collected from a variety of sources from a patient including a body fluid sample, cell sample, or a tissue sample comprising nucleic acids and/or proteins. “Body fluids” refer to fluids that are excreted or secreted from the body as well as fluids that are normally not (e.g., amniotic fluid, aqueous humor, bile, blood and blood plasma, cerebrospinal fluid, cerumen and earwax, cowper's fluid or pre-ejaculatory fluid, chyle, chyme, stool, female ejaculate,

interstitial fluid, intracellular fluid, lymph, menses, breast milk, mucus, pleural fluid, pus, saliva, sebum, semen, serum, sweat, synovial fluid, tears, urine, vaginal lubrication, vitreous humor, vomit). In a preferred embodiment, the subject and/or control sample is selected from the group consisting of cells, cell lines, histological slides, paraffin embedded tissues, biopsies, whole blood, nipple aspirate, serum, plasma, buccal scrape, saliva, cerebrospinal fluid, urine, stool, and bone marrow. In one embodiment, the sample is serum, plasma, or urine. In another embodiment, the sample is serum.

(148) The samples can be collected from individuals repeatedly over a longitudinal period of time (e.g., once or more on the order of days, weeks, months, annually, biannually, etc.). Obtaining numerous samples from an individual over a period of time can be used to verify results from earlier detections and/or to identify an alteration in biological pattern as a result of, for example, disease progression, drug treatment, etc. For example, subject samples can be taken and monitored every month, every two months, or combinations of one, two, or three month intervals according to the present invention. In addition, the biomarker amount and/or activity measurements of the subject obtained over time can be conveniently compared with each other, as well as with those of normal controls during the monitoring period, thereby providing the subject's own values, as an internal, or personal, control for long-term monitoring.

(149) Sample preparation and separation can involve any of the procedures, depending on the type of sample collected and/or analysis of biomarker measurement(s). Such procedures include, by way of example only, concentration, dilution, adjustment of pH, removal of high abundance polypeptides (e.g., albumin, gamma globulin, and transferrin, etc.), addition of preservatives and calibrants, addition of protease inhibitors, addition of denaturants, desalting of samples, concentration of sample proteins, extraction and purification of lipids.

(150) The sample preparation can also isolate molecules that are bound in non-covalent complexes to other protein (e.g., carrier proteins). This process may isolate those molecules bound to a specific carrier protein (e.g., albumin), or use a more general process, such as the release of bound molecules from all carrier proteins via protein denaturation, for example using an acid, followed by removal of the carrier proteins.

(151) Removal of undesired proteins (e.g., high abundance, uninformative, or undetectable proteins) from a sample can be achieved using high affinity reagents, high molecular weight filters, ultracentrifugation and/or electrodialysis. High affinity reagents include antibodies or other reagents (e.g., aptamers) that selectively bind to high abundance proteins. Sample preparation could also include ion exchange chromatography, metal ion affinity chromatography, gel filtration, hydrophobic chromatography, chromatofocusing, adsorption chromatography, isoelectric focusing and related techniques. Molecular weight filters include membranes that separate molecules on the basis of size and molecular weight. Such filters may further employ reverse osmosis, nanofiltration, ultrafiltration and microfiltration.

(152) Ultracentrifugation is a method for removing undesired polypeptides from a sample. Ultracentrifugation is the centrifugation of a sample at about 15,000-60,000 rpm while monitoring with an optical system the sedimentation (or lack thereof) of particles. Electrodialysis is a procedure which uses an electromembrane or semipermeable membrane in a process in which ions are transported through semi-permeable membranes from one solution to another under the influence of a potential gradient. Since the membranes used in electrodialysis may have the ability to selectively transport ions having positive or negative charge, reject ions of the opposite charge, or to allow species to migrate through a semipermeable membrane based on size and charge, it renders electrodialysis useful for concentration, removal, or separation of electrolytes.

(153) Separation and purification in the present invention may include any procedure known in the art, such as capillary electrophoresis (e.g., in capillary or on-chip) or chromatography (e.g., in capillary, column or on a chip). Electrophoresis is a method which can be used to separate ionic molecules under the influence of an electric field. Electrophoresis can be conducted in a gel,

capillary, or in a microchannel on a chip. Examples of gels used for electrophoresis include starch, acrylamide, polyethylene oxides, agarose, or combinations thereof. A gel can be modified by its cross-linking, addition of detergents, or denaturants, immobilization of enzymes or antibodies (affinity electrophoresis) or substrates (zymography) and incorporation of a pH gradient. Examples of capillaries used for electrophoresis include capillaries that interface with an electrospray.

(154) Capillary electrophoresis (CE) is preferred for separating complex hydrophilic molecules and highly charged solutes. CE technology can also be implemented on microfluidic chips. Depending on the types of capillary and buffers used, CE can be further segmented into separation techniques such as capillary zone electrophoresis (CZE), capillary isoelectric focusing (CIEF), capillary isotachophoresis (cITP) and capillary electrochromatography (CEC). An embodiment to couple CE techniques to electrospray ionization involves the use of volatile solutions, for example, aqueous mixtures containing a volatile acid and/or base and an organic such as an alcohol or acetonitrile.

(155) Capillary isotachophoresis (cITP) is a technique in which the analytes move through the capillary at a constant speed but are nevertheless separated by their respective mobilities. Capillary zone electrophoresis (CZE), also known as free-solution CE (FSCE), is based on differences in the electrophoretic mobility of the species, determined by the charge on the molecule, and the frictional resistance the molecule encounters during migration which is often directly proportional to the size of the molecule. Capillary isoelectric focusing (CIEF) allows weakly-ionizable amphoteric molecules, to be separated by electrophoresis in a pH gradient. CEC is a hybrid technique between traditional high performance liquid chromatography (HPLC) and CE.

(156) Separation and purification techniques used in the present invention include any chromatography procedures known in the art. Chromatography can be based on the differential adsorption and elution of certain analytes or partitioning of analytes between mobile and stationary phases. Different examples of chromatography include, but not limited to, liquid chromatography (LC), gas chromatography (GC), high performance liquid chromatography (HPLC), etc.

IV. Biomarker Nucleic Acids and Polypeptides

(157) One aspect of the present invention pertains to the use of isolated nucleic acid molecules that correspond to biomarker nucleic acids that encode a biomarker polypeptide or a portion of such a polypeptide. As used herein, the term “nucleic acid molecule” is intended to include DNA molecules (e.g., cDNA or genomic DNA) and RNA molecules (e.g., mRNA) and analogs of the DNA or RNA generated using nucleotide analogs. The nucleic acid molecule can be single-stranded or double-stranded, but preferably is double-stranded DNA.

(158) An “isolated” nucleic acid molecule is one which is separated from other nucleic acid molecules which are present in the natural source of the nucleic acid molecule. Preferably, an “isolated” nucleic acid molecule is free of sequences (preferably protein-encoding sequences) which naturally flank the nucleic acid (i.e., sequences located at the 5' and 3' ends of the nucleic acid) in the genomic DNA of the organism from which the nucleic acid is derived. For example, in various embodiments, the isolated nucleic acid molecule can contain less than about 5 kB, 4 kB, 3 kB, 2 kB, 1 kB, 0.5 kB or 0.1 kB of nucleotide sequences which naturally flank the nucleic acid molecule in genomic DNA of the cell from which the nucleic acid is derived. Moreover, an “isolated” nucleic acid molecule, such as a cDNA molecule, can be substantially free of other cellular material or culture medium when produced by recombinant techniques, or substantially free of chemical precursors or other chemicals when chemically synthesized.

(159) A biomarker nucleic acid molecule of the present invention can be isolated using standard molecular biology techniques and the sequence information in the database records described herein. Using all or a portion of such nucleic acid sequences, nucleic acid molecules of the present invention can be isolated using standard hybridization and cloning techniques (e.g., as described in Sambrook et al., ed., *Molecular Cloning: A Laboratory Manual*, 2nd ed., Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY, 1989).

(160) A nucleic acid molecule of the present invention can be amplified using cDNA, mRNA, or

genomic DNA as a template and appropriate oligonucleotide primers according to standard PCR amplification techniques. The nucleic acid molecules so amplified can be cloned into an appropriate vector and characterized by DNA sequence analysis. Furthermore, oligonucleotides corresponding to all or a portion of a nucleic acid molecule of the present invention can be prepared by standard synthetic techniques, e.g., using an automated DNA synthesizer.

(161) Moreover, a nucleic acid molecule of the present invention can comprise only a portion of a nucleic acid sequence, wherein the full length nucleic acid sequence comprises a marker of the present invention or which encodes a polypeptide corresponding to a marker of the present invention. Such nucleic acid molecules can be used, for example, as a probe or primer. The probe/primer typically is used as one or more substantially purified oligonucleotides. The oligonucleotide typically comprises a region of nucleotide sequence that hybridizes under stringent conditions to at least about 7, preferably about 15, more preferably about 25, 50, 75, 100, 125, 150, 175, 200, 250, 300, 350, or 400 or more consecutive nucleotides of a biomarker nucleic acid sequence. Probes based on the sequence of a biomarker nucleic acid molecule can be used to detect transcripts or genomic sequences corresponding to one or more markers of the present invention. The probe comprises a label group attached thereto, e.g., a radioisotope, a fluorescent compound, an enzyme, or an enzyme co-factor.

(162) A biomarker nucleic acid molecules that differ, due to degeneracy of the genetic code, from the nucleotide sequence of nucleic acid molecules encoding a protein which corresponds to the biomarker, and thus encode the same protein, are also contemplated.

(163) In addition, it will be appreciated by those skilled in the art that DNA sequence polymorphisms that lead to changes in the amino acid sequence can exist within a population (e.g., the human population). Such genetic polymorphisms can exist among individuals within a population due to natural allelic variation. An allele is one of a group of genes which occur alternatively at a given genetic locus. In addition, it will be appreciated that DNA polymorphisms that affect RNA expression levels can also exist that may affect the overall expression level of that gene (e.g., by affecting regulation or degradation).

(164) The term “allele,” which is used interchangeably herein with “allelic variant,” refers to alternative forms of a gene or portions thereof. Alleles occupy the same locus or position on homologous chromosomes. When a subject has two identical alleles of a gene, the subject is said to be homozygous for the gene or allele. When a subject has two different alleles of a gene, the subject is said to be heterozygous for the gene or allele. For example, biomarker alleles can differ from each other in a single nucleotide, or several nucleotides, and can include substitutions, deletions, and insertions of nucleotides. An allele of a gene can also be a form of a gene containing one or more mutations.

(165) The term “allelic variant of a polymorphic region of gene” or “allelic variant”, used interchangeably herein, refers to an alternative form of a gene having one of several possible nucleotide sequences found in that region of the gene in the population. As used herein, allelic variant is meant to encompass functional allelic variants, non-functional allelic variants, SNPs, mutations and polymorphisms.

(166) The term “single nucleotide polymorphism” (SNP) refers to a polymorphic site occupied by a single nucleotide, which is the site of variation between allelic sequences. The site is usually preceded by and followed by highly conserved sequences of the allele (e.g., sequences that vary in less than 1/100 or 1/1000 members of a population). A SNP usually arises due to substitution of one nucleotide for another at the polymorphic site. SNPs can also arise from a deletion of a nucleotide or an insertion of a nucleotide relative to a reference allele. Typically the polymorphic site is occupied by a base other than the reference base. For example, where the reference allele contains the base “T” (thymidine) at the polymorphic site, the altered allele can contain a “C” (cytidine), “G” (guanine), or “A” (adenine) at the polymorphic site. SNP's may occur in protein-coding nucleic acid sequences, in which case they may give rise to a defective or otherwise variant protein,

or genetic disease. Such a SNP may alter the coding sequence of the gene and therefore specify another amino acid (a “missense” SNP) or a SNP may introduce a stop codon (a “nonsense” SNP). When a SNP does not alter the amino acid sequence of a protein, the SNP is called “silent.” SNP's may also occur in noncoding regions of the nucleotide sequence. This may result in defective protein expression, e.g., as a result of alternative splicing, or it may have no effect on the function of the protein.

(167) As used herein, the terms “gene” and “recombinant gene” refer to nucleic acid molecules comprising an open reading frame encoding a polypeptide corresponding to a marker of the present invention. Such natural allelic variations can typically result in 1-5% variance in the nucleotide sequence of a given gene. Alternative alleles can be identified by sequencing the gene of interest in a number of different individuals. This can be readily carried out by using hybridization probes to identify the same genetic locus in a variety of individuals. Any and all such nucleotide variations and resulting amino acid polymorphisms or variations that are the result of natural allelic variation and that do not alter the functional activity are intended to be within the scope of the present invention.

(168) In another embodiment, a biomarker nucleic acid molecule is at least 7, 15, 20, 25, 30, 40, 60, 80, 100, 150, 200, 250, 300, 350, 400, 450, 550, 650, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 1900, 2000, 2200, 2400, 2600, 2800, 3000, 3500, 4000, 4500, or more nucleotides in length and hybridizes under stringent conditions to a nucleic acid molecule corresponding to a marker of the present invention or to a nucleic acid molecule encoding a protein corresponding to a marker of the present invention. As used herein, the term “hybridizes under stringent conditions” is intended to describe conditions for hybridization and washing under which nucleotide sequences at least 60% (65%, 70%, 75%, 80%, preferably 85%) identical to each other typically remain hybridized to each other. Such stringent conditions are known to those skilled in the art and can be found in sections 6.3.1-6.3.6 of *Current Protocols in Molecular Biology*, John Wiley & Sons, N.Y. (1989). A preferred, non-limiting example of stringent hybridization conditions are hybridization in 6× sodium chloride/sodium citrate (SSC) at about 45° C., followed by one or more washes in 0.2×SSC, 0.1% SDS at 50-65° C.

(169) In addition to naturally-occurring allelic variants of a nucleic acid molecule of the present invention that can exist in the population, the skilled artisan will further appreciate that sequence changes can be introduced by mutation thereby leading to changes in the amino acid sequence of the encoded protein, without altering the biological activity of the protein encoded thereby. For example, one can make nucleotide substitutions leading to amino acid substitutions at “non-essential” amino acid residues. A “non-essential” amino acid residue is a residue that can be altered from the wild-type sequence without altering the biological activity, whereas an “essential” amino acid residue is required for biological activity. For example, amino acid residues that are not conserved or only semi-conserved among homologs of various species may be non-essential for activity and thus would be likely targets for alteration. Alternatively, amino acid residues that are conserved among the homologs of various species (e.g., murine and human) may be essential for activity and thus would not be likely targets for alteration.

(170) Accordingly, another aspect of the present invention pertains to nucleic acid molecules encoding a polypeptide of the present invention that contain changes in amino acid residues that are not essential for activity. Such polypeptides differ in amino acid sequence from the naturally-occurring proteins which correspond to the markers of the present invention, yet retain biological activity. In one embodiment, a biomarker protein has an amino acid sequence that is at least about 40% identical, 50%, 60%, 70%, 75%, 80%, 83%, 85%, 87.5%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99% or identical to the amino acid sequence of a biomarker protein described herein.

(171) An isolated nucleic acid molecule encoding a variant protein can be created by introducing one or more nucleotide substitutions, additions or deletions into the nucleotide sequence of nucleic

acids of the present invention, such that one or more amino acid residue substitutions, additions, or deletions are introduced into the encoded protein. Mutations can be introduced by standard techniques, such as site-directed mutagenesis and PCR-mediated mutagenesis. Preferably, conservative amino acid substitutions are made at one or more predicted non-essential amino acid residues. A “conservative amino acid substitution” is one in which the amino acid residue is replaced with an amino acid residue having a similar side chain. Families of amino acid residues having similar side chains have been defined in the art. These families include amino acids with basic side chains (e.g., lysine, arginine, histidine), acidic side chains (e.g., aspartic acid, glutamic acid), uncharged polar side chains (e.g., glycine, asparagine, glutamine, serine, threonine, tyrosine, cysteine), non-polar side chains (e.g., alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine, tryptophan), beta-branched side chains (e.g., threonine, valine, isoleucine) and aromatic side chains (e.g., tyrosine, phenylalanine, tryptophan, histidine). Alternatively, mutations can be introduced randomly along all or part of the coding sequence, such as by saturation mutagenesis, and the resultant mutants can be screened for biological activity to identify mutants that retain activity. Following mutagenesis, the encoded protein can be expressed recombinantly and the activity of the protein can be determined.

(172) In some embodiments, the present invention further contemplates the use of anti-biomarker antisense nucleic acid molecules, i.e., molecules which are complementary to a sense nucleic acid of the present invention, e.g., complementary to the coding strand of a double-stranded cDNA molecule corresponding to a marker of the present invention or complementary to an mRNA sequence corresponding to a marker of the present invention. Accordingly, an antisense nucleic acid molecule of the present invention can hydrogen bond to (i.e. anneal with) a sense nucleic acid of the present invention. The antisense nucleic acid can be complementary to an entire coding strand, or to only a portion thereof, e.g., all or part of the protein coding region (or open reading frame). An antisense nucleic acid molecule can also be antisense to all or part of a non-coding region of the coding strand of a nucleotide sequence encoding a polypeptide of the present invention. The non-coding regions (“5′ and 3′ untranslated regions”) are the 5′ and 3′ sequences which flank the coding region and are not translated into amino acids.

(173) An antisense oligonucleotide can be, for example, about 5, 10, 15, 20, 25, 30, 35, 40, 45, or 50 or more nucleotides in length. An antisense nucleic acid can be constructed using chemical synthesis and enzymatic ligation reactions using procedures known in the art. For example, an antisense nucleic acid (e.g., an antisense oligonucleotide) can be chemically synthesized using naturally occurring nucleotides or variously modified nucleotides designed to increase the biological stability of the molecules or to increase the physical stability of the duplex formed between the antisense and sense nucleic acids, e.g., phosphorothioate derivatives and acridine substituted nucleotides can be used. Examples of modified nucleotides which can be used to generate the antisense nucleic acid include 5-fluorouracil, 5-bromouracil, 5-chlorouracil, 5-iodouracil, hypoxanthine, xanthine, 4-acetylcytosine, 5-(carboxyhydroxymethyl) uracil, 5-carboxymethylaminomethyl-2-thiouridine, 5-carboxymethylaminomethyluracil, dihydrouracil, beta-D-galactosylqueosine, inosine, N6-isopentenyladenine, 1-methylguanine, 1-methylinosine, 2,2-dimethylguanine, 2-methyladenine, 2-methylguanine, 3-methylcytosine, 5-methylcytosine, N6-adenine, 7-methylguanine, 5-methylaminomethyluracil, 5-methoxyaminomethyl-2-thiouracil, beta-D-mannosylqueosine, 5′-methoxycarboxymethyluracil, 5-methoxyuracil, 2-methylthio-N6-isopentenyladenine, uracil-5-oxyacetic acid (v), wybutoxosine, pseudouracil, queosine, 2-thiocytosine, 5-methyl-2-thiouracil, 2-thiouracil, 4-thiouracil, 5-methyluracil, uracil-5-oxyacetic acid methylester, uracil-5-oxy acetic acid (v), 5-methyl-2-thiouracil, 3-(3-amino-3-N-2-carboxypropyl) uracil, (acp3)w, and 2,6-diaminopurine. Alternatively, the antisense nucleic acid can be produced biologically using an expression vector into which a nucleic acid has been sub-cloned in an antisense orientation (i.e., RNA transcribed from the inserted nucleic acid will be of an antisense orientation to a target nucleic acid of interest, described further in the following

subsection).

(174) The antisense nucleic acid molecules of the present invention are typically administered to a subject or generated in situ such that they hybridize with or bind to cellular mRNA and/or genomic DNA encoding a polypeptide corresponding to a selected marker of the present invention to thereby inhibit expression of the marker, e.g., by inhibiting transcription and/or translation. The hybridization can be by conventional nucleotide complementarity to form a stable duplex, or, for example, in the case of an antisense nucleic acid molecule which binds to DNA duplexes, through specific interactions in the major groove of the double helix. Examples of a route of administration of antisense nucleic acid molecules of the present invention includes direct injection at a tissue site or infusion of the antisense nucleic acid into a blood- or bone marrow-associated body fluid. Alternatively, antisense nucleic acid molecules can be modified to target selected cells and then administered systemically. For example, for systemic administration, antisense molecules can be modified such that they specifically bind to receptors or antigens expressed on a selected cell surface, e.g., by linking the antisense nucleic acid molecules to peptides or antibodies which bind to cell surface receptors or antigens. The antisense nucleic acid molecules can also be delivered to cells using the vectors described herein. To achieve sufficient intracellular concentrations of the antisense molecules, vector constructs in which the antisense nucleic acid molecule is placed under the control of a strong pol II or pol III promoter are preferred.

(175) An antisense nucleic acid molecule of the present invention can be an α -anomeric nucleic acid molecule. An α -anomeric nucleic acid molecule forms specific double-stranded hybrids with complementary RNA in which, contrary to the usual α -units, the strands run parallel to each other (Gaultier et al. (1987) *Nucleic Acids Res.* 15:6625-6641). The antisense nucleic acid molecule can also comprise a 2'-o-methylribonucleotide (Inoue et al. (1987) *Nucleic Acids Res.* 15:6131-6148) or a chimeric RNA-DNA analogue (Inoue et al. (1987) *FEBS Lett.* 215:327-330).

(176) The present invention also encompasses ribozymes. Ribozymes are catalytic RNA molecules with ribonuclease activity which are capable of cleaving a single-stranded nucleic acid, such as an mRNA, to which they have a complementary region. Thus, ribozymes (e.g., hammerhead ribozymes as described in Haselhoff and Gerlach (1988) *Nature* 334:585-591) can be used to catalytically cleave mRNA transcripts to thereby inhibit translation of the protein encoded by the mRNA. A ribozyme having specificity for a nucleic acid molecule encoding a polypeptide corresponding to a marker of the present invention can be designed based upon the nucleotide sequence of a cDNA corresponding to the marker. For example, a derivative of a Tetrahymena L-19 IVS RNA can be constructed in which the nucleotide sequence of the active site is complementary to the nucleotide sequence to be cleaved (see Cech et al. U.S. Pat. No. 4,987,071; and Cech et al. U.S. Pat. No. 5,116,742). Alternatively, an mRNA encoding a polypeptide of the present invention can be used to select a catalytic RNA having a specific ribonuclease activity from a pool of RNA molecules (see, e.g., Bartel and Szostak (1993) *Science* 261:1411-1418).

(177) The present invention also encompasses nucleic acid molecules which form triple helical structures. For example, expression of a biomarker protein can be inhibited by targeting nucleotide sequences complementary to the regulatory region of the gene encoding the polypeptide (e.g., the promoter and/or enhancer) to form triple helical structures that prevent transcription of the gene in target cells. See generally Helene (1991) *Anticancer Drug Des.* 6(6):569-84; Helene (1992) *Ann. N. Y. Acad. Sci.* 660:27-36; and Maher (1992) *Bioassays* 14(12):807-15.

(178) In various embodiments, the nucleic acid molecules of the present invention can be modified at the base moiety, sugar moiety or phosphate backbone to improve, e.g., the stability, hybridization, or solubility of the molecule. For example, the deoxyribose phosphate backbone of the nucleic acid molecules can be modified to generate peptide nucleic acid molecules (see Hyrup et al. (1996) *Bioorganic & Medicinal Chemistry* 4(1): 5-23). As used herein, the terms "peptide nucleic acids" or "PNAs" refer to nucleic acid mimics, e.g., DNA mimics, in which the deoxyribose phosphate backbone is replaced by a pseudopeptide backbone and only the four

natural nucleobases are retained. The neutral backbone of PNAs has been shown to allow for specific hybridization to DNA and RNA under conditions of low ionic strength. The synthesis of PNA oligomers can be performed using standard solid phase peptide synthesis protocols as described in Hyrup et al. (1996), supra; Perry-O'Keefe et al. (1996) *Proc. Natl. Acad. Sci. USA* 93:14670-675.

(179) PNAs can be used in therapeutic and diagnostic applications. For example, PNAs can be used as antisense or antigene agents for sequence-specific modulation of gene expression by, e.g., inducing transcription or translation arrest or inhibiting replication. PNAs can also be used, e.g., in the analysis of single base pair mutations in a gene by, e.g., PNA directed PCR clamping; as artificial restriction enzymes when used in combination with other enzymes, e.g., S1 nucleases (Hyrup (1996), supra; or as probes or primers for DNA sequence and hybridization (Hyrup (1996), supra; Perry-O'Keefe et al. (1996) *Proc. Natl. Acad. Sci. USA* 93:14670-14675).

(180) In another embodiment, PNAs can be modified, e.g., to enhance their stability or cellular uptake, by attaching lipophilic or other helper groups to PNA, by the formation of PNA-DNA chimeras, or by the use of liposomes or other techniques of drug delivery known in the art. For example, PNA-DNA chimeras can be generated which can combine the advantageous properties of PNA and DNA. Such chimeras allow DNA recognition enzymes, e.g., RNASE H and DNA polymerases, to interact with the DNA portion while the PNA portion would provide high binding affinity and specificity. PNA-DNA chimeras can be linked using linkers of appropriate lengths selected in terms of base stacking, number of bonds between the nucleobases, and orientation (Hyrup (1996), supra). The synthesis of PNA-DNA chimeras can be performed as described in Hyrup (1996), supra, and Finn et al. (1996) *Nucleic Acids Res.* 24(17):3357-3363. For example, a DNA chain can be synthesized on a solid support using standard phosphoramidite coupling chemistry and modified nucleoside analogs. Compounds such as 5'-(4-methoxytrityl)amino-5'-deoxy-thymidine phosphoramidite can be used as a link between the PNA and the 5' end of DNA (Mag et al. (1989) *Nucleic Acids Res.* 17:5973-5988). PNA monomers are then coupled in a step-wise manner to produce a chimeric molecule with a 5' PNA segment and a 3' DNA segment (Finn et al. (1996) *Nucleic Acids Res.* 24:3357-3363). Alternatively, chimeric molecules can be synthesized with a 5' DNA segment and a 3' PNA segment (Peterser et al. (1975) *Bioorganic Med. Chem. Lett.* 5:1119-11124).

(181) In other embodiments, the oligonucleotide can include other appended groups such as peptides (e.g., for targeting host cell receptors in vivo), or agents facilitating transport across the cell membrane (see, e.g., Letsinger et al. (1989) *Proc. Natl. Acad. Sci. USA* 86:6553-6556; Lemaitre et al. (1987) *Proc. Natl. Acad. Sci. USA* 84:648-652; PCT Publication No. WO 88/09810) or the blood-brain barrier (see, e.g., PCT Publication No. WO 89/10134). In addition, oligonucleotides can be modified with hybridization-triggered cleavage agents (see, e.g., Krol et al. (1988) *Bio Techniques* 6:958-976) or intercalating agents (see, e.g., Zon (1988) *Pharm. Res.* 5:539-549). To this end, the oligonucleotide can be conjugated to another molecule, e.g., a peptide, hybridization triggered cross-linking agent, transport agent, hybridization-triggered cleavage agent, etc.

(182) Another aspect of the present invention pertains to the use of biomarker proteins and biologically active portions thereof. In one embodiment, the native polypeptide corresponding to a marker can be isolated from cells or tissue sources by an appropriate purification scheme using standard protein purification techniques. In another embodiment, polypeptides corresponding to a marker of the present invention are produced by recombinant DNA techniques. Alternative to recombinant expression, a polypeptide corresponding to a marker of the present invention can be synthesized chemically using standard peptide synthesis techniques.

(183) An "isolated" or "purified" protein or biologically active portion thereof is substantially free of cellular material or other contaminating proteins from the cell or tissue source from which the protein is derived, or substantially free of chemical precursors or other chemicals when chemically

synthesized. The language “substantially free of cellular material” includes preparations of protein in which the protein is separated from cellular components of the cells from which it is isolated or recombinantly produced. Thus, protein that is substantially free of cellular material includes preparations of protein having less than about 30%, 20%, 10%, or 5% (by dry weight) of heterologous protein (also referred to herein as a “contaminating protein”). When the protein or biologically active portion thereof is recombinantly produced, it is also preferably substantially free of culture medium, i.e., culture medium represents less than about 20%, 10%, or 5% of the volume of the protein preparation. When the protein is produced by chemical synthesis, it is preferably substantially free of chemical precursors or other chemicals, i.e., it is separated from chemical precursors or other chemicals which are involved in the synthesis of the protein. Accordingly such preparations of the protein have less than about 30%, 20%, 10%, 5% (by dry weight) of chemical precursors or compounds other than the polypeptide of interest.

(184) Biologically active portions of a biomarker polypeptide include polypeptides comprising amino acid sequences sufficiently identical to or derived from a biomarker protein amino acid sequence described herein, but which includes fewer amino acids than the full length protein, and exhibit at least one activity of the corresponding full-length protein. Typically, biologically active portions comprise a domain or motif with at least one activity of the corresponding protein. A biologically active portion of a protein of the present invention can be a polypeptide which is, for example, 10, 25, 50, 100 or more amino acids in length. Moreover, other biologically active portions, in which other regions of the protein are deleted, can be prepared by recombinant techniques and evaluated for one or more of the functional activities of the native form of a polypeptide of the present invention.

(185) Preferred polypeptides have an amino acid sequence of a biomarker protein encoded by a nucleic acid molecule described herein. Other useful proteins are substantially identical (e.g., at least about 40%, preferably 50%, 60%, 70%, 75%, 80%, 83%, 85%, 88%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%) to one of these sequences and retain the functional activity of the protein of the corresponding naturally-occurring protein yet differ in amino acid sequence due to natural allelic variation or mutagenesis.

(186) To determine the percent identity of two amino acid sequences or of two nucleic acids, the sequences are aligned for optimal comparison purposes (e.g., gaps can be introduced in the sequence of a first amino acid or nucleic acid sequence for optimal alignment with a second amino or nucleic acid sequence). The amino acid residues or nucleotides at corresponding amino acid positions or nucleotide positions are then compared. When a position in the first sequence is occupied by the same amino acid residue or nucleotide as the corresponding position in the second sequence, then the molecules are identical at that position. The percent identity between the two sequences is a function of the number of identical positions shared by the sequences (i.e., % identity=# of identical positions/total # of positions (e.g., overlapping positions)×100). In one embodiment the two sequences are the same length.

(187) The determination of percent identity between two sequences can be accomplished using a mathematical algorithm. A preferred, non-limiting example of a mathematical algorithm utilized for the comparison of two sequences is the algorithm of Karlin and Altschul (1990) *Proc. Natl. Acad. Sci. USA* 87:2264-2268, modified as in Karlin and Altschul (1993) *Proc. Natl. Acad. Sci. USA* 90:5873-5877. Such an algorithm is incorporated into the NBLAST and XBLAST programs of Altschul, et al. (1990) *J Mol. Biol.* 215:403-410. BLAST nucleotide searches can be performed with the NBLAST program, score=100, wordlength=12 to obtain nucleotide sequences homologous to a nucleic acid molecules of the present invention. BLAST protein searches can be performed with the XBLAST program, score=50, wordlength=3 to obtain amino acid sequences homologous to a protein molecules of the present invention. To obtain gapped alignments for comparison purposes, Gapped BLAST can be utilized as described in Altschul et al. (1997) *Nucleic Acids Res.* 25:3389-3402. Alternatively, PSI-Blast can be used to perform an iterated search which

detects distant relationships between molecules. When utilizing BLAST, Gapped BLAST, and PSI-Blast programs, the default parameters of the respective programs (e.g., XBLAST and NBLAST) can be used. See <http://www.ncbi.nlm.nih.gov>. Another preferred, non-limiting example of a mathematical algorithm utilized for the comparison of sequences is the algorithm of Myers and Miller, (1988) *Comput Appl Biosci*, 4:11-7. Such an algorithm is incorporated into the ALIGN program (version 2.0) which is part of the GCG sequence alignment software package. When utilizing the ALIGN program for comparing amino acid sequences, a PAM120 weight residue table, a gap length penalty of 12, and a gap penalty of 4 can be used. Yet another useful algorithm for identifying regions of local sequence similarity and alignment is the FASTA algorithm as described in Pearson and Lipman (1988) *Proc. Natl. Acad. Sci. USA* 85:2444-2448. When using the FASTA algorithm for comparing nucleotide or amino acid sequences, a PAM120 weight residue table can, for example, be used with a k-tuple value of 2.

(188) The percent identity between two sequences can be determined using techniques similar to those described above, with or without allowing gaps. In calculating percent identity, only exact matches are counted.

(189) The present invention also provides chimeric or fusion proteins corresponding to a biomarker protein. As used herein, a “chimeric protein” or “fusion protein” comprises all or part (preferably a biologically active part) of a polypeptide corresponding to a marker of the present invention operably linked to a heterologous polypeptide (i.e., a polypeptide other than the polypeptide corresponding to the marker). Within the fusion protein, the term “operably linked” is intended to indicate that the polypeptide of the present invention and the heterologous polypeptide are fused in-frame to each other. The heterologous polypeptide can be fused to the amino-terminus or the carboxyl-terminus of the polypeptide of the present invention.

(190) One useful fusion protein is a GST fusion protein in which a polypeptide corresponding to a marker of the present invention is fused to the carboxyl terminus of GST sequences. Such fusion proteins can facilitate the purification of a recombinant polypeptide of the present invention.

(191) In another embodiment, the fusion protein contains a heterologous signal sequence, immunoglobulin fusion protein, toxin, or other useful protein sequence. Chimeric and fusion proteins of the present invention can be produced by standard recombinant DNA techniques. In another embodiment, the fusion gene can be synthesized by conventional techniques including automated DNA synthesizers. Alternatively, PCR amplification of gene fragments can be carried out using anchor primers which give rise to complementary overhangs between two consecutive gene fragments which can subsequently be annealed and re-amplified to generate a chimeric gene sequence (see, e.g., Ausubel et al., supra). Moreover, many expression vectors are commercially available that already encode a fusion moiety (e.g., a GST polypeptide). A nucleic acid encoding a polypeptide of the present invention can be cloned into such an expression vector such that the fusion moiety is linked in-frame to the polypeptide of the present invention.

(192) A signal sequence can be used to facilitate secretion and isolation of the secreted protein or other proteins of interest. Signal sequences are typically characterized by a core of hydrophobic amino acids which are generally cleaved from the mature protein during secretion in one or more cleavage events. Such signal peptides contain processing sites that allow cleavage of the signal sequence from the mature proteins as they pass through the secretory pathway. Thus, the present invention pertains to the described polypeptides having a signal sequence, as well as to polypeptides from which the signal sequence has been proteolytically cleaved (i.e., the cleavage products). In one embodiment, a nucleic acid sequence encoding a signal sequence can be operably linked in an expression vector to a protein of interest, such as a protein which is ordinarily not secreted or is otherwise difficult to isolate. The signal sequence directs secretion of the protein, such as from a eukaryotic host into which the expression vector is transformed, and the signal sequence is subsequently or concurrently cleaved. The protein can then be readily purified from the extracellular medium by art recognized methods. Alternatively, the signal sequence can be linked to

the protein of interest using a sequence which facilitates purification, such as with a GST domain.

(193) The present invention also pertains to variants of the biomarker polypeptides described herein. Such variants have an altered amino acid sequence which can function as either agonists (mimetics) or as antagonists. Variants can be generated by mutagenesis, e.g., discrete point mutation or truncation. An agonist can retain substantially the same, or a subset, of the biological activities of the naturally occurring form of the protein. An antagonist of a protein can inhibit one or more of the activities of the naturally occurring form of the protein by, for example, competitively binding to a downstream or upstream member of a cellular signaling cascade which includes the protein of interest. Thus, specific biological effects can be elicited by treatment with a variant of limited function. Treatment of a subject with a variant having a subset of the biological activities of the naturally occurring form of the protein can have fewer side effects in a subject relative to treatment with the naturally occurring form of the protein.

(194) Variants of a biomarker protein which function as either agonists (mimetics) or as antagonists can be identified by screening combinatorial libraries of mutants, e.g., truncation mutants, of the protein of the present invention for agonist or antagonist activity. In one embodiment, a variegated library of variants is generated by combinatorial mutagenesis at the nucleic acid level and is encoded by a variegated gene library. A variegated library of variants can be produced by, for example, enzymatically ligating a mixture of synthetic oligonucleotides into gene sequences such that a degenerate set of potential protein sequences is expressible as individual polypeptides, or alternatively, as a set of larger fusion proteins (e.g., for phage display). There are a variety of methods which can be used to produce libraries of potential variants of the polypeptides of the present invention from a degenerate oligonucleotide sequence. Methods for synthesizing degenerate oligonucleotides are known in the art (see, e.g., Narang (1983) *Tetrahedron* 39:3; Itakura et al. (1984) *Annu. Rev. Biochem.* 53:323; Itakura et al. (1984) *Science* 198:1056; Ike et al. (1983) *Nucleic Acid Res.* 11:477).

(195) In addition, libraries of fragments of the coding sequence of a polypeptide corresponding to a marker of the present invention can be used to generate a variegated population of polypeptides for screening and subsequent selection of variants. For example, a library of coding sequence fragments can be generated by treating a double stranded PCR fragment of the coding sequence of interest with a nuclease under conditions wherein nicking occurs only about once per molecule, denaturing the double stranded DNA, renaturing the DNA to form double stranded DNA which can include sense/antisense pairs from different nicked products, removing single stranded portions from reformed duplexes by treatment with S1 nuclease, and ligating the resulting fragment library into an expression vector. By this method, an expression library can be derived which encodes amino terminal and internal fragments of various sizes of the protein of interest.

(196) Several techniques are known in the art for screening gene products of combinatorial libraries made by point mutations or truncation, and for screening cDNA libraries for gene products having a selected property. The most widely used techniques, which are amenable to high throughput analysis, for screening large gene libraries typically include cloning the gene library into replicable expression vectors, transforming appropriate cells with the resulting library of vectors, and expressing the combinatorial genes under conditions in which detection of a desired activity facilitates isolation of the vector encoding the gene whose product was detected. Recursive ensemble mutagenesis (REM), a technique which enhances the frequency of functional mutants in the libraries, can be used in combination with the screening assays to identify variants of a protein of the present invention (Arkin and Yourvan (1992) *Proc. Natl. Acad. Sci. USA* 89:7811-7815; Delgrave et al. 1993) *Protein Engineering* 6(3):327-331).

(197) The production and use of biomarker nucleic acid and/or biomarker polypeptide molecules described herein can be facilitated by using standard recombinant techniques. In some embodiments, such techniques use vectors, preferably expression vectors, containing a nucleic acid encoding a biomarker polypeptide or a portion of such a polypeptide. As used herein, the term

“vector” refers to a nucleic acid molecule capable of transporting another nucleic acid to which it has been linked. One type of vector is a “plasmid”, which refers to a circular double stranded DNA loop into which additional DNA segments can be ligated. Another type of vector is a viral vector, wherein additional DNA segments can be ligated into the viral genome. Certain vectors are capable of autonomous replication in a host cell into which they are introduced (e.g., bacterial vectors having a bacterial origin of replication and episomal mammalian vectors). Other vectors (e.g., non-episomal mammalian vectors) are integrated into the genome of a host cell upon introduction into the host cell, and thereby are replicated along with the host genome. Moreover, certain vectors, namely expression vectors, are capable of directing the expression of genes to which they are operably linked. In general, expression vectors of utility in recombinant DNA techniques are often in the form of plasmids (vectors). However, the present invention is intended to include such other forms of expression vectors, such as viral vectors (e.g., replication defective retroviruses, adenoviruses and adeno-associated viruses), which serve equivalent functions.

(198) The recombinant expression vectors of the present invention comprise a nucleic acid of the present invention in a form suitable for expression of the nucleic acid in a host cell. This means that the recombinant expression vectors include one or more regulatory sequences, selected on the basis of the host cells to be used for expression, which is operably linked to the nucleic acid sequence to be expressed. Within a recombinant expression vector, “operably linked” is intended to mean that the nucleotide sequence of interest is linked to the regulatory sequence(s) in a manner which allows for expression of the nucleotide sequence (e.g., in an in vitro transcription/translation system or in a host cell when the vector is introduced into the host cell). The term “regulatory sequence” is intended to include promoters, enhancers and other expression control elements (e.g., polyadenylation signals). Such regulatory sequences are described, for example, in Goeddel, *Methods in Enzymology: Gene Expression Technology* vol. 185, Academic Press, San Diego, CA (1991). Regulatory sequences include those which direct constitutive expression of a nucleotide sequence in many types of host cell and those which direct expression of the nucleotide sequence only in certain host cells (e.g., tissue-specific regulatory sequences). It will be appreciated by those skilled in the art that the design of the expression vector can depend on such factors as the choice of the host cell to be transformed, the level of expression of protein desired, and the like. The expression vectors of the present invention can be introduced into host cells to thereby produce proteins or peptides, including fusion proteins or peptides, encoded by nucleic acids as described herein.

(199) The recombinant expression vectors for use in the present invention can be designed for expression of a polypeptide corresponding to a marker of the present invention in prokaryotic (e.g., *E. coli*) or eukaryotic cells (e.g., insect cells {using baculovirus expression vectors}, yeast cells or mammalian cells). Suitable host cells are discussed further in Goeddel, supra. Alternatively, the recombinant expression vector can be transcribed and translated in vitro, for example using T7 promoter regulatory sequences and T7 polymerase.

(200) Expression of proteins in prokaryotes is most often carried out in *E. coli* with vectors containing constitutive or inducible promoters directing the expression of either fusion or non-fusion proteins. Fusion vectors add a number of amino acids to a protein encoded therein, usually to the amino terminus of the recombinant protein. Such fusion vectors typically serve three purposes: 1) to increase expression of recombinant protein; 2) to increase the solubility of the recombinant protein; and 3) to aid in the purification of the recombinant protein by acting as a ligand in affinity purification. Often, in fusion expression vectors, a proteolytic cleavage site is introduced at the junction of the fusion moiety and the recombinant protein to enable separation of the recombinant protein from the fusion moiety subsequent to purification of the fusion protein. Such enzymes, and their cognate recognition sequences, include Factor Xa, thrombin and enterokinase. Typical fusion expression vectors include pGEX (Pharmacia Biotech Inc; Smith and Johnson, 1988, *Gene* 67:31-40), pMAL (New England Biolabs, Beverly, MA) and pRIT5

(Pharmacia, Piscataway, NJ) which fuse glutathione S-transferase (GST), maltose E binding protein, or protein A, respectively, to the target recombinant protein.

(201) Examples of suitable inducible non-fusion *E. coli* expression vectors include pTrc (Amann et al. (1988) *Gene* 69:301-315) and pET 11d (Studier et al., p. 60-89, In *Gene Expression Technology: Methods in Enzymology* vol. 185, Academic Press, San Diego, CA, 1991). Target biomarker nucleic acid expression from the pTrc vector relies on host RNA polymerase transcription from a hybrid trp-lac fusion promoter. Target biomarker nucleic acid expression from the pET 11d vector relies on transcription from a T7 gn10-lac fusion promoter mediated by a co-expressed viral RNA polymerase (T7 gn1). This viral polymerase is supplied by host strains BL21 (DE3) or HMS174(DE3) from a resident prophage harboring a T7 gn1 gene under the transcriptional control of the lacUV 5 promoter.

(202) One strategy to maximize recombinant protein expression in *E. coli* is to express the protein in a host bacterium with an impaired capacity to proteolytically cleave the recombinant protein (Gottesman, p. 119-128, In *Gene Expression Technology: Methods in Enzymology* vol. 185, Academic Press, San Diego, CA, 1990. Another strategy is to alter the nucleic acid sequence of the nucleic acid to be inserted into an expression vector so that the individual codons for each amino acid are those preferentially utilized in *E. coli* (Wada et al., (1992) *Nucleic Acids Res.* 20:2111-2118). Such alteration of nucleic acid sequences of the present invention can be carried out by standard DNA synthesis techniques.

(203) In another embodiment, the expression vector is a yeast expression vector. Examples of vectors for expression in yeast *S. cerevisiae* include pYepSec1 (Baldari et al. (1987) *EMBO J.* 6:229-234), pMFa (Kurjan and Herskowitz (1982) *Cell* 30:933-943), pJRY88 (Schultz et al. (1987) *Gene* 54:113-123), pYES2 (Invitrogen Corporation, San Diego, CA), and pPicZ (Invitrogen Corp, San Diego, CA).

(204) Alternatively, the expression vector is a baculovirus expression vector. Baculovirus vectors available for expression of proteins in cultured insect cells (e.g., Sf 9 cells) include the pAc series (Smith et al. (1983) *Mol. Cell Biol.* 3:2156-2165) and the pVL series (Lucklow and Summers (1989) *Virology* 170:31-39).

(205) In yet another embodiment, a nucleic acid of the present invention is expressed in mammalian cells using a mammalian expression vector. Examples of mammalian expression vectors include pCDM8 (Seed (1987) *Nature* 329:840) and pMT2PC (Kaufman et al. (1987) *EMBO J.* 6:187-195). When used in mammalian cells, the expression vector's control functions are often provided by viral regulatory elements. For example, commonly used promoters are derived from polyoma, Adenovirus 2, cytomegalovirus and Simian Virus 40. For other suitable expression systems for both prokaryotic and eukaryotic cells see chapters 16 and 17 of Sambrook et al., supra.

(206) In another embodiment, the recombinant mammalian expression vector is capable of directing expression of the nucleic acid preferentially in a particular cell type (e.g., tissue-specific regulatory elements are used to express the nucleic acid). Tissue-specific regulatory elements are known in the art. Non-limiting examples of suitable tissue-specific promoters include the albumin promoter (liver-specific; Pinkert et al. (1987) *Genes Dev.* 1:268-277), lymphoid-specific promoters (Calame and Eaton (1988) *Adv. Immunol.* 43:235-275), in particular promoters of T cell receptors (Winoto and Baltimore (1989) *EMBO J.* 8:729-733) and immunoglobulins (Banerji et al. (1983) *Cell* 33:729-740; Queen and Baltimore (1983) *Cell* 33:741-748), neuron-specific promoters (e.g., the neurofilament promoter; Byrne and Ruddle (1989) *Proc. Natl. Acad. Sci. USA* 86:5473-5477), pancreas-specific promoters (Edlund et al. (1985) *Science* 230:912-916), and mammary gland-specific promoters (e.g., milk whey promoter; U.S. Pat. No. 4,873,316 and European Application Publication No. 264,166). Developmentally-regulated promoters are also encompassed, for example the murine hox promoters (Kessel and Gruss (1990) *Science* 249:374-379) and the α -fetoprotein promoter (Camper and Tilghman (1989) *Genes Dev.* 3:537-546).

(207) The present invention further provides a recombinant expression vector comprising a DNA

molecule cloned into the expression vector in an antisense orientation. That is, the DNA molecule is operably linked to a regulatory sequence in a manner which allows for expression (by transcription of the DNA molecule) of an RNA molecule which is antisense to the mRNA encoding a polypeptide of the present invention. Regulatory sequences operably linked to a nucleic acid cloned in the antisense orientation can be chosen which direct the continuous expression of the antisense RNA molecule in a variety of cell types, for instance viral promoters and/or enhancers, or regulatory sequences can be chosen which direct constitutive, tissue-specific or cell type specific expression of antisense RNA. The antisense expression vector can be in the form of a recombinant plasmid, phagemid, or attenuated virus in which antisense nucleic acids are produced under the control of a high efficiency regulatory region, the activity of which can be determined by the cell type into which the vector is introduced. For a discussion of the regulation of gene expression using antisense genes (see Weintraub et al. (1986) *Trends in Genetics*, Vol. 1(1)).

(208) Another aspect of the present invention pertains to host cells into which a recombinant expression vector of the present invention has been introduced. The terms “host cell” and “recombinant host cell” are used interchangeably herein. It is understood that such terms refer not only to the particular subject cell but to the progeny or potential progeny of such a cell. Because certain modifications may occur in succeeding generations due to either mutation or environmental influences, such progeny may not, in fact, be identical to the parent cell, but are still included within the scope of the term as used herein.

(209) A host cell can be any prokaryotic (e.g., *E. coli*) or eukaryotic cell (e.g., insect cells, yeast or mammalian cells).

(210) Vector DNA can be introduced into prokaryotic or eukaryotic cells via conventional transformation or transfection techniques. As used herein, the terms “transformation” and “transfection” are intended to refer to a variety of art-recognized techniques for introducing foreign nucleic acid into a host cell, including calcium phosphate or calcium chloride co-precipitation, DEAE-dextran-mediated transfection, lipofection, or electroporation. Suitable methods for transforming or transfecting host cells can be found in Sambrook, et al. (supra), and other laboratory manuals.

(211) For stable transfection of mammalian cells, it is known that, depending upon the expression vector and transfection technique used, only a small fraction of cells may integrate the foreign DNA into their genome. In order to identify and select these integrants, a gene that encodes a selectable marker (e.g., for resistance to antibiotics) is generally introduced into the host cells along with the gene of interest. Preferred selectable markers include those which confer resistance to drugs, such as G418, hygromycin and methotrexate. Cells stably transfected with the introduced nucleic acid can be identified by drug selection (e.g., cells that have incorporated the selectable marker gene will survive, while the other cells die).

V. Analyzing Biomarker Nucleic Acids and Polypeptides

(212) Biomarker nucleic acids and/or biomarker polypeptides can be analyzed according to the methods described herein and techniques known to the skilled artisan to identify such genetic or expression alterations useful for the present invention including, but not limited to, 1) an alteration in the level of a biomarker transcript or polypeptide, 2) a deletion or addition of one or more nucleotides from a biomarker gene, 4) a substitution of one or more nucleotides of a biomarker gene, 5) aberrant modification of a biomarker gene, such as an expression regulatory region, and the like.

(213) a. Methods for Detection of Copy Number

(214) Methods of evaluating the copy number of a biomarker nucleic acid are well known to those of skill in the art. The presence or absence of chromosomal gain or loss can be evaluated simply by a determination of copy number of the regions or markers identified herein.

(215) In one embodiment, a biological sample is tested for the presence of copy number changes in genomic loci containing the genomic marker. A copy number of at least 3, 4, 5, 6, 7, 8, 9, or 10 is

predictive of poorer outcome of anti-immune checkpoint treatment.

(216) Methods of evaluating the copy number of a biomarker locus include, but are not limited to, hybridization-based assays. Hybridization-based assays include, but are not limited to, traditional “direct probe” methods, such as Southern blots, in situ hybridization (e.g., FISH and FISH plus SKY) methods, and “comparative probe” methods, such as comparative genomic hybridization (CGH), e.g., cDNA-based or oligonucleotide-based CGH. The methods can be used in a wide variety of formats including, but not limited to, substrate (e.g. membrane or glass) bound methods or array-based approaches.

(217) In one embodiment, evaluating the biomarker gene copy number in a sample involves a Southern Blot. In a Southern Blot, the genomic DNA (typically fragmented and separated on an electrophoretic gel) is hybridized to a probe specific for the target region. Comparison of the intensity of the hybridization signal from the probe for the target region with control probe signal from analysis of normal genomic DNA (e.g., a non-amplified portion of the same or related cell, tissue, organ, etc.) provides an estimate of the relative copy number of the target nucleic acid. Alternatively, a Northern blot may be utilized for evaluating the copy number of encoding nucleic acid in a sample. In a Northern blot, mRNA is hybridized to a probe specific for the target region. Comparison of the intensity of the hybridization signal from the probe for the target region with control probe signal from analysis of normal RNA (e.g., a non-amplified portion of the same or related cell, tissue, organ, etc.) provides an estimate of the relative copy number of the target nucleic acid. Alternatively, other methods well known in the art to detect RNA can be used, such that higher or lower expression relative to an appropriate control (e.g., a non-amplified portion of the same or related cell tissue, organ, etc.) provides an estimate of the relative copy number of the target nucleic acid.

(218) An alternative means for determining genomic copy number is in situ hybridization (e.g., Angerer (1987)*Meth. Enzymol* 152: 649). Generally, in situ hybridization comprises the following steps: (1) fixation of tissue or biological structure to be analyzed; (2) prehybridization treatment of the biological structure to increase accessibility of target DNA, and to reduce nonspecific binding; (3) hybridization of the mixture of nucleic acids to the nucleic acid in the biological structure or tissue; (4) post-hybridization washes to remove nucleic acid fragments not bound in the hybridization and (5) detection of the hybridized nucleic acid fragments. The reagent used in each of these steps and the conditions for use vary depending on the particular application. In a typical in situ hybridization assay, cells are fixed to a solid support, typically a glass slide. If a nucleic acid is to be probed, the cells are typically denatured with heat or alkali. The cells are then contacted with a hybridization solution at a moderate temperature to permit annealing of labeled probes specific to the nucleic acid sequence encoding the protein. The targets (e.g., cells) are then typically washed at a predetermined stringency or at an increasing stringency until an appropriate signal to noise ratio is obtained. The probes are typically labeled, e.g., with radioisotopes or fluorescent reporters. In one embodiment, probes are sufficiently long so as to specifically hybridize with the target nucleic acid(s) under stringent conditions. Probes generally range in length from about 200 bases to about 1000 bases. In some applications it is necessary to block the hybridization capacity of repetitive sequences. Thus, in some embodiments, tRNA, human genomic DNA, or Cot-I DNA is used to block non-specific hybridization.

(219) An alternative means for determining genomic copy number is comparative genomic hybridization. In general, genomic DNA is isolated from normal reference cells, as well as from test cells (e.g., tumor cells) and amplified, if necessary. The two nucleic acids are differentially labeled and then hybridized in situ to metaphase chromosomes of a reference cell. The repetitive sequences in both the reference and test DNAs are either removed or their hybridization capacity is reduced by some means, for example by prehybridization with appropriate blocking nucleic acids and/or including such blocking nucleic acid sequences for said repetitive sequences during said hybridization. The bound, labeled DNA sequences are then rendered in a visualizable form, if

necessary. Chromosomal regions in the test cells which are at increased or decreased copy number can be identified by detecting regions where the ratio of signal from the two DNAs is altered. For example, those regions that have decreased in copy number in the test cells will show relatively lower signal from the test DNA than the reference compared to other regions of the genome. Regions that have been increased in copy number in the test cells will show relatively higher signal from the test DNA. Where there are chromosomal deletions or multiplications, differences in the ratio of the signals from the two labels will be detected and the ratio will provide a measure of the copy number. In another embodiment of CGH, array CGH (aCGH), the immobilized chromosome element is replaced with a collection of solid support bound target nucleic acids on an array, allowing for a large or complete percentage of the genome to be represented in the collection of solid support bound targets. Target nucleic acids may comprise cDNAs, genomic DNAs, oligonucleotides (e.g., to detect single nucleotide polymorphisms) and the like. Array-based CGH may also be performed with single-color labeling (as opposed to labeling the control and the possible tumor sample with two different dyes and mixing them prior to hybridization, which will yield a ratio due to competitive hybridization of probes on the arrays). In single color CGH, the control is labeled and hybridized to one array and absolute signals are read, and the possible tumor sample is labeled and hybridized to a second array (with identical content) and absolute signals are read. Copy number difference is calculated based on absolute signals from the two arrays. Methods of preparing immobilized chromosomes or arrays and performing comparative genomic hybridization are well known in the art (see, e.g., U.S. Pat. Nos. 6,335,167; 6,197,501; 5,830,645; and 5,665,549 and Albertson (1984) *EMBO J.* 3: 1227-1234; Pinkel (1988) *Proc. Natl. Acad. Sci. USA* 85: 9138-9142; EPO Pub. No. 430,402; *Methods in Molecular Biology*, Vol. 33: In situ Hybridization Protocols, Choo, ed., Humana Press, Totowa, N.J. (1994), etc.). In another embodiment, the hybridization protocol of Pinkel, et al. (1998) *Nature Genetics* 20: 207-211, or of Kallioniemi (1992) *Proc. Natl Acad Sci USA* 89:5321-5325 (1992) is used.

(220) In still another embodiment, amplification-based assays can be used to measure copy number. In such amplification-based assays, the nucleic acid sequences act as a template in an amplification reaction (e.g., Polymerase Chain Reaction (PCR)). In a quantitative amplification, the amount of amplification product will be proportional to the amount of template in the original sample. Comparison to appropriate controls, e.g. healthy tissue, provides a measure of the copy number.

(221) Methods of “quantitative” amplification are well known to those of skill in the art. For example, quantitative PCR involves simultaneously co-amplifying a known quantity of a control sequence using the same primers. This provides an internal standard that may be used to calibrate the PCR reaction. Detailed protocols for quantitative PCR are provided in Innis, et al. (1990) *PCR Protocols, A Guide to Methods and Applications*, Academic Press, Inc. N.Y.). Measurement of DNA copy number at microsatellite loci using quantitative PCR analysis is described in Ginzinger, et al. (2000) *Cancer Research* 60:5405-5409. The known nucleic acid sequence for the genes is sufficient to enable one of skill in the art to routinely select primers to amplify any portion of the gene. Fluorogenic quantitative PCR may also be used in the methods of the present invention. In fluorogenic quantitative PCR, quantitation is based on amount of fluorescence signals, e.g., TaqMan and SYBR green.

(222) Other suitable amplification methods include, but are not limited to, ligase chain reaction (LCR) (see Wu and Wallace (1989) *Genomics* 4: 560, Landegren, et al. (1988) *Science* 241:1077, and Barringer et al. (1990) *Gene* 89: 117), transcription amplification (Kwoh, et al. (1989) *Proc. Natl. Acad. Sci. USA* 86: 1173), self-sustained sequence replication (Guatelli, et al. (1990) *Proc. Nat. Acad. Sci. USA* 87: 1874), dot PCR, and linker adapter PCR, etc.

(223) Loss of heterozygosity (LOH) and major copy proportion (MCP) mapping (Wang, Z. C., et al. (2004) *Cancer Res* 64(1):64-71; Seymour, A. B., et al. (1994) *Cancer Res* 54, 2761-4; Hahn, S. A., et al. (1995) *Cancer Res* 55, 4670-5; Kimura, M., et al. (1996) *Genes Chromosomes Cancer* 17,

88-93; Li et al., (2008) *MBC Bioinform.* 9, 204-219) may also be used to identify regions of amplification or deletion.

(224) b. Methods for Detection of Biomarker Nucleic Acid Expression

(225) Biomarker expression may be assessed by any of a wide variety of well known methods for detecting expression of a transcribed molecule or protein. Non-limiting examples of such methods include immunological methods for detection of secreted, cell-surface, cytoplasmic, or nuclear proteins, protein purification methods, protein function or activity assays, nucleic acid hybridization methods, nucleic acid reverse transcription methods, and nucleic acid amplification methods.

(226) In preferred embodiments, activity of a particular gene is characterized by a measure of gene transcript (e.g. mRNA), by a measure of the quantity of translated protein, or by a measure of gene product activity. Marker expression can be monitored in a variety of ways, including by detecting mRNA levels, protein levels, or protein activity, any of which can be measured using standard techniques. Detection can involve quantification of the level of gene expression (e.g., genomic DNA, cDNA, mRNA, protein, or enzyme activity), or, alternatively, can be a qualitative assessment of the level of gene expression, in particular in comparison with a control level. The type of level being detected will be clear from the context.

(227) In another embodiment, detecting or determining expression levels of a biomarker and functionally similar homologs thereof, including a fragment or genetic alteration thereof (e.g., in regulatory or promoter regions thereof) comprises detecting or determining RNA levels for the marker of interest. In one embodiment, one or more cells from the subject to be tested are obtained and RNA is isolated from the cells. In a preferred embodiment, a sample of breast tissue cells is obtained from the subject.

(228) In one embodiment, RNA is obtained from a single cell. For example, a cell can be isolated from a tissue sample by laser capture microdissection (LCM). Using this technique, a cell can be isolated from a tissue section, including a stained tissue section, thereby assuring that the desired cell is isolated (see, e.g., Bonner et al. (1997) *Science* 278:1481; Emmert-Buck et al. (1996) *Science* 274:998; Fend et al. (1999) *Am. J. Path.* 154:61 and Murakami et al. (2000) *Kidney Int.* 58:1346). For example, Murakami et al., supra, describe isolation of a cell from a previously immunostained tissue section.

(229) It is also possible to obtain cells from a subject and culture the cells in vitro, such as to obtain a larger population of cells from which RNA can be extracted. Methods for establishing cultures of non-transformed cells, i.e., primary cell cultures, are known in the art.

(230) When isolating RNA from tissue samples or cells from individuals, it may be important to prevent any further changes in gene expression after the tissue or cells has been removed from the subject. Changes in expression levels are known to change rapidly following perturbations, e.g., heat shock or activation with lipopolysaccharide (LPS) or other reagents. In addition, the RNA in the tissue and cells may quickly become degraded. Accordingly, in a preferred embodiment, the tissue or cells obtained from a subject is snap frozen as soon as possible.

(231) RNA can be extracted from the tissue sample by a variety of methods, e.g., the guanidium thiocyanate lysis followed by CsCl centrifugation (Chirgwin et al. (1979) *Biochemistry* 18:5294-5299). RNA from single cells can be obtained as described in methods for preparing cDNA libraries from single cells, such as those described in Dulac, C. (1998) *Curr. Top. Dev. Biol.* 36:245 and Jena et al. (1996) *J. Immunol. Methods* 190:199. Care to avoid RNA degradation must be taken, e.g., by inclusion of RNAsin.

(232) The RNA sample can then be enriched in particular species. In one embodiment, poly(A)+RNA is isolated from the RNA sample. In general, such purification takes advantage of the poly-A tails on mRNA. In particular and as noted above, poly-T oligonucleotides may be immobilized within on a solid support to serve as affinity ligands for mRNA. Kits for this purpose are commercially available, e.g., the MessageMaker kit (Life Technologies, Grand Island, NY).

(233) In a preferred embodiment, the RNA population is enriched in marker sequences. Enrichment can be undertaken, e.g., by primer-specific cDNA synthesis, or multiple rounds of linear amplification based on cDNA synthesis and template-directed in vitro transcription (see, e.g., Wang et al. (1989) *PNAS* 86, 9717; Dulac et al., *supra*, and Jena et al., *supra*).

(234) The population of RNA, enriched or not in particular species or sequences, can further be amplified. As defined herein, an “amplification process” is designed to strengthen, increase, or augment a molecule within the RNA. For example, where RNA is mRNA, an amplification process such as RT-PCR can be utilized to amplify the mRNA, such that a signal is detectable or detection is enhanced. Such an amplification process is beneficial particularly when the biological, tissue, or tumor sample is of a small size or volume.

(235) Various amplification and detection methods can be used. For example, it is within the scope of the present invention to reverse transcribe mRNA into cDNA followed by polymerase chain reaction (RT-PCR); or, to use a single enzyme for both steps as described in U.S. Pat. No. 5,322,770, or reverse transcribe mRNA into cDNA followed by symmetric gap ligase chain reaction (RT-AGLCR) as described by R. L. Marshall, et al., *PCR Methods and Applications* 4: 80-84 (1994). Real time PCR may also be used.

(236) Other known amplification methods which can be utilized herein include but are not limited to the so-called “NASBA” or “3SR” technique described in *PNAS USA* 87: 1874-1878 (1990) and also described in *Nature* 350 (No. 6313): 91-92 (1991); Q-beta amplification as described in published European Patent Application (EPA) No. 4544610; strand displacement amplification (as described in G. T. Walker et al., *Clin. Chem.* 42: 9-13 (1996) and European Patent Application No. 684315; target mediated amplification, as described by PCT Publication WO9322461; PCR; ligase chain reaction (LCR) (see, e.g., Wu and Wallace, *Genomics* 4, 560 (1989), Landegren et al., *Science* 241, 1077 (1988)); self-sustained sequence replication (SSR) (see, e.g., Guatelli et al., *Proc. Nat. Acad. Sci. USA*, 87, 1874 (1990)); and transcription amplification (see, e.g., Kwoh et al., *Proc. Natl. Acad. Sci. USA* 86, 1173 (1989)).

(237) Many techniques are known in the state of the art for determining absolute and relative levels of gene expression, commonly used techniques suitable for use in the present invention include Northern analysis, RNase protection assays (RPA), microarrays and PCR-based techniques, such as quantitative PCR and differential display PCR. For example, Northern blotting involves running a preparation of RNA on a denaturing agarose gel, and transferring it to a suitable support, such as activated cellulose, nitrocellulose or glass or nylon membranes. Radiolabeled cDNA or RNA is then hybridized to the preparation, washed and analyzed by autoradiography.

(238) In situ hybridization visualization may also be employed, wherein a radioactively labeled antisense RNA probe is hybridized with a thin section of a biopsy sample, washed, cleaved with RNase and exposed to a sensitive emulsion for autoradiography. The samples may be stained with hematoxylin to demonstrate the histological composition of the sample, and dark field imaging with a suitable light filter shows the developed emulsion. Non-radioactive labels such as digoxigenin may also be used.

(239) Alternatively, mRNA expression can be detected on a DNA array, chip or a microarray. Labeled nucleic acids of a test sample obtained from a subject may be hybridized to a solid surface comprising biomarker DNA. Positive hybridization signal is obtained with the sample containing biomarker transcripts. Methods of preparing DNA arrays and their use are well known in the art (see, e.g., U.S. Pat. Nos. 6,618,679; 6,379,897; 6,664,377; 6,451,536; 5,482,257; U.S. 20030157485 and Schena et al. (1995) *Science* 20, 467-470; Gerhold et al. (1999) *Trends In Biochem. Sci.* 24, 168-173; and Lennon et al. (2000) *Drug Discovery Today* 5, 59-65, which are herein incorporated by reference in their entirety). Serial Analysis of Gene Expression (SAGE) can also be performed (See for example U.S. Patent Application 20030215858).

(240) To monitor mRNA levels, for example, mRNA is extracted from the biological sample to be tested, reverse transcribed, and fluorescently-labeled cDNA probes are generated. The microarrays

capable of hybridizing to marker cDNA are then probed with the labeled cDNA probes, the slides scanned and fluorescence intensity measured. This intensity correlates with the hybridization intensity and expression levels.

(241) Types of probes that can be used in the methods described herein include cDNA, riboprobes, synthetic oligonucleotides and genomic probes. The type of probe used will generally be dictated by the particular situation, such as riboprobes for in situ hybridization, and cDNA for Northern blotting, for example. In one embodiment, the probe is directed to nucleotide regions unique to the RNA. The probes may be as short as is required to differentially recognize marker mRNA transcripts, and may be as short as, for example, 15 bases; however, probes of at least 17, 18, 19 or 20 or more bases can be used. In one embodiment, the primers and probes hybridize specifically under stringent conditions to a DNA fragment having the nucleotide sequence corresponding to the marker. As herein used, the term “stringent conditions” means hybridization will occur only if there is at least 95% identity in nucleotide sequences. In another embodiment, hybridization under “stringent conditions” occurs when there is at least 97% identity between the sequences.

(242) The form of labeling of the probes may be any that is appropriate, such as the use of radioisotopes, for example, ^{32}P and ^{35}S . Labeling with radioisotopes may be achieved, whether the probe is synthesized chemically or biologically, by the use of suitably labeled bases.

(243) In one embodiment, the biological sample contains polypeptide molecules from the test subject. Alternatively, the biological sample can contain mRNA molecules from the test subject or genomic DNA molecules from the test subject.

(244) In another embodiment, the methods further involve obtaining a control biological sample from a control subject, contacting the control sample with a compound or agent capable of detecting marker polypeptide, mRNA, genomic DNA, or fragments thereof, such that the presence of the marker polypeptide, mRNA, genomic DNA, or fragments thereof, is detected in the biological sample, and comparing the presence of the marker polypeptide, mRNA, genomic DNA, or fragments thereof, in the control sample with the presence of the marker polypeptide, mRNA, genomic DNA, or fragments thereof in the test sample.

(245) c. Methods for Detection of Biomarker Protein Expression

(246) The activity or level of a biomarker protein can be detected and/or quantified by detecting or quantifying the expressed polypeptide. The polypeptide can be detected and quantified by any of a number of means well known to those of skill in the art. Aberrant levels of polypeptide expression of the polypeptides encoded by a biomarker nucleic acid and functionally similar homologs thereof, including a fragment or genetic alteration thereof (e.g., in regulatory or promoter regions thereof) are associated with the likelihood of response of a cancer to an immune checkpoint therapy. Any method known in the art for detecting polypeptides can be used. Such methods include, but are not limited to, immunodiffusion, immunoelectrophoresis, radioimmunoassay (RIA), enzyme-linked immunosorbent assays (ELISAs), immunofluorescent assays, Western blotting, binder-ligand assays, immunohistochemical techniques, agglutination, complement assays, high performance liquid chromatography (HPLC), thin layer chromatography (TLC), hyperdiffusion chromatography, and the like (e.g., Basic and Clinical Immunology, Sites and Terr, eds., Appleton and Lange, Norwalk, Conn. pp 217-262, 1991 which is incorporated by reference). Preferred are binder-ligand immunoassay methods including reacting antibodies with an epitope or epitopes and competitively displacing a labeled polypeptide or derivative thereof.

(247) For example, ELISA and RIA procedures may be conducted such that a desired biomarker protein standard is labeled (with a radioisotope such as ^{125}I or ^{35}S , or an assayable enzyme, such as horseradish peroxidase or alkaline phosphatase), and, together with the unlabelled sample, brought into contact with the corresponding antibody, whereon a second antibody is used to bind the first, and radioactivity or the immobilized enzyme assayed (competitive assay). Alternatively, the biomarker protein in the sample is allowed to react with the corresponding immobilized antibody, radioisotope- or enzyme-labeled anti-biomarker protein antibody is allowed

to react with the system, and radioactivity or the enzyme assayed (ELISA-sandwich assay). Other conventional methods may also be employed as suitable.

(248) The above techniques may be conducted essentially as a “one-step” or “two-step” assay. A “one-step” assay involves contacting antigen with immobilized antibody and, without washing, contacting the mixture with labeled antibody. A “two-step” assay involves washing before contacting the mixture with labeled antibody. Other conventional methods may also be employed as suitable.

(249) In one embodiment, a method for measuring biomarker protein levels comprises the steps of: contacting a biological specimen with an antibody or variant (e.g., fragment) thereof which selectively binds the biomarker protein, and detecting whether said antibody or variant thereof is bound to said sample and thereby measuring the levels of the biomarker protein.

(250) Enzymatic and radiolabeling of biomarker protein and/or the antibodies may be effected by conventional means. Such means will generally include covalent linking of the enzyme to the antigen or the antibody in question, such as by glutaraldehyde, specifically so as not to adversely affect the activity of the enzyme, by which is meant that the enzyme must still be capable of interacting with its substrate, although it is not necessary for all of the enzyme to be active, provided that enough remains active to permit the assay to be effected. Indeed, some techniques for binding enzyme are non-specific (such as using formaldehyde), and will only yield a proportion of active enzyme.

(251) It is usually desirable to immobilize one component of the assay system on a support, thereby allowing other components of the system to be brought into contact with the component and readily removed without laborious and time-consuming labor. It is possible for a second phase to be immobilized away from the first, but one phase is usually sufficient.

(252) It is possible to immobilize the enzyme itself on a support, but if solid-phase enzyme is required, then this is generally best achieved by binding to antibody and affixing the antibody to a support, models and systems for which are well-known in the art. Simple polyethylene may provide a suitable support.

(253) Enzymes employable for labeling are not particularly limited, but may be selected from the members of the oxidase group, for example. These catalyze production of hydrogen peroxide by reaction with their substrates, and glucose oxidase is often used for its good stability, ease of availability and cheapness, as well as the ready availability of its substrate (glucose). Activity of the oxidase may be assayed by measuring the concentration of hydrogen peroxide formed after reaction of the enzyme-labeled antibody with the substrate under controlled conditions well-known in the art.

(254) Other techniques may be used to detect biomarker protein according to a practitioner's preference based upon the present disclosure. One such technique is Western blotting (Towbin et al., *Proc. Nat. Acad. Sci.* 76:4350 (1979)), wherein a suitably treated sample is run on an SDS-PAGE gel before being transferred to a solid support, such as a nitrocellulose filter. Anti-biomarker protein antibodies (unlabeled) are then brought into contact with the support and assayed by a secondary immunological reagent, such as labeled protein A or anti-immunoglobulin (suitable labels including ^{125}I , horseradish peroxidase and alkaline phosphatase). Chromatographic detection may also be used.

(255) Immunohistochemistry may be used to detect expression of biomarker protein, e.g., in a biopsy sample. A suitable antibody is brought into contact with, for example, a thin layer of cells, washed, and then contacted with a second, labeled antibody. Labeling may be by fluorescent markers, enzymes, such as peroxidase, avidin, or radiolabelling. The assay is scored visually, using microscopy.

(256) Anti-biomarker protein antibodies, such as intrabodies, may also be used for imaging purposes, for example, to detect the presence of biomarker protein in cells and tissues of a subject. Suitable labels include radioisotopes, iodine (^{125}I , ^{121}I), carbon (^{14}C), sulphur

(.sup.35S), tritium (.sup.3H), indium (.sup.112In), and technetium (.sup.99mTc), fluorescent labels, such as fluorescein and rhodamine, and biotin.

(257) For in vivo imaging purposes, antibodies are not detectable, as such, from outside the body, and so must be labeled, or otherwise modified, to permit detection. Markers for this purpose may be any that do not substantially interfere with the antibody binding, but which allow external detection. Suitable markers may include those that may be detected by X-radiography, NMR or MRI. For X-radiographic techniques, suitable markers include any radioisotope that emits detectable radiation but that is not overtly harmful to the subject, such as barium or cesium, for example. Suitable markers for NMR and MRI generally include those with a detectable characteristic spin, such as deuterium, which may be incorporated into the antibody by suitable labeling of nutrients for the relevant hybridoma, for example.

(258) The size of the subject, and the imaging system used, will determine the quantity of imaging moiety needed to produce diagnostic images. In the case of a radioisotope moiety, for a human subject, the quantity of radioactivity injected will normally range from about 5 to 20 millicuries of technetium-99. The labeled antibody or antibody fragment will then preferentially accumulate at the location of cells which contain biomarker protein. The labeled antibody or antibody fragment can then be detected using known techniques.

(259) Antibodies that may be used to detect biomarker protein include any antibody, whether natural or synthetic, full length or a fragment thereof, monoclonal or polyclonal, that binds sufficiently strongly and specifically to the biomarker protein to be detected. An antibody may have a K.sub.d of at most about 10.sup.-6M, 10.sup.-7M, 10.sup.-8M, 10.sup.-9M, 10.sup.-10M, 10.sup.-11M, 10.sup.-12M. The phrase "specifically binds" refers to binding of, for example, an antibody to an epitope or antigen or antigenic determinant in such a manner that binding can be displaced or competed with a second preparation of identical or similar epitope, antigen or antigenic determinant. An antibody may bind preferentially to the biomarker protein relative to other proteins, such as related proteins.

(260) Antibodies are commercially available or may be prepared according to methods known in the art.

(261) Antibodies and derivatives thereof that may be used encompass polyclonal or monoclonal antibodies, chimeric, human, humanized, primatized (CDR-grafted), veneered or single-chain antibodies as well as functional fragments, i.e., biomarker protein binding fragments, of antibodies. For example, antibody fragments capable of binding to a biomarker protein or portions thereof, including, but not limited to, Fv, Fab, Fab' and F(ab')₂ fragments can be used. Such fragments can be produced by enzymatic cleavage or by recombinant techniques. For example, papain or pepsin cleavage can generate Fab or F(ab')₂ fragments, respectively. Other proteases with the requisite substrate specificity can also be used to generate Fab or F(ab')₂ fragments. Antibodies can also be produced in a variety of truncated forms using antibody genes in which one or more stop codons have been introduced upstream of the natural stop site. For example, a chimeric gene encoding a F(ab')₂ heavy chain portion can be designed to include DNA sequences encoding the CH₁ domain and hinge region of the heavy chain.

(262) Synthetic and engineered antibodies are described in, e.g., Cabilly et al., U.S. Pat. No. 4,816,567; Cabilly et al., European Patent No. 0,125,023 B1; Boss et al., U.S. Pat. No. 4,816,397; Boss et al., European Patent No. 0,120,694 B1; Neuberger, M. S. et al., WO 86/01533; Neuberger, M. S. et al., European Patent No. 0,194,276 B1; Winter, U.S. Pat. No. 5,225,539; Winter, European Patent No. 0,239,400 B1; Queen et al., European Patent No. 0,451,216 B1; and Padlan, E. A. et al., EP 0,519,596 A1. See also, Newman, R. et al., BioTechnology, 10: 1455-1460 (1992), regarding primatized antibody, and Ladner et al., U.S. Pat. No. 4,946,778 and Bird, R. E. et al., Science, 242: 423-426 (1988)) regarding single-chain antibodies. Antibodies produced from a library, e.g., phage display library, may also be used.

(263) In some embodiments, agents that specifically bind to a biomarker protein other than

antibodies are used, such as peptides. Peptides that specifically bind to a biomarker protein can be identified by any means known in the art. For example, specific peptide binders of a biomarker protein can be screened for using peptide phage display libraries.

(264) d. Methods for Detection of Biomarker Structural Alterations

(265) The following illustrative methods can be used to identify the presence of a structural alteration in a biomarker nucleic acid and/or biomarker polypeptide molecule in order to, for example, identify PBRM1 (or ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, EGFR, and the like) proteins that having mutations such as described herein.

(266) In certain embodiments, detection of the alteration involves the use of a probe/primer in a polymerase chain reaction (PCR) (see, e.g., U.S. Pat. Nos. 4,683,195 and 4,683,202), such as anchor PCR or RACE PCR, or, alternatively, in a ligation chain reaction (LCR) (see, e.g., Landegran et al. (1988) *Science* 241:1077-1080; and Nakazawa et al. (1994) *Proc. Natl. Acad. Sci. USA* 91:360-364), the latter of which can be particularly useful for detecting point mutations in a biomarker nucleic acid such as a biomarker gene (see Abravaya et al. (1995) *Nucleic Acids Res.* 23:675-682). This method can include the steps of collecting a sample of cells from a subject, isolating nucleic acid (e.g., genomic, mRNA or both) from the cells of the sample, contacting the nucleic acid sample with one or more primers which specifically hybridize to a biomarker gene under conditions such that hybridization and amplification of the biomarker gene (if present) occurs, and detecting the presence or absence of an amplification product, or detecting the size of the amplification product and comparing the length to a control sample. It is anticipated that PCR and/or LCR may be desirable to use as a preliminary amplification step in conjunction with any of the techniques used for detecting mutations described herein.

(267) Alternative amplification methods include: self sustained sequence replication (Guatelli, J. C. et al. (1990) *Proc. Natl. Acad. Sci. USA* 87:1874-1878), transcriptional amplification system (Kwoh, D. Y. et al. (1989) *Proc. Natl. Acad. Sci. USA* 86:1173-1177), Q-Beta Replicase (Lizardi, P. M. et al. (1988) *Bio-Technology* 6:1197), or any other nucleic acid amplification method, followed by the detection of the amplified molecules using techniques well known to those of skill in the art. These detection schemes are especially useful for the detection of nucleic acid molecules if such molecules are present in very low numbers.

(268) In an alternative embodiment, mutations in a biomarker nucleic acid from a sample cell can be identified by alterations in restriction enzyme cleavage patterns. For example, sample and control DNA is isolated, amplified (optionally), digested with one or more restriction endonucleases, and fragment length sizes are determined by gel electrophoresis and compared. Differences in fragment length sizes between sample and control DNA indicates mutations in the sample DNA. Moreover, the use of sequence specific ribozymes (see, for example, U.S. Pat. No. 5,498,531) can be used to score for the presence of specific mutations by development or loss of a ribozyme cleavage site.

(269) In other embodiments, genetic mutations in biomarker nucleic acid can be identified by hybridizing a sample and control nucleic acids, e.g., DNA or RNA, to high density arrays containing hundreds or thousands of oligonucleotide probes (Cronin, M. T. et al. (1996) *Hum. Mutat.* 7:244-255; Kozal, M. J. et al. (1996) *Nat. Med.* 2:753-759). For example, biomarker genetic mutations can be identified in two dimensional arrays containing light-generated DNA probes as described in Cronin et al. (1996) *supra*. Briefly, a first hybridization array of probes can be used to scan through long stretches of DNA in a sample and control to identify base changes between the sequences by making linear arrays of sequential, overlapping probes. This step allows the identification of point mutations. This step is followed by a second hybridization array that allows the characterization of specific mutations by using smaller, specialized probe arrays complementary to all variants or mutations detected. Each mutation array is composed of parallel probe sets, one complementary to the wild-type gene and the other complementary to the mutant gene. Such biomarker genetic mutations can be identified in a variety of contexts, including, for

example, germline and somatic mutations.

(270) In yet another embodiment, any of a variety of sequencing reactions known in the art can be used to directly sequence a biomarker gene and detect mutations by comparing the sequence of the sample biomarker with the corresponding wild-type (control) sequence. Examples of sequencing reactions include those based on techniques developed by Maxam and Gilbert (1977) *Proc. Natl. Acad. Sci. USA* 74:560 or Sanger (1977) *Proc. Natl. Acad. Sci. USA* 74:5463. It is also contemplated that any of a variety of automated sequencing procedures can be utilized when performing the diagnostic assays (Naeve (1995) *Biotechniques* 19:448-53), including sequencing by mass spectrometry (see, e.g., PCT International Publication No. WO 94/16101; Cohen et al. (1996) *Adv. Chromatogr.* 36:127-162; and Griffin et al. (1993) *Appl. Biochem. Biotechnol.* 38:147-159).

(271) Other methods for detecting mutations in a biomarker gene include methods in which protection from cleavage agents is used to detect mismatched bases in RNA/RNA or RNA/DNA heteroduplexes (Myers et al. (1985) *Science* 230:1242). In general, the art technique of “mismatch cleavage” starts by providing heteroduplexes formed by hybridizing (labeled) RNA or DNA containing the wild-type biomarker sequence with potentially mutant RNA or DNA obtained from a tissue sample. The double-stranded duplexes are treated with an agent which cleaves single-stranded regions of the duplex such as which will exist due to base pair mismatches between the control and sample strands. For instance, RNA/DNA duplexes can be treated with RNase and DNA/DNA hybrids treated with SI nuclease to enzymatically digest the mismatched regions. In other embodiments, either DNA/DNA or RNA/DNA duplexes can be treated with hydroxylamine or osmium tetroxide and with piperidine in order to digest mismatched regions. After digestion of the mismatched regions, the resulting material is then separated by size on denaturing polyacrylamide gels to determine the site of mutation. See, for example, Cotton et al. (1988) *Proc. Natl. Acad. Sci. USA* 85:4397 and Saleeba et al. (1992) *Methods Enzymol.* 217:286-295. In a preferred embodiment, the control DNA or RNA can be labeled for detection.

(272) In still another embodiment, the mismatch cleavage reaction employs one or more proteins that recognize mismatched base pairs in double-stranded DNA (so called “DNA mismatch repair” enzymes) in defined systems for detecting and mapping point mutations in biomarker cDNAs obtained from samples of cells. For example, the mutY enzyme of *E. coli* cleaves A at G/A mismatches and the thymidine DNA glycosylase from HeLa cells cleaves T at G/T mismatches (Hsu et al. (1994) *Carcinogenesis* 15:1657-1662). According to an exemplary embodiment, a probe based on a biomarker sequence, e.g., a wild-type biomarker treated with a DNA mismatch repair enzyme, and the cleavage products, if any, can be detected from electrophoresis protocols or the like (e.g., U.S. Pat. No. 5,459,039.)

(273) In other embodiments, alterations in electrophoretic mobility can be used to identify mutations in biomarker genes. For example, single strand conformation polymorphism (SSCP) may be used to detect differences in electrophoretic mobility between mutant and wild type nucleic acids (Orita et al. (1989) *Proc Natl. Acad. Sci USA* 86:2766; see also Cotton (1993) *Mutat. Res.* 285:125-144 and Hayashi (1992) *Genet. Anal. Tech. Appl.* 9:73-79). Single-stranded DNA fragments of sample and control biomarker nucleic acids will be denatured and allowed to renature. The secondary structure of single-stranded nucleic acids varies according to sequence, the resulting alteration in electrophoretic mobility enables the detection of even a single base change. The DNA fragments may be labeled or detected with labeled probes. The sensitivity of the assay may be enhanced by using RNA (rather than DNA), in which the secondary structure is more sensitive to a change in sequence. In a preferred embodiment, the subject method utilizes heteroduplex analysis to separate double stranded heteroduplex molecules on the basis of changes in electrophoretic mobility (Keen et al. (1991) *Trends Genet.* 7:5).

(274) In yet another embodiment the movement of mutant or wild-type fragments in polyacrylamide gels containing a gradient of denaturant is assayed using denaturing gradient gel electrophoresis (DGGE) (Myers et al. (1985) *Nature* 313:495). When DGGE is used as the method

of analysis, DNA will be modified to ensure that it does not completely denature, for example by adding a GC clamp of approximately 40 bp of high-melting GC-rich DNA by PCR. In a further embodiment, a temperature gradient is used in place of a denaturing gradient to identify differences in the mobility of control and sample DNA (Rosenbaum and Reissner (1987) *Biophys. Chem.* 265:12753).

(275) Examples of other techniques for detecting point mutations include, but are not limited to, selective oligonucleotide hybridization, selective amplification, or selective primer extension. For example, oligonucleotide primers may be prepared in which the known mutation is placed centrally and then hybridized to target DNA under conditions which permit hybridization only if a perfect match is found (Saiki et al. (1986) *Nature* 324:163; Saiki et al. (1989) *Proc. Natl. Acad. Sci. USA* 86:6230). Such allele specific oligonucleotides are hybridized to PCR amplified target DNA or a number of different mutations when the oligonucleotides are attached to the hybridizing membrane and hybridized with labeled target DNA.

(276) Alternatively, allele specific amplification technology which depends on selective PCR amplification may be used in conjunction with the instant invention. Oligonucleotides used as primers for specific amplification may carry the mutation of interest in the center of the molecule (so that amplification depends on differential hybridization) (Gibbs et al. (1989) *Nucleic Acids Res.* 17:2437-2448) or at the extreme 3' end of one primer where, under appropriate conditions, mismatch can prevent, or reduce polymerase extension (Prossner (1993) *Tibtech* 11:238). In addition it may be desirable to introduce a novel restriction site in the region of the mutation to create cleavage-based detection (Gasparini et al. (1992) *Mol. Cell Probes* 6:1). It is anticipated that in certain embodiments amplification may also be performed using Taq ligase for amplification (Barany (1991) *Proc. Natl. Acad. Sci. USA* 88:189). In such cases, ligation will occur only if there is a perfect match at the 3' end of the 5' sequence making it possible to detect the presence of a known mutation at a specific site by looking for the presence or absence of amplification.

(277) 3. Anti-Cancer Therapies

(278) The efficacy of immune checkpoint therapy is predicted according to biomarker amount and/or activity associated with a cancer in a subject according to the methods described herein. In one embodiment, such immune checkpoint therapy or combinations of therapies (e.g., anti-PD-1 antibodies) can be administered once a subject is indicated as being a likely responder to immune checkpoint therapy. In another embodiment, such immune checkpoint therapy can be avoided once a subject is indicated as not being a likely responder to immune checkpoint therapy and an alternative treatment regimen, such as targeted and/or untargeted anti-cancer therapies can be administered. Combination therapies are also contemplated and can comprise, for example, one or more chemotherapeutic agents and radiation, one or more chemotherapeutic agents and immunotherapy, or one or more chemotherapeutic agents, radiation and chemotherapy, each combination of which can be with immune checkpoint therapy.

(279) The term “targeted therapy” refers to administration of agents that selectively interact with a chosen biomolecule to thereby treat cancer. For example, anti-PBRM1 agents (or anti-ARID2 agents, anti-BRD7 agents, anti-PHF10 agents, anti-KDM6A agents, etc.), such as therapeutic monoclonal blocking antibodies, which are well-known in the art and described above, can be used to target tumor microenvironments and cells expressing unwanted PBRM1 (or ARID2, BRD7, PHF10, KDM6A, ARID1A, ARID1B, BRG1, BRM, CRB1, EGFR, and the like). Similarly, nivolumab (Opdivo®) is a human IgG4 anti-PD-1 monoclonal antibody that blocks PD-1 activity (see, for example, Wang et al. (2014) *Cancer Immunol. Res.* 2:846-856; Johnson et al. (2015) *Ther. Adv. Med. Oncol.* 7:97-106; and Sundar et al. (2015) *Ther. Adv. Med. Oncol.* 7:85-96).

(280) Immunotherapy is one form of targeted therapy that may comprise, for example, the use of cancer vaccines and/or sensitized antigen presenting cells. For example, an oncolytic virus is a virus that is able to infect and lyse cancer cells, while leaving normal cells unharmed, making them potentially useful in cancer therapy. Replication of oncolytic viruses both facilitates tumor cell

destruction and also produces dose amplification at the tumor site. They may also act as vectors for anticancer genes, allowing them to be specifically delivered to the tumor site. The immunotherapy can involve passive immunity for short-term protection of a host, achieved by the administration of pre-formed antibody directed against a cancer antigen or disease antigen (e.g., administration of a monoclonal antibody, optionally linked to a chemotherapeutic agent or toxin, to a tumor antigen). Immunotherapy can also focus on using the cytotoxic lymphocyte-recognized epitopes of cancer cell lines. Alternatively, antisense polynucleotides, ribozymes, RNA interference molecules, triple helix polynucleotides and the like, can be used to selectively modulate biomolecules that are linked to the initiation, progression, and/or pathology of a tumor or cancer.

(281) The term “untargeted therapy” refers to administration of agents that do not selectively interact with a chosen biomolecule yet treat cancer. Representative examples of untargeted therapies include, without limitation, chemotherapy, gene therapy, and radiation therapy.

(282) In one embodiment, chemotherapy is used. Chemotherapy includes the administration of a chemotherapeutic agent. Such a chemotherapeutic agent may be, but is not limited to, those selected from among the following groups of compounds: platinum compounds, cytotoxic antibiotics, antimetabolites, anti-mitotic agents, alkylating agents, arsenic compounds, DNA topoisomerase inhibitors, taxanes, nucleoside analogues, plant alkaloids, and toxins; and synthetic derivatives thereof. Exemplary compounds include, but are not limited to, alkylating agents: cisplatin, treosulfan, and trofosfamide; plant alkaloids: vinblastine, paclitaxel, docetaxol; DNA topoisomerase inhibitors: teniposide, crisnatol, and mitomycin; anti-folates: methotrexate, mycophenolic acid, and hydroxyurea; pyrimidine analogs: 5-fluorouracil, doxifluridine, and cytosine arabinoside; purine analogs: mercaptopurine and thioguanine; DNA antimetabolites: 2'-deoxy-5-fluorouridine, aphidicolin glycinate, and pyrazoloimidazole; and antimitotic agents: halichondrin, colchicine, and rhizoxin. Compositions comprising one or more chemotherapeutic agents (e.g., FLAG, CHOP) may also be used. FLAG comprises fludarabine, cytosine arabinoside (Ara-C) and G-CSF. CHOP comprises cyclophosphamide, vincristine, doxorubicin, and prednisone. In another embodiment, PARP (e.g., PARP-1 and/or PARP-2) inhibitors are used and such inhibitors are well known in the art (e.g., Olaparib, ABT-888, BSI-201, BGP-15 (N-Gene Research Laboratories, Inc.); INO-1001 (Inotek Pharmaceuticals Inc.); PJ34 (Soriano et al., 2001; Pacher et al., 2002b); 3-aminobenzamide (Trevigen); 4-amino-1,8-naphthalimide; (Trevigen); 6(5H)-phenanthridinone (Trevigen); benzamide (U.S. Pat. Re. 36,397); and NU1025 (Bowman et al.)). The mechanism of action is generally related to the ability of PARP inhibitors to bind PARP and decrease its activity. PARP catalyzes the conversion of .beta.-nicotinamide adenine dinucleotide (NAD⁺) into nicotinamide and poly-ADP-ribose (PAR). Both poly (ADP-ribose) and PARP have been linked to regulation of transcription, cell proliferation, genomic stability, and carcinogenesis (Bouchard V. J. et. al. *Experimental Hematology*, Volume 31, Number 6, June 2003, pp. 446-454(9); Herceg Z.; Wang Z.-Q. *Mutation Research/Fundamental and Molecular Mechanisms of Mutagenesis*, Volume 477, Number 1, 2 Jun. 2001, pp. 97-110(14)). Poly(ADP-ribose) polymerase 1 (PARP1) is a key molecule in the repair of DNA single-strand breaks (SSBs) (de Murcia J. et al. 1997. *Proc Natl Acad Sci USA* 94:7303-7307; Schreiber V, Dantzer F, Ame J C, de Murcia G (2006) *Nat Rev Mol Cell Biol* 7:517-528; Wang Z Q, et al. (1997) *Genes Dev* 11:2347-2358). Knockout of SSB repair by inhibition of PARP1 function induces DNA double-strand breaks (DSBs) that can trigger synthetic lethality in cancer cells with defective homology-directed DSB repair (Bryant H E, et al. (2005) *Nature* 434:913-917; Farmer H, et al. (2005) *Nature* 434:917-921). The foregoing examples of chemotherapeutic agents are illustrative, and are not intended to be limiting.

(283) In another embodiment, radiation therapy is used. The radiation used in radiation therapy can be ionizing radiation. Radiation therapy can also be gamma rays, X-rays, or proton beams. Examples of radiation therapy include, but are not limited to, external-beam radiation therapy, interstitial implantation of radioisotopes (1-125, palladium, iridium), radioisotopes such as

strontium-89, thoracic radiation therapy, intraperitoneal P-32 radiation therapy, and/or total abdominal and pelvic radiation therapy. For a general overview of radiation therapy, see Hellman, Chapter 16: Principles of Cancer Management: Radiation Therapy, 6th edition, 2001, DeVita et al., eds., J. B. Lippencott Company, Philadelphia. The radiation therapy can be administered as external beam radiation or teletherapy wherein the radiation is directed from a remote source. The radiation treatment can also be administered as internal therapy or brachytherapy wherein a radioactive source is placed inside the body close to cancer cells or a tumor mass. Also encompassed is the use of photodynamic therapy comprising the administration of photosensitizers, such as hematoporphyrin and its derivatives, Vertoporphin (BPD-MA), phthalocyanine, photosensitizer Pc4, demethoxy-hypocrellin A; and 2BA-2-DMHA.

(284) In another embodiment, hormone therapy is used. Hormonal therapeutic treatments can comprise, for example, hormonal agonists, hormonal antagonists (e.g., flutamide, bicalutamide, tamoxifen, raloxifene, leuprolide acetate (LUPRON), LH-RH antagonists), inhibitors of hormone biosynthesis and processing, and steroids (e.g., dexamethasone, retinoids, deltoids, betamethasone, cortisol, cortisone, prednisone, dehydrotestosterone, glucocorticoids, mineralocorticoids, estrogen, testosterone, progestins), vitamin A derivatives (e.g., all-trans retinoic acid (ATRA)); vitamin D3 analogs; antigestagens (e.g., mifepristone, onapristone), or antiandrogens (e.g., cyproterone acetate).

(285) In another embodiment, hyperthermia, a procedure in which body tissue is exposed to high temperatures (up to 106° F.) is used. Heat may help shrink tumors by damaging cells or depriving them of substances they need to live. Hyperthermia therapy can be local, regional, and whole-body hyperthermia, using external and internal heating devices. Hyperthermia is almost always used with other forms of therapy (e.g., radiation therapy, chemotherapy, and biological therapy) to try to increase their effectiveness. Local hyperthermia refers to heat that is applied to a very small area, such as a tumor. The area may be heated externally with high-frequency waves aimed at a tumor from a device outside the body. To achieve internal heating, one of several types of sterile probes may be used, including thin, heated wires or hollow tubes filled with warm water; implanted microwave antennae; and radiofrequency electrodes. In regional hyperthermia, an organ or a limb is heated. Magnets and devices that produce high energy are placed over the region to be heated. In another approach, called perfusion, some of the patient's blood is removed, heated, and then pumped (perfused) into the region that is to be heated internally. Whole-body heating is used to treat metastatic cancer that has spread throughout the body. It can be accomplished using warm-water blankets, hot wax, inductive coils (like those in electric blankets), or thermal chambers (similar to large incubators). Hyperthermia does not cause any marked increase in radiation side effects or complications. Heat applied directly to the skin, however, can cause discomfort or even significant local pain in about half the patients treated. It can also cause blisters, which generally heal rapidly.

(286) In still another embodiment, photodynamic therapy (also called PDT, photoradiation therapy, phototherapy, or photochemotherapy) is used for the treatment of some types of cancer. It is based on the discovery that certain chemicals known as photosensitizing agents can kill one-celled organisms when the organisms are exposed to a particular type of light. PDT destroys cancer cells through the use of a fixed-frequency laser light in combination with a photosensitizing agent. In PDT, the photosensitizing agent is injected into the bloodstream and absorbed by cells all over the body. The agent remains in cancer cells for a longer time than it does in normal cells. When the treated cancer cells are exposed to laser light, the photosensitizing agent absorbs the light and produces an active form of oxygen that destroys the treated cancer cells. Light exposure must be timed carefully so that it occurs when most of the photosensitizing agent has left healthy cells but is still present in the cancer cells. The laser light used in PDT can be directed through a fiber-optic (a very thin glass strand). The fiber-optic is placed close to the cancer to deliver the proper amount of light. The fiber-optic can be directed through a bronchoscope into the lungs for the treatment of

lung cancer or through an endoscope into the esophagus for the treatment of esophageal cancer. An advantage of PDT is that it causes minimal damage to healthy tissue. However, because the laser light currently in use cannot pass through more than about 3 centimeters of tissue (a little more than one and an eighth inch), PDT is mainly used to treat tumors on or just under the skin or on the lining of internal organs. Photodynamic therapy makes the skin and eyes sensitive to light for 6 weeks or more after treatment. Patients are advised to avoid direct sunlight and bright indoor light for at least 6 weeks. If patients must go outdoors, they need to wear protective clothing, including sunglasses. Other temporary side effects of PDT are related to the treatment of specific areas and can include coughing, trouble swallowing, abdominal pain, and painful breathing or shortness of breath. In December 1995, the U.S. Food and Drug Administration (FDA) approved a photosensitizing agent called porfimer sodium, or Photofrin®, to relieve symptoms of esophageal cancer that is causing an obstruction and for esophageal cancer that cannot be satisfactorily treated with lasers alone. In January 1998, the FDA approved porfimer sodium for the treatment of early nonsmall cell lung cancer in patients for whom the usual treatments for lung cancer are not appropriate. The National Cancer Institute and other institutions are supporting clinical trials (research studies) to evaluate the use of photodynamic therapy for several types of cancer, including cancers of the bladder, brain, larynx, and oral cavity.

(287) In yet another embodiment, laser therapy is used to harness high-intensity light to destroy cancer cells. This technique is often used to relieve symptoms of cancer such as bleeding or obstruction, especially when the cancer cannot be cured by other treatments. It may also be used to treat cancer by shrinking or destroying tumors. The term “laser” stands for light amplification by stimulated emission of radiation. Ordinary light, such as that from a light bulb, has many wavelengths and spreads in all directions. Laser light, on the other hand, has a specific wavelength and is focused in a narrow beam. This type of high-intensity light contains a lot of energy. Lasers are very powerful and may be used to cut through steel or to shape diamonds. Lasers also can be used for very precise surgical work, such as repairing a damaged retina in the eye or cutting through tissue (in place of a scalpel). Although there are several different kinds of lasers, only three kinds have gained wide use in medicine: Carbon dioxide (CO.sub.2) laser—This type of laser can remove thin layers from the skin's surface without penetrating the deeper layers. This technique is particularly useful in treating tumors that have not spread deep into the skin and certain precancerous conditions. As an alternative to traditional scalpel surgery, the CO.sub.2 laser is also able to cut the skin. The laser is used in this way to remove skin cancers. Neodymium:yttrium-aluminum-garnet (Nd:YAG) laser—Light from this laser can penetrate deeper into tissue than light from the other types of lasers, and it can cause blood to clot quickly. It can be carried through optical fibers to less accessible parts of the body. This type of laser is sometimes used to treat throat cancers. Argon laser—This laser can pass through only superficial layers of tissue and is therefore useful in dermatology and in eye surgery. It also is used with light-sensitive dyes to treat tumors in a procedure known as photodynamic therapy (PDT). Lasers have several advantages over standard surgical tools, including: Lasers are more precise than scalpels. Tissue near an incision is protected, since there is little contact with surrounding skin or other tissue. The heat produced by lasers sterilizes the surgery site, thus reducing the risk of infection. Less operating time may be needed because the precision of the laser allows for a smaller incision. Healing time is often shortened; since laser heat seals blood vessels, there is less bleeding, swelling, or scarring. Laser surgery may be less complicated. For example, with fiber optics, laser light can be directed to parts of the body without making a large incision. More procedures may be done on an outpatient basis. Lasers can be used in two ways to treat cancer: by shrinking or destroying a tumor with heat, or by activating a chemical—known as a photosensitizing agent—that destroys cancer cells. In PDT, a photosensitizing agent is retained in cancer cells and can be stimulated by light to cause a reaction that kills cancer cells. CO.sub.2 and Nd:YAG lasers are used to shrink or destroy tumors. They may be used with endoscopes, tubes that allow physicians to see into certain areas of the body, such as

the bladder. The light from some lasers can be transmitted through a flexible endoscope fitted with fiber optics. This allows physicians to see and work in parts of the body that could not otherwise be reached except by surgery and therefore allows very precise aiming of the laser beam. Lasers also may be used with low-power microscopes, giving the doctor a clear view of the site being treated. Used with other instruments, laser systems can produce a cutting area as small as 200 microns in diameter—less than the width of a very fine thread. Lasers are used to treat many types of cancer. Laser surgery is a standard treatment for certain stages of glottis (vocal cord), cervical, skin, lung, vaginal, vulvar, and penile cancers. In addition to its use to destroy the cancer, laser surgery is also used to help relieve symptoms caused by cancer (palliative care). For example, lasers may be used to shrink or destroy a tumor that is blocking a patient's trachea (windpipe), making it easier to breathe. It is also sometimes used for palliation in colorectal and anal cancer. Laser-induced interstitial thermotherapy (LITT) is one of the most recent developments in laser therapy. LITT uses the same idea as a cancer treatment called hyperthermia; that heat may help shrink tumors by damaging cells or depriving them of substances they need to live. In this treatment, lasers are directed to interstitial areas (areas between organs) in the body. The laser light then raises the temperature of the tumor, which damages or destroys cancer cells.

(288) The duration and/or dose of treatment with anti-immune checkpoint therapies may vary according to the particular anti-immune checkpoint agent or combination thereof. An appropriate treatment time for a particular cancer therapeutic agent will be appreciated by the skilled artisan. The present invention contemplates the continued assessment of optimal treatment schedules for each cancer therapeutic agent, where the phenotype of the cancer of the subject as determined by the methods of the present invention is a factor in determining optimal treatment doses and schedules.

(289) Any means for the introduction of a polynucleotide into mammals, human or non-human, or cells thereof may be adapted to the practice of this invention for the delivery of the various constructs of the present invention into the intended recipient. In one embodiment of the present invention, the DNA constructs are delivered to cells by transfection, i.e., by delivery of “naked” DNA or in a complex with a colloidal dispersion system. A colloidal system includes macromolecule complexes, nanocapsules, microspheres, beads, and lipid-based systems including oil-in-water emulsions, micelles, mixed micelles, and liposomes. The preferred colloidal system of this invention is a lipid-complexed or liposome-formulated DNA. In the former approach, prior to formulation of DNA, e.g., with lipid, a plasmid containing a transgene bearing the desired DNA constructs may first be experimentally optimized for expression (e.g., inclusion of an intron in the 5' untranslated region and elimination of unnecessary sequences (Felgner, et al., *Ann NY Acad Sci* 126-139, 1995). Formulation of DNA, e.g. with various lipid or liposome materials, may then be effected using known methods and materials and delivered to the recipient mammal. See, e.g., Canonico et al, *Am J Respir Cell Mol Biol* 10:24-29, 1994; Tsan et al, *Am J Physiol* 268; Alton et al., *Nat Genet.* 5:135-142, 1993 and U.S. Pat. No. 5,679,647 by Carson et al.

(290) The targeting of liposomes can be classified based on anatomical and mechanistic factors. Anatomical classification is based on the level of selectivity, for example, organ-specific, cell-specific, and organelle-specific. Mechanistic targeting can be distinguished based upon whether it is passive or active. Passive targeting utilizes the natural tendency of liposomes to distribute to cells of the reticulo-endothelial system (RES) in organs, which contain sinusoidal capillaries. Active targeting, on the other hand, involves alteration of the liposome by coupling the liposome to a specific ligand such as a monoclonal antibody, sugar, glycolipid, or protein, or by changing the composition or size of the liposome in order to achieve targeting to organs and cell types other than the naturally occurring sites of localization.

(291) The surface of the targeted delivery system may be modified in a variety of ways. In the case of a liposomal targeted delivery system, lipid groups can be incorporated into the lipid bilayer of the liposome in order to maintain the targeting ligand in stable association with the liposomal

bilayer. Various linking groups can be used for joining the lipid chains to the targeting ligand. Naked DNA or DNA associated with a delivery vehicle, e.g., liposomes, can be administered to several sites in a subject (see below).

(292) Nucleic acids can be delivered in any desired vector. These include viral or non-viral vectors, including adenovirus vectors, adeno-associated virus vectors, retrovirus vectors, lentivirus vectors, and plasmid vectors. Exemplary types of viruses include HSV (herpes simplex virus), AAV (adeno associated virus), HIV (human immunodeficiency virus), BIV (bovine immunodeficiency virus), and MLV (murine leukemia virus). Nucleic acids can be administered in any desired format that provides sufficiently efficient delivery levels, including in virus particles, in liposomes, in nanoparticles, and complexed to polymers.

(293) The nucleic acids encoding a protein or nucleic acid of interest may be in a plasmid or viral vector, or other vector as is known in the art. Such vectors are well known and any can be selected for a particular application. In one embodiment of the present invention, the gene delivery vehicle comprises a promoter and a demethylase coding sequence. Preferred promoters are tissue-specific promoters and promoters which are activated by cellular proliferation, such as the thymidine kinase and thymidylate synthase promoters. Other preferred promoters include promoters which are activatable by infection with a virus, such as the α - and β -interferon promoters, and promoters which are activatable by a hormone, such as estrogen. Other promoters which can be used include the Moloney virus LTR, the CMV promoter, and the mouse albumin promoter. A promoter may be constitutive or inducible.

(294) In another embodiment, naked polynucleotide molecules are used as gene delivery vehicles, as described in WO 90/11092 and U.S. Pat. No. 5,580,859. Such gene delivery vehicles can be either growth factor DNA or RNA and, in certain embodiments, are linked to killed adenovirus. Curiel et al., Hum. Gene. Ther. 3:147-154, 1992. Other vehicles which can optionally be used include DNA-ligand (Wu et al., J. Biol. Chem. 264:16985-16987, 1989), lipid-DNA combinations (Felgner et al., Proc. Natl. Acad. Sci. USA 84:7413-7417, 1989), liposomes (Wang et al., Proc. Natl. Acad. Sci. 84:7851-7855, 1987) and microprojectiles (Williams et al., Proc. Natl. Acad. Sci. 88:2726-2730, 1991).

(295) A gene delivery vehicle can optionally comprise viral sequences such as a viral origin of replication or packaging signal. These viral sequences can be selected from viruses such as astrovirus, coronavirus, orthomyxovirus, papovavirus, paramyxovirus, parvovirus, picornavirus, poxvirus, retrovirus, togavirus or adenovirus. In a preferred embodiment, the growth factor gene delivery vehicle is a recombinant retroviral vector. Recombinant retroviruses and various uses thereof have been described in numerous references including, for example, Mann et al., Cell 33:153, 1983, Cane and Mulligan, Proc. Nat'l. Acad. Sci. USA 81:6349, 1984, Miller et al., Human Gene Therapy 1:5-14, 1990, U.S. Pat. Nos. 4,405,712, 4,861,719, and 4,980,289, and PCT Application Nos. WO 89/02,468, WO 89/05,349, and WO 90/02,806. Numerous retroviral gene delivery vehicles can be utilized in the present invention, including for example those described in EP 0,415,731; WO 90/07936; WO 94/03622; WO 93/25698; WO 93/25234; U.S. Pat. No. 5,219,740; WO 9311230; WO 9310218; Vile and Hart, Cancer Res. 53:3860-3864, 1993; Vile and Hart, Cancer Res. 53:962-967, 1993; Ram et al., Cancer Res. 53:83-88, 1993; Takamiya et al., J. Neurosci. Res. 33:493-503, 1992; Baba et al., J. Neurosurg. 79:729-735, 1993 (U.S. Pat. No. 4,777,127, GB 2,200,651, EP 0,345,242 and WO91/02805).

(296) Other viral vector systems that can be used to deliver a polynucleotide of the present invention have been derived from herpes virus, e.g., Herpes Simplex Virus (U.S. Pat. No. 5,631,236 by Woo et al., issued May 20, 1997 and WO 00/08191 by Neurovex), vaccinia virus (Ridgeway (1988) Ridgeway, "Mammalian expression vectors," In: Rodriguez R L, Denhardt D T, ed. Vectors: A survey of molecular cloning vectors and their uses. Stoneham: Butterworth; Baichwal and Sugden (1986) "Vectors for gene transfer derived from animal DNA viruses: Transient and stable expression of transferred genes," In: Kucherlapati R, ed. Gene transfer. New

York: Plenum Press; Coupar et al. (1988) *Gene*, 68:1-10), and several RNA viruses. Preferred viruses include an alphavirus, a poxvirus, an arena virus, a vaccinia virus, a polio virus, and the like. They offer several attractive features for various mammalian cells (Friedmann (1989) *Science*, 244:1275-1281; Ridgeway, 1988, *supra*; Baichwal and Sugden, 1986, *supra*; Coupar et al., 1988; Horwich et al. (1990) *J. Virol.*, 64:642-650).

(297) In other embodiments, target DNA in the genome can be manipulated using well-known methods in the art. For example, the target DNA in the genome can be manipulated by deletion, insertion, and/or mutation are retroviral insertion, artificial chromosome techniques, gene insertion, random insertion with tissue specific promoters, gene targeting, transposable elements and/or any other method for introducing foreign DNA or producing modified DNA/modified nuclear DNA. Other modification techniques include deleting DNA sequences from a genome and/or altering nuclear DNA sequences. Nuclear DNA sequences, for example, may be altered by site-directed mutagenesis.

(298) In other embodiments, recombinant biomarker polypeptides, and fragments thereof, can be administered to subjects. In some embodiments, fusion proteins can be constructed and administered which have enhanced biological properties. In addition, the biomarker polypeptides, and fragment thereof, can be modified according to well-known pharmacological methods in the art (e.g., pegylation, glycosylation, oligomerization, etc.) in order to further enhance desirable biological activities, such as increased bioavailability and decreased proteolytic degradation.

(299) 4. Clinical Efficacy

(300) Clinical efficacy can be measured by any method known in the art. For example, the response to a therapy, such as anti-immune checkpoint therapies, relates to any response of the cancer, e.g., a tumor, to the therapy, preferably to a change in tumor mass and/or volume after initiation of neoadjuvant or adjuvant chemotherapy. Tumor response may be assessed in a neoadjuvant or adjuvant situation where the size of a tumor after systemic intervention can be compared to the initial size and dimensions as measured by CT, PET, mammogram, ultrasound or palpation and the cellularity of a tumor can be estimated histologically and compared to the cellularity of a tumor biopsy taken before initiation of treatment. Response may also be assessed by caliper measurement or pathological examination of the tumor after biopsy or surgical resection. Response may be recorded in a quantitative fashion like percentage change in tumor volume or cellularity or using a semi-quantitative scoring system such as residual cancer burden (Symmans et al., *J. Clin. Oncol.* (2007) 25:4414-4422) or Miller-Payne score (Ogston et al., (2003) *Breast* (Edinburgh, Scotland) 12:320-327) in a qualitative fashion like “pathological complete response” (pCR), “clinical complete remission” (cCR), “clinical partial remission” (cPR), “clinical stable disease” (cSD), “clinical progressive disease” (cPD) or other qualitative criteria. Assessment of tumor response may be performed early after the onset of neoadjuvant or adjuvant therapy, e.g., after a few hours, days, weeks or preferably after a few months. A typical endpoint for response assessment is upon termination of neoadjuvant chemotherapy or upon surgical removal of residual tumor cells and/or the tumor bed.

(301) In some embodiments, clinical efficacy of the therapeutic treatments described herein may be determined by measuring the clinical benefit rate (CBR). The clinical benefit rate is measured by determining the sum of the percentage of patients who are in complete remission (CR), the number of patients who are in partial remission (PR) and the number of patients having stable disease (SD) at a time point at least 6 months out from the end of therapy. The shorthand for this formula is $CBR = CR + PR + SD$ over 6 months. In some embodiments, the CBR for a particular anti-immune checkpoint therapeutic regimen is at least 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, or more.

(302) Additional criteria for evaluating the response to anti-immune checkpoint therapies are related to “survival,” which includes all of the following: survival until mortality, also known as overall survival (wherein said mortality may be either irrespective of cause or tumor related);

“recurrence-free survival” (wherein the term recurrence shall include both localized and distant recurrence); metastasis free survival; disease free survival (wherein the term disease shall include cancer and diseases associated therewith). The length of said survival may be calculated by reference to a defined start point (e.g., time of diagnosis or start of treatment) and end point (e.g., death, recurrence or metastasis). In addition, criteria for efficacy of treatment can be expanded to include response to chemotherapy, probability of survival, probability of metastasis within a given time period, and probability of tumor recurrence.

(303) For example, in order to determine appropriate threshold values, a particular anti-immune checkpoint therapeutic regimen can be administered to a population of subjects and the outcome can be correlated to biomarker measurements that were determined prior to administration of any immune checkpoint therapy. The outcome measurement may be pathologic response to therapy given in the neoadjuvant setting. Alternatively, outcome measures, such as overall survival and disease-free survival can be monitored over a period of time for subjects following immune checkpoint therapy for whom biomarker measurement values are known. In certain embodiments, the same doses of anti-immune checkpoint agents are administered to each subject. In related embodiments, the doses administered are standard doses known in the art for anti-immune checkpoint agents. The period of time for which subjects are monitored can vary. For example, subjects may be monitored for at least 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 25, 30, 35, 40, 45, 50, 55, or 60 months. Biomarker measurement threshold values that correlate to outcome of an immune checkpoint therapy can be determined using methods such as those described in the Examples section.

(304) 5. Further Uses and Methods of the Present Invention

(305) The methods described herein can be used in a variety of diagnostic, prognostic, and therapeutic applications. In any method described herein, such as a diagnostic method, prognostic method, therapeutic method, or combination thereof, all steps of the method can be performed by a single actor or, alternatively, by more than one actor. For example, diagnosis can be performed directly by the actor providing therapeutic treatment. Alternatively, a person providing a therapeutic agent can request that a diagnostic assay be performed. The diagnostician and/or the therapeutic interventionist can interpret the diagnostic assay results to determine a therapeutic strategy. Similarly, such alternative processes can apply to other assays, such as prognostic assays. The compositions described herein can also be used in a variety of diagnostic, prognostic, and therapeutic applications regarding biomarkers described herein, such as those listed in Table 1. Moreover, any method of diagnosis, prognosis, prevention, and the like described herein can be applied to a therapy or test agent of interest, such as immune checkpoint therapies, EGFR therapies, anti-cancer therapies, and the like.

(306) a. Screening Methods

(307) One aspect of the present invention relates to screening assays, including non-cell based assays. In one embodiment, the assays provide a method for identifying whether a cancer is likely to respond to immune checkpoint therapy and/or whether an agent can inhibit the growth of or kill a cancer cell that is unlikely to respond to immune checkpoint therapy.

(308) In one embodiment, the present invention relates to assays for screening test agents which bind to, or modulate the biological activity of, at least one biomarker listed in Table 1. In one embodiment, a method for identifying such an agent entails determining the ability of the agent to modulate, e.g. inhibit, the at least one biomarker listed in Table 1.

(309) In one embodiment, an assay is a cell-free or cell-based assay, comprising contacting at least one biomarker listed in Table 1, with a test agent, and determining the ability of the test agent to modulate (e.g. inhibit) the enzymatic activity of the biomarker, such as by measuring direct binding of substrates or by measuring indirect parameters as described below.

(310) For example, in a direct binding assay, biomarker protein (or their respective target polypeptides or molecules) can be coupled with a radioisotope or enzymatic label such that binding

can be determined by detecting the labeled protein or molecule in a complex. For example, the targets can be labeled with ¹²⁵I, ³⁵S, ¹⁴C, or ³H, either directly or indirectly, and the radioisotope detected by direct counting of radioemission or by scintillation counting. Alternatively, the targets can be enzymatically labeled with, for example, horseradish peroxidase, alkaline phosphatase, or luciferase, and the enzymatic label detected by determination of conversion of an appropriate substrate to product. Determining the interaction between biomarker and substrate can also be accomplished using standard binding or enzymatic analysis assays. In one or more embodiments of the above described assay methods, it may be desirable to immobilize polypeptides or molecules to facilitate separation of complexed from uncomplexed forms of one or both of the proteins or molecules, as well as to accommodate automation of the assay.

(311) Binding of a test agent to a target can be accomplished in any vessel suitable for containing the reactants. Non-limiting examples of such vessels include microtiter plates, test tubes, and micro-centrifuge tubes. Immobilized forms of the antibodies of the present invention can also include antibodies bound to a solid phase like a porous, microporous (with an average pore diameter less than about one micron) or macroporous (with an average pore diameter of more than about 10 microns) material, such as a membrane, cellulose, nitrocellulose, or glass fibers; a bead, such as that made of agarose or polyacrylamide or latex; or a surface of a dish, plate, or well, such as one made of polystyrene.

(312) In an alternative embodiment, determining the ability of the agent to modulate the interaction between the biomarker and a substrate or a biomarker and its natural binding partner can be accomplished by determining the ability of the test agent to modulate the activity of a polypeptide or other product that functions downstream or upstream of its position within the signaling pathway (e.g., feedback loops). Such feedback loops are well-known in the art (see, for example, Chen and Guillemin (2009) *Int. J Tryptophan Res.* 2:1-19).

(313) The present invention further pertains to novel agents identified by the above-described screening assays. Accordingly, it is within the scope of this invention to further use an agent identified as described herein in an appropriate animal model. For example, an agent identified as described herein can be used in an animal model to determine the efficacy, toxicity, or side effects of treatment with such an agent. Alternatively, an antibody identified as described herein can be used in an animal model to determine the mechanism of action of such an agent.

(314) b. Predictive Medicine

(315) The present invention also pertains to the field of predictive medicine in which diagnostic assays, prognostic assays, and monitoring clinical trials are used for prognostic (predictive) purposes to thereby treat an individual prophylactically. Accordingly, one aspect of the present invention relates to diagnostic assays for determining the amount and/or activity level of a biomarker listed in Table 1 in the context of a biological sample (e.g., blood, serum, cells, or tissue) to thereby determine whether an individual afflicted with a cancer is likely to respond to immune checkpoint therapy, whether in an original or recurrent cancer. Such assays can be used for prognostic or predictive purpose to thereby prophylactically treat an individual prior to the onset or after recurrence of a disorder characterized by or associated with biomarker polypeptide, nucleic acid expression or activity. The skilled artisan will appreciate that any method can use one or more (e.g., combinations) of biomarkers listed in Table 1.

(316) Another aspect of the present invention pertains to monitoring the influence of agents (e.g., drugs, compounds, and small nucleic acid-based molecules) on the expression or activity of a biomarker listed in Table 1. These and other agents are described in further detail in the following sections.

(317) The skilled artisan will also appreciate that, in certain embodiments, the methods of the present invention implement a computer program and computer system. For example, a computer program can be used to perform the algorithms described herein. A computer system can also store and manipulate data generated by the methods of the present invention which comprises a plurality

of biomarker signal changes/profiles which can be used by a computer system in implementing the methods of this invention. In certain embodiments, a computer system receives biomarker expression data; (ii) stores the data; and (iii) compares the data in any number of ways described herein (e.g., analysis relative to appropriate controls) to determine the state of informative biomarkers from cancerous or pre-cancerous tissue. In other embodiments, a computer system (i) compares the determined expression biomarker level to a threshold value; and (ii) outputs an indication of whether said biomarker level is significantly modulated (e.g., above or below) the threshold value, or a phenotype based on said indication.

(318) In certain embodiments, such computer systems are also considered part of the present invention. Numerous types of computer systems can be used to implement the analytic methods of this invention according to knowledge possessed by a skilled artisan in the bioinformatics and/or computer arts. Several software components can be loaded into memory during operation of such a computer system. The software components can comprise both software components that are standard in the art and components that are special to the present invention (e.g., dCHIP software described in Lin et al. (2004) *Bioinformatics* 20, 1233-1240; radial basis machine learning algorithms (RBM) known in the art).

(319) The methods of the present invention can also be programmed or modeled in mathematical software packages that allow symbolic entry of equations and high-level specification of processing, including specific algorithms to be used, thereby freeing a user of the need to procedurally program individual equations and algorithms. Such packages include, e.g., Matlab from Mathworks (Natick, Mass.), Mathematica from Wolfram Research (Champaign, Ill.) or S-Plus from MathSoft (Seattle, Wash.).

(320) In certain embodiments, the computer comprises a database for storage of biomarker data. Such stored profiles can be accessed and used to perform comparisons of interest at a later point in time. For example, biomarker expression profiles of a sample derived from the non-cancerous tissue of a subject and/or profiles generated from population-based distributions of informative loci of interest in relevant populations of the same species can be stored and later compared to that of a sample derived from the cancerous tissue of the subject or tissue suspected of being cancerous of the subject.

(321) In addition to the exemplary program structures and computer systems described herein, other, alternative program structures and computer systems will be readily apparent to the skilled artisan. Such alternative systems, which do not depart from the above described computer system and programs structures either in spirit or in scope, are therefore intended to be comprehended within the accompanying claims.

(322) c. Diagnostic Assays

(323) The present invention provides, in part, methods, systems, and code for accurately classifying whether a biological sample is associated with a cancer that is likely to respond to immune checkpoint therapy. In some embodiments, the present invention is useful for classifying a sample (e.g., from a subject) as associated with or at risk for responding to or not responding to immune checkpoint therapy using a statistical algorithm and/or empirical data (e.g., the amount or activity of a biomarker listed in Table 1).

(324) An exemplary method for detecting the amount or activity of a biomarker listed in Table 1, and thus useful for classifying whether a sample is likely or unlikely to respond to immune checkpoint therapy involves obtaining a biological sample from a test subject and contacting the biological sample with an agent, such as a protein-binding agent like an antibody or antigen-binding fragment thereof, or a nucleic acid-binding agent like an oligonucleotide, capable of detecting the amount or activity of the biomarker in the biological sample. In some embodiments, at least one antibody or antigen-binding fragment thereof is used, wherein two, three, four, five, six, seven, eight, nine, ten, or more such antibodies or antibody fragments can be used in combination (e.g., in sandwich ELISAs) or in serial. In certain instances, the statistical algorithm is a single

learning statistical classifier system. For example, a single learning statistical classifier system can be used to classify a sample as a based upon a prediction or probability value and the presence or level of the biomarker. The use of a single learning statistical classifier system typically classifies the sample as, for example, a likely immune checkpoint therapy responder or progressor sample with a sensitivity, specificity, positive predictive value, negative predictive value, and/or overall accuracy of at least about 75%, 76%, 77%, 78%, 79%, 80%, 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99%.

(325) Other suitable statistical algorithms are well known to those of skill in the art. For example, learning statistical classifier systems include a machine learning algorithmic technique capable of adapting to complex data sets (e.g., panel of markers of interest) and making decisions based upon such data sets. In some embodiments, a single learning statistical classifier system such as a classification tree (e.g., random forest) is used. In other embodiments, a combination of 2, 3, 4, 5, 6, 7, 8, 9, 10, or more learning statistical classifier systems are used, preferably in tandem. Examples of learning statistical classifier systems include, but are not limited to, those using inductive learning (e.g., decision/classification trees such as random forests, classification and regression trees (C&RT), boosted trees, etc.), Probably Approximately Correct (PAC) learning, connectionist learning (e.g., neural networks (NN), artificial neural networks (ANN), neuro fuzzy networks (NFN), network structures, perceptrons such as multi-layer perceptrons, multi-layer feed-forward networks, applications of neural networks, Bayesian learning in belief networks, etc.), reinforcement learning (e.g., passive learning in a known environment such as naive learning, adaptive dynamic learning, and temporal difference learning, passive learning in an unknown environment, active learning in an unknown environment, learning action-value functions, applications of reinforcement learning, etc.), and genetic algorithms and evolutionary programming. Other learning statistical classifier systems include support vector machines (e.g., Kernel methods), multivariate adaptive regression splines (MARS), Levenberg-Marquardt algorithms, Gauss-Newton algorithms, mixtures of Gaussians, gradient descent algorithms, and learning vector quantization (LVQ). In certain embodiments, the method of the present invention further comprises sending the sample classification results to a clinician, e.g., an oncologist.

(326) In another embodiment, the diagnosis of a subject is followed by administering to the individual a therapeutically effective amount of a defined treatment based upon the diagnosis.

(327) In one embodiment, the methods further involve obtaining a control biological sample (e.g., biological sample from a subject who does not have a cancer or whose cancer is susceptible to immune checkpoint therapy), a biological sample from the subject during remission, or a biological sample from the subject during treatment for developing a cancer progressing despite immune checkpoint therapy.

(328) d. Prognostic Assays

(329) The diagnostic methods described herein can furthermore be utilized to identify subjects having or at risk of developing a cancer that is likely or unlikely to be responsive to immune checkpoint therapy. The assays described herein, such as the preceding diagnostic assays or the following assays, can be utilized to identify a subject having or at risk of developing a disorder associated with a misregulation of the amount or activity of at least one biomarker described in Table 1, such as in cancer. Alternatively, the prognostic assays can be utilized to identify a subject having or at risk for developing a disorder associated with a misregulation of the at least one biomarker described in Table 1, such as in cancer. Furthermore, the prognostic assays described herein can be used to determine whether a subject can be administered an agent (e.g., an agonist, antagonist, peptidomimetic, polypeptide, peptide, nucleic acid, small molecule, or other drug candidate) to treat a disease or disorder associated with the aberrant biomarker expression or activity.

(330) e. Treatment Methods

(331) The compositions described herein (including dual binding antibodies and derivatives and

conjugates thereof) can be used in a variety of in vitro and in vivo therapeutic applications using the formulations and/or combinations described herein. In one embodiment, anti-immune checkpoint agents can be used to treat cancers determined to be responsive thereto. For example, antibodies that block the interaction between PD-L1, PD-L2, and/or CTLA-4 and their receptors (e.g., PD-L1 binding to PD-1, PD-L2 binding to PD-1, and the like) can be used to treat cancer in subjects identified as likely responding thereto.

(332) 6. Pharmaceutical Compositions

(333) In another aspect, the present invention provides pharmaceutically acceptable compositions which comprise a therapeutically-effective amount of an agent that modulates (e.g., decreases) biomarker expression and/or activity, formulated together with one or more pharmaceutically acceptable carriers (additives) and/or diluents. As described in detail below, the pharmaceutical compositions of the present invention may be specially formulated for administration in solid or liquid form, including those adapted for the following: (1) oral administration, for example, drenches (aqueous or non-aqueous solutions or suspensions), tablets, boluses, powders, granules, pastes; (2) parenteral administration, for example, by subcutaneous, intramuscular or intravenous injection as, for example, a sterile solution or suspension; (3) topical application, for example, as a cream, ointment or spray applied to the skin; (4) intravaginally or intrarectally, for example, as a pessary, cream or foam; or (5) aerosol, for example, as an aqueous aerosol, liposomal preparation or solid particles containing the compound.

(334) The phrase “therapeutically-effective amount” as used herein means that amount of an agent that modulates (e.g., inhibits) biomarker expression and/or activity, or expression and/or activity of the complex, or composition comprising an agent that modulates (e.g., inhibits) biomarker expression and/or activity, or expression and/or activity of the complex, which is effective for producing some desired therapeutic effect, e.g., cancer treatment, at a reasonable benefit/risk ratio.

(335) The phrase “pharmaceutically acceptable” is employed herein to refer to those agents, materials, compositions, and/or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of human beings and animals without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio.

(336) The phrase “pharmaceutically-acceptable carrier” as used herein means a pharmaceutically-acceptable material, composition or vehicle, such as a liquid or solid filler, diluent, excipient, solvent or encapsulating material, involved in carrying or transporting the subject chemical from one organ, or portion of the body, to another organ, or portion of the body. Each carrier must be “acceptable” in the sense of being compatible with the other ingredients of the formulation and not injurious to the subject. Some examples of materials which can serve as pharmaceutically-acceptable carriers include: (1) sugars, such as lactose, glucose and sucrose; (2) starches, such as corn starch and potato starch; (3) cellulose, and its derivatives, such as sodium carboxymethyl cellulose, ethyl cellulose and cellulose acetate; (4) powdered tragacanth; (5) malt; (6) gelatin; (7) talc; (8) excipients, such as cocoa butter and suppository waxes; (9) oils, such as peanut oil, cottonseed oil, safflower oil, sesame oil, olive oil, corn oil and soybean oil; (10) glycols, such as propylene glycol; (11) polyols, such as glycerin, sorbitol, mannitol and polyethylene glycol; (12) esters, such as ethyl oleate and ethyl laurate; (13) agar; (14) buffering agents, such as magnesium hydroxide and aluminum hydroxide; (15) alginic acid; (16) pyrogen-free water; (17) isotonic saline; (18) Ringer's solution; (19) ethyl alcohol; (20) phosphate buffer solutions; and (21) other non-toxic compatible substances employed in pharmaceutical formulations.

(337) The term “pharmaceutically-acceptable salts” refers to the relatively non-toxic, inorganic and organic acid addition salts of the agents that modulates (e.g., inhibits) biomarker expression and/or activity, or expression and/or activity of the complex encompassed by the present invention. These salts can be prepared in situ during the final isolation and purification of the respiration uncoupling agents, or by separately reacting a purified respiration uncoupling agent in its free base form with a

suitable organic or inorganic acid, and isolating the salt thus formed. ReRepresentative salts include the hydrobromide, hydrochloride, sulfate, bisulfate, phosphate, nitrate, acetate, valerate, oleate, palmitate, stearate, laurate, benzoate, lactate, phosphate, tosylate, citrate, maleate, fumarate, succinate, tartrate, naphthylate, mesylate, glucoheptonate, lactobionate, and laurylsulphonate salts and the like (See, for example, Berge et al. (1977) "Pharmaceutical Salts", *J. Pharm. Sci.* 66:1-19). (338) In other cases, the agents useful in the methods of the present invention may contain one or more acidic functional groups and, thus, are capable of forming pharmaceutically-acceptable salts with pharmaceutically-acceptable bases. The term "pharmaceutically-acceptable salts" in these instances refers to the relatively non-toxic, inorganic and organic base addition salts of agents that modulates (e.g., inhibits) biomarker expression and/or activity, or expression and/or activity of the complex. These salts can likewise be prepared in situ during the final isolation and purification of the respiration uncoupling agents, or by separately reacting the purified respiration uncoupling agent in its free acid form with a suitable base, such as the hydroxide, carbonate or bicarbonate of a pharmaceutically-acceptable metal cation, with ammonia, or with a pharmaceutically-acceptable organic primary, secondary or tertiary amine. ReRepresentative alkali or alkaline earth salts include the lithium, sodium, potassium, calcium, magnesium, and aluminum salts and the like. ReRepresentative organic amines useful for the formation of base addition salts include ethylamine, diethylamine, ethylenediamine, ethanolamine, diethanolamine, piperazine and the like (see, for example, Berge et al., supra).

(339) Wetting agents, emulsifiers and lubricants, such as sodium lauryl sulfate and magnesium stearate, as well as coloring agents, release agents, coating agents, sweetening, flavoring and perfuming agents, preservatives and antioxidants can also be present in the compositions.

(340) Examples of pharmaceutically-acceptable antioxidants include: (1) water soluble antioxidants, such as ascorbic acid, cysteine hydrochloride, sodium bisulfate, sodium metabisulfite, sodium sulfite and the like; (2) oil-soluble antioxidants, such as ascorbyl palmitate, butylated hydroxyanisole (BHA), butylated hydroxytoluene (BHT), lecithin, propyl gallate, alpha-tocopherol, and the like; and (3) metal chelating agents, such as citric acid, ethylenediamine tetraacetic acid (EDTA), sorbitol, tartaric acid, phosphoric acid, and the like.

(341) Formulations useful in the methods of the present invention include those suitable for oral, nasal, topical (including buccal and sublingual), rectal, vaginal, aerosol and/or parenteral administration. The formulations may conveniently be presented in unit dosage form and may be prepared by any methods well known in the art of pharmacy. The amount of active ingredient which can be combined with a carrier material to produce a single dosage form will vary depending upon the host being treated, the particular mode of administration. The amount of active ingredient, which can be combined with a carrier material to produce a single dosage form will generally be that amount of the compound which produces a therapeutic effect. Generally, out of one hundred percent, this amount will range from about 1 percent to about ninety-nine percent of active ingredient, preferably from about 5 percent to about 70 percent, most preferably from about 10 percent to about 30 percent.

(342) Methods of preparing these formulations or compositions include the step of bringing into association an agent that modulates (e.g., inhibits) biomarker expression and/or activity, with the carrier and, optionally, one or more accessory ingredients. In general, the formulations are prepared by uniformly and intimately bringing into association a respiration uncoupling agent with liquid carriers, or finely divided solid carriers, or both, and then, if necessary, shaping the product.

(343) Formulations suitable for oral administration may be in the form of capsules, cachets, pills, tablets, lozenges (using a flavored basis, usually sucrose and acacia or tragacanth), powders, granules, or as a solution or a suspension in an aqueous or non-aqueous liquid, or as an oil-in-water or water-in-oil liquid emulsion, or as an elixir or syrup, or as pastilles (using an inert base, such as gelatin and glycerin, or sucrose and acacia) and/or as mouth washes and the like, each containing a predetermined amount of a respiration uncoupling agent as an active ingredient. A compound may

also be administered as a bolus, electuary or paste.

(344) In solid dosage forms for oral administration (capsules, tablets, pills, dragees, powders, granules and the like), the active ingredient is mixed with one or more pharmaceutically-acceptable carriers, such as sodium citrate or dicalcium phosphate, and/or any of the following: (1) fillers or extenders, such as starches, lactose, sucrose, glucose, mannitol, and/or silicic acid; (2) binders, such as, for example, carboxymethylcellulose, alginates, gelatin, polyvinyl pyrrolidone, sucrose and/or acacia; (3) humectants, such as glycerol; (4) disintegrating agents, such as agar-agar, calcium carbonate, potato or tapioca starch, alginic acid, certain silicates, and sodium carbonate; (5) solution retarding agents, such as paraffin; (6) absorption accelerators, such as quaternary ammonium compounds; (7) wetting agents, such as, for example, acetyl alcohol and glycerol monostearate; (8) absorbents, such as kaolin and bentonite clay; (9) lubricants, such as talc, calcium stearate, magnesium stearate, solid polyethylene glycols, sodium lauryl sulfate, and mixtures thereof; and (10) coloring agents. In the case of capsules, tablets and pills, the pharmaceutical compositions may also comprise buffering agents. Solid compositions of a similar type may also be employed as fillers in soft and hard-filled gelatin capsules using such excipients as lactose or milk sugars, as well as high molecular weight polyethylene glycols and the like.

(345) A tablet may be made by compression or molding, optionally with one or more accessory ingredients. Compressed tablets may be prepared using binder (for example, gelatin or hydroxypropylmethyl cellulose), lubricant, inert diluent, preservative, disintegrant (for example, sodium starch glycolate or cross-linked sodium carboxymethyl cellulose), surface-active or dispersing agent. Molded tablets may be made by molding in a suitable machine a mixture of the powdered peptide or peptidomimetic moistened with an inert liquid diluent.

(346) Tablets, and other solid dosage forms, such as dragees, capsules, pills and granules, may optionally be scored or prepared with coatings and shells, such as enteric coatings and other coatings well known in the pharmaceutical-formulating art. They may also be formulated so as to provide slow or controlled release of the active ingredient therein using, for example, hydroxypropylmethyl cellulose in varying proportions to provide the desired release profile, other polymer matrices, liposomes and/or microspheres. They may be sterilized by, for example, filtration through a bacteria-retaining filter, or by incorporating sterilizing agents in the form of sterile solid compositions, which can be dissolved in sterile water, or some other sterile injectable medium immediately before use. These compositions may also optionally contain opacifying agents and may be of a composition that they release the active ingredient(s) only, or preferentially, in a certain portion of the gastrointestinal tract, optionally, in a delayed manner. Examples of embedding compositions, which can be used include polymeric substances and waxes. The active ingredient can also be in micro-encapsulated form, if appropriate, with one or more of the above-described excipients.

(347) Liquid dosage forms for oral administration include pharmaceutically acceptable emulsions, microemulsions, solutions, suspensions, syrups and elixirs. In addition to the active ingredient, the liquid dosage forms may contain inert diluents commonly used in the art, such as, for example, water or other solvents, solubilizing agents and emulsifiers, such as ethyl alcohol, isopropyl alcohol, ethyl carbonate, ethyl acetate, benzyl alcohol, benzyl benzoate, propylene glycol, 1,3-butylene glycol, oils (in particular, cottonseed, groundnut, corn, germ, olive, castor and sesame oils), glycerol, tetrahydrofuryl alcohol, polyethylene glycols and fatty acid esters of sorbitan, and mixtures thereof.

(348) Besides inert diluents, the oral compositions can also include adjuvants such as wetting agents, emulsifying and suspending agents, sweetening, flavoring, coloring, perfuming and preservative agents.

(349) Suspensions, in addition to the active agent may contain suspending agents as, for example, ethoxylated isostearyl alcohols, polyoxyethylene sorbitol and sorbitan esters, microcrystalline cellulose, aluminum metahydroxide, bentonite, agar-agar and tragacanth, and mixtures thereof.

(350) Formulations for rectal or vaginal administration may be presented as a suppository, which may be prepared by mixing one or more respiration uncoupling agents with one or more suitable nonirritating excipients or carriers comprising, for example, cocoa butter, polyethylene glycol, a suppository wax or a salicylate, and which is solid at room temperature, but liquid at body temperature and, therefore, will melt in the rectum or vaginal cavity and release the active agent.

(351) Formulations which are suitable for vaginal administration also include pessaries, tampons, creams, gels, pastes, foams or spray formulations containing such carriers as are known in the art to be appropriate.

(352) Dosage forms for the topical or transdermal administration of an agent that modulates (e.g., inhibits) biomarker expression and/or activity include powders, sprays, ointments, pastes, creams, lotions, gels, solutions, patches and inhalants. The active component may be mixed under sterile conditions with a pharmaceutically-acceptable carrier, and with any preservatives, buffers, or propellants which may be required.

(353) The ointments, pastes, creams and gels may contain, in addition to a respiration uncoupling agent, excipients, such as animal and vegetable fats, oils, waxes, paraffins, starch, tragacanth, cellulose derivatives, polyethylene glycols, silicones, bentonites, silicic acid, talc and zinc oxide, or mixtures thereof.

(354) Powders and sprays can contain, in addition to an agent that modulates (e.g., inhibits) biomarker expression and/or activity, excipients such as lactose, talc, silicic acid, aluminum hydroxide, calcium silicates and polyamide powder, or mixtures of these substances. Sprays can additionally contain customary propellants, such as chlorofluorohydrocarbons and volatile unsubstituted hydrocarbons, such as butane and propane.

(355) The agent that modulates (e.g., inhibits) biomarker expression and/or activity, can be alternatively administered by aerosol. This is accomplished by preparing an aqueous aerosol, liposomal preparation or solid particles containing the compound. A nonaqueous (e.g., fluorocarbon propellant) suspension could be used. Sonic nebulizers are preferred because they minimize exposing the agent to shear, which can result in degradation of the compound.

(356) Ordinarily, an aqueous aerosol is made by formulating an aqueous solution or suspension of the agent together with conventional pharmaceutically acceptable carriers and stabilizers. The carriers and stabilizers vary with the requirements of the particular compound, but typically include nonionic surfactants (Tweens, Pluronic, or polyethylene glycol), innocuous proteins like serum albumin, sorbitan esters, oleic acid, lecithin, amino acids such as glycine, buffers, salts, sugars or sugar alcohols. Aerosols generally are prepared from isotonic solutions.

(357) Transdermal patches have the added advantage of providing controlled delivery of a respiration uncoupling agent to the body. Such dosage forms can be made by dissolving or dispersing the agent in the proper medium. Absorption enhancers can also be used to increase the flux of the peptidomimetic across the skin. The rate of such flux can be controlled by either providing a rate controlling membrane or dispersing the peptidomimetic in a polymer matrix or gel.

(358) Ophthalmic formulations, eye ointments, powders, solutions and the like, are also contemplated as being within the scope of this invention.

(359) Pharmaceutical compositions of this invention suitable for parenteral administration comprise one or more respiration uncoupling agents in combination with one or more pharmaceutically-acceptable sterile isotonic aqueous or nonaqueous solutions, dispersions, suspensions or emulsions, or sterile powders which may be reconstituted into sterile injectable solutions or dispersions just prior to use, which may contain antioxidants, buffers, bacteriostats, solutes which render the formulation isotonic with the blood of the intended recipient or suspending or thickening agents.

(360) Examples of suitable aqueous and nonaqueous carriers which may be employed in the pharmaceutical compositions of the present invention include water, ethanol, polyols (such as glycerol, propylene glycol, polyethylene glycol, and the like), and suitable mixtures thereof, vegetable oils, such as olive oil, and injectable organic esters, such as ethyl oleate. Proper fluidity

can be maintained, for example, by the use of coating materials, such as lecithin, by the maintenance of the required particle size in the case of dispersions, and by the use of surfactants. (361) These compositions may also contain adjuvants such as preservatives, wetting agents, emulsifying agents and dispersing agents. Prevention of the action of microorganisms may be ensured by the inclusion of various antibacterial and antifungal agents, for example, paraben, chlorobutanol, phenol sorbic acid, and the like. It may also be desirable to include isotonic agents, such as sugars, sodium chloride, and the like into the compositions. In addition, prolonged absorption of the injectable pharmaceutical form may be brought about by the inclusion of agents which delay absorption such as aluminum monostearate and gelatin.

(362) In some cases, in order to prolong the effect of a drug, it is desirable to slow the absorption of the drug from subcutaneous or intramuscular injection. This may be accomplished by the use of a liquid suspension of crystalline or amorphous material having poor water solubility. The rate of absorption of the drug then depends upon its rate of dissolution, which, in turn, may depend upon crystal size and crystalline form. Alternatively, delayed absorption of a parenterally-administered drug form is accomplished by dissolving or suspending the drug in an oil vehicle.

(363) Injectable depot forms are made by forming microencapsule matrices of an agent that modulates (e.g., inhibits) biomarker expression and/or activity, in biodegradable polymers such as polylactide-polyglycolide. Depending on the ratio of drug to polymer, and the nature of the particular polymer employed, the rate of drug release can be controlled. Examples of other biodegradable polymers include poly(orthoesters) and poly(anhydrides). Depot injectable formulations are also prepared by entrapping the drug in liposomes or microemulsions, which are compatible with body tissue.

(364) When the respiration uncoupling agents of the present invention are administered as pharmaceuticals, to humans and animals, they can be given per se or as a pharmaceutical composition containing, for example, 0.1 to 99.5% (more preferably, 0.5 to 90%) of active ingredient in combination with a pharmaceutically acceptable carrier.

(365) Actual dosage levels of the active ingredients in the pharmaceutical compositions of this invention may be determined by the methods of the present invention so as to obtain an amount of the active ingredient, which is effective to achieve the desired therapeutic response for a particular subject, composition, and mode of administration, without being toxic to the subject.

(366) The nucleic acid molecules of the present invention can be inserted into vectors and used as gene therapy vectors. Gene therapy vectors can be delivered to a subject by, for example, intravenous injection, local administration (see U.S. Pat. No. 5,328,470) or by stereotactic injection (see e.g., Chen et al. (1994) *Proc. Nat. Acad. Sci. USA* 91:3054 3057). The pharmaceutical preparation of the gene therapy vector can include the gene therapy vector in an acceptable diluent, or can comprise a slow release matrix in which the gene delivery vehicle is imbedded.

Alternatively, where the complete gene delivery vector can be produced intact from recombinant cells, e.g., retroviral vectors, the pharmaceutical preparation can include one or more cells which produce the gene delivery system.

(367) The present invention also encompasses kits for detecting and/or modulating biomarkers described herein. A kit of the present invention may also include instructional materials disclosing or describing the use of the kit or an antibody of the disclosed invention in a method of the disclosed invention as provided herein. A kit may also include additional components to facilitate the particular application for which the kit is designed. For example, a kit may additionally contain means of detecting the label (e.g., enzyme substrates for enzymatic labels, filter sets to detect fluorescent labels, appropriate secondary labels such as a sheep anti-mouse-HRP, etc.) and reagents necessary for controls (e.g., control biological samples or standards). A kit may additionally include buffers and other reagents recognized for use in a method of the disclosed invention. Non-limiting examples include agents to reduce non-specific binding, such as a carrier protein or a detergent.

EXEMPLIFICATION

(368) This invention is further illustrated by the following examples, which should not be construed as limiting.

Example 1: Materials and Methods for Examples 2-3

(369) a. Meta-Analysis: Cohort Consolidation

(370) Mutation annotation files, HLA types, and clinical annotations were obtained from published data. Standardized pipelines for somatic variant calling, mutational signature deconvolution, and neoantigen prediction were used. Patients were stratified into clinical benefit (CB) and no clinical benefit (NCB) as described in Van Allen et al. (2015) *Science* 350:207-211. Analyses were repeated using two other published response metrics (CB=PFS>6 months; CB=CR or PR; see at least Rizvi et al. (2015) *Science* 348:124-128 and Snyder et al. (2014) *N. Engl. J Med.* 371:2189-2199).

(371) Specifically, pre-treatment tumors from patients who received immune checkpoint blockade for metastatic cancer at the DFCI and partner institutions were sequenced at the Broad Institute (FIG. 1). Data sources listed in FIG. 1 include data from Zaretsky et al. (2016) *New Engl. J Med.* 375:819-829 disclosing 4 patients treated with anti-PD-1 therapy for metastatic melanoma (Zaretsky); Van Allen et al. (2015) *Science* 350:207-211 disclosing treatment of metastatic melanoma patients across Germany with anti-CTLA-4 ipilimumab therapy (Schadendorf); Hugo et al. (2016) *Cell* 165:35-44 (Hugo); Rizvi et al. (2015) *Science* 348:124-128 (Rizvi); Snyder et al. (2014) *N. Engl. J Med.* 371:2189-2199 (Synder); and samples obtained from patients treated with immune checkpoint therapies with clinical monitoring at the Dana-Farber Cancer Institute (DFCI, CANSEQ, BroadNext10, and Rizwan Haq). For example, samples from 2 patients for WES obtained from Dana-Farber Cancer Institute attending physician, Rizwan Haq (Rizwan Haq). Cancer types included melanoma, non-small-cell lung cancer, bladder cancer, anal cancer, sarcoma, head and neck squamous cell carcinoma, including one previously published cohort in Van Allen et al. (2015) *Science* 350:207-211. Quality controls were included based on exclusion criteria and inclusion criteria (FIG. 2).

(372) In addition, whole exome data from published clinical cohorts of patients who received immune checkpoint therapy for metastatic non-small-cell lung cancer (Rizvi et al. (2015) *Science* 348:124-128) and melanoma (Snyder et al. (2014) *N. Engl. J. Med.* 371:2189-2199; Hugo et al. (2016) *Cell* 165:35-44) were used in the meta-analysis.

(373) In separate meta-analyses described in Example 3 below, mutation annotation files, HLA types, and clinical annotations were obtained from published data. In particular, published mutation annotation files (MAFs), clinical annotations, and neoantigen files were taken from the supplemental information of Rizvi et al. (2015) *Science* 348:124-128; Van Allen et al. (2015) *Science* 350:207-211; and Snyder et al. (2014) *N. Engl. J Med.* 371:2189-2199. These files were joined to form the meta-analysis cohort. Enrichment in mutations in specific genes was assessed in this joined cohort using Fisher's exact test. The joined MAF was also processed through MutSigCV to identify genes significantly mutated throughout the cohort. Thus, standardized pipelines for somatic variant calling, mutational signature deconvolution, and neoantigen prediction were used. Patients were stratified into clinical benefit (CB) and no clinical benefit (NCB) as described in Van Allen et al. (2015) *Science* 350:207-211. Analyses were repeated using two other published response metrics (CB=PFS>6 months; CB=CR or PR).

(374) b. Sample Preparation, DNA and RNA Extraction, and Sequencing

(375) Sample preparation, DNA and RNA extraction, and sequencing information vary slightly between the studies described in Rizvi et al. (2015) *Science* 348:124-128; Van Allen et al. (2015) *Science* 350:207-211; and Snyder et al. (2014) *N. Engl. J Med.* 371:2189-2199, and particular details of such methods used can be found in the supplemental information files of these three publications.

(376) For other samples, such as metastatic melanoma samples described in Example 3, or unless otherwise described, sample preparation, DNA and RNA extraction, and sequencing information proceeded as described below. Briefly, after fixation and mounting, 5-10 10 µm slices from

formalin-fixed, paraffin-embedded (FFPE) tumor blocks were obtained, and tumor-enriched tissue was macrodissected. Paraffin was removed from FFPE sections and cores using CitriSolv™ (Fisher Scientific, Hampton, NH), followed by ethanol washes and tissue lysis overnight at 56° C. Samples were then incubated at 90° C. to remove DNA crosslinks, and DNA- and when possible, RNA-extraction was performed using Qiagen AllPrep DNA/RNA Mini Kit (#51306, Qiagen, Hilden, Germany). Germline DNA was obtained from adjacent PBMCs.

(377) Whole exome and whole transcriptome sequencing of tumor and germline samples were performed as previously described (Van Allen et al. (2015) *Science* 350:207-211; Van Allen et al. (2014) *Nat. Med.* 20:682-688). All samples in the training cohort were sequenced using the Illumina exome, while a portion of the samples in the validation cohort were sequenced using the Agilent exome (Table 3A). The Illumina exome uses Illumina's in-solution DNA probe based hybrid selection method to target approximately 37.7 Mb of mainly exonic territory, using similar principles as the Broad Institute-Agilent Technologies developed in-solution RNA probe based hybrid selection method (Agilent SureSelect™ All Exon V2) (Fisher et al. (2011) *Genome Biol.* 12:R1; Gnirke et al. (2009) *Nat. Biotechnol.* 27:182-189) to generate Illumina exome sequencing libraries.

(378) Pooled libraries were normalized to 2 nM and denatured using 0.2 N NaOH prior to sequencing. Flowcell cluster amplification and sequencing were performed according to the manufacturer's protocols using either the HiSeq 2000 v3 or HiSeq 2500. Each run was a 76 bp paired-end with a dual eight-base index barcode read. Data was analyzed using the Broad Picard Pipeline, which includes de-multiplexing and data aggregation.

(379) Exome sequence data processing was performed using established analytical pipelines at the Broad Institute. A BAM file was produced using the Picard pipeline (at the World Wide Web address of picard.sourceforge.net), which aligns the tumor and normal sequences to the hg19 human genome build using Illumina sequencing reads. The BAM was uploaded into the Firehose pipeline (at the World Wide Web address of broadinstitute.org/cancer/cga/Firehose), which manages input and output files to be executed by GenePattern (Reich et al. (2006) *Nat. Genet.* 38:500-501). Samples with mean target coverage less than 25× in the tumor and less than 15× in matched normal were excluded.

(380) Quality control modules within Firehose were applied to all sequencing data for comparison of the origin of tumor and normal genotypes and to assess fingerprinting concordance. Cross-contamination of samples was estimated using ContEst (Cibulskis et al. (2011) *Bioinformatics* 27:2601-2602). Samples with ContEst estimates exceeding 5% were excluded from analysis.

c. Whole Exome and Whole Transcriptome Analyses

(381) MuTect was applied to identify somatic single-nucleotide variants (Cibulskis et al. (2013) *Nat. Biotechnol.* 31:213-219). Strelka was used to identify somatic insertions and deletions (Saunders et al. (2012) *Bioinformatics* 28:1811-1817) across the whole exome. Indelocator, which detects small insertions and deletions after local realignment of tumor and normal sequences, was additionally applied to provide further sensitivity to detect indels in PBRM1 (Cancer Genome Atlas Research (2011) *Nature* 474:609-615). The union of indels called by Strelka and Indelocator was used for final analysis. Artifacts introduced by DNA oxidation during sequencing were computationally removed using a filter-based method (Costello et al. (2013) *Nuc. Acids Res.* 41:e67). All somatic mutations detected by whole-exome sequencing were analyzed for potential false positive calls by performing a comparison to mutation calls from a panel of 2,500 germline DNA samples (Stachler et al. (2015) *Nat. Genet.* 47:1047-1055). Mutations found in germline samples were removed from analysis. Annotation of identified variants was done using Oncotator (available at the World Wide Web address of www.broadinstitute.org/cancer/cga/oncotator). All nonsynonymous alterations in PBRM1 were manually reviewed in Integrated Genomics Viewer (IGV_2.3.57) for sequencing quality (Thorvaldsdottir et al. (2013) *Brief Bioinform* 14:178-192).

(382) Copy ratios were calculated for each captured target by dividing the tumor coverage by the

median coverage obtained in a set of reference normal samples. The resulting copy ratios were segmented using the circular binary segmentation algorithm (Olshen et al. (2004) *Biostatistics* 5:557-572). Allelic copy number alterations were called while taking into account sample-specific overall chromosomal aberrations (focality) (Brastianos et al. (2015) *Cancer Discov.* 5:1164-1177). Inference of mutational clonality, tumor purity, and tumor ploidy was accomplished with ABSOLUTE (Carter et al. (2012) *Nat Biotechnol.* 30:413-421). Samples had to have estimated tumor purity greater than 10% to be included in the final analysis. As a final quality control metric to ensure adequate sequencing coverage and tumor purity to detect relevant oncogenic events, all samples had to have at least one nonsynonymous mutation in at least one high confidence or candidate cancer driver gene to be included in the final analysis (Tamborero et al. (2013) *Sci. Rep.* 3:2650).

(383) The 4-digit HLA type for each sample was inferred using Polysolver (Shukla et al. (2015) *Nat. Biotechnol.* 33:1152-1158). Neo-epitopes were predicted for each patient by defining all novel amino acid 9mers and 10mers resulting from each single nucleotide polymorphism and indel and determining whether the predicted binding affinity to the patient's germline HLA alleles was <500 nM using NetMHCpan (v2.4) (Hoof et al. (2009) *Immunogenetics* 61:1-13; Karosiene et al. (2013) *Immunogenetics* 65:711-724; Nielsen et al. (2007) *PLoS One* 2:e796).

(384) Statistical analyses were also applied and are described further herein.

Example 2: Meta-Analysis of Genomic Predictors of Response to Immune Checkpoint Therapy Across a Variety of Cancers

(385) A large cohort of whole-exome-sequenced pre-treatment tumors were gathered from subjects who received immune checkpoint therapies for various cancers, were gathered and assessed for mutations associated with differential response to immune checkpoint therapies. Tumor types included in this discovery cohort included bladder cancer, renal cell carcinoma, lung cancer, and head and neck squamous cell carcinoma. Tumor-specific (somatic) mutations were assessed in all sequenced tumors to generate mutation annotation files (MAFs), which were later joined with MAFs from previously published studies in clinical cohorts in melanoma (Van Allen et al. (2015) *Science* 350:207-211; Snyder et al. (2014) *N. Engl. J. Med.* 371:2189-2199; Hugo et al. (2016) *Cell* 165:35-44) and non-small cell lung cancer (Rizvi et al. (2015) *Science* 348:124-128).

(386) In particular, pre-treatment samples from cancer patients (N=268) were tested for their genetic sequences in relation to results of immune checkpoint therapy experienced by the cancer patients (FIG. 3). The results are shown in Table 2 as response by cancer type.

(387) TABLE-US-00003 TABLE 2 Melanoma Lung HNSCC Sarcoma Bladder Clinical benefit 66 (38.6%) 20 (29.4%) 4 (28.6%) 1 (100%) 6 (46.2%) Intermediate benefit 6 (3.5%) 23 (33.8%) 3 (21.4%) 0 1 (7.7%) No clinical benefit 96 (56.1%) 25 (36.8%) 7 (50.0%) 0 4 (30.8%) Mixed response or not 3 (1.8%) 0 0 0 2 (1.5%) evaluable Total 171 68 14 1 13

(388) Nonsynonymous mutational burden is a strong predictor of clinical outcomes across cancer types. As shown in FIG. 4, patients experiencing clinical benefit to immune checkpoint therapy had significantly more nonsynonymous mutations than those experiencing no clinical benefit ($p=5.52e-06$; Wilcoxon rank-sum). Such relationship is strong for patients with lung cancer ($p=5.2e-05$) and also significant for patients with bladder cancer ($p=0.019$) and melanoma ($p=0.0016$). No relationship between clinical outcome and mutational burden in HNSCC ($p=0.45$) was found. In addition, complete responders in sarcoma and anal cancer had unexceptional mutational burdens (below and near median, respectively).

(389) In order to identify biomarkers associated with response or nonresponse to immune checkpoint blockade, the following procedure was conducted: Split all patients into responders and non-responders Version A: N=98 CB vs. N=132 NCB Version B: N=98 CB vs. N=165 NCB or IB Calculate significance value for enrichment of mutations in a given gene in responder or non-responders (Fisher's exact test) Version A: All nonsynonymous mutations Version B: Only truncating alterations (frame-shift insertions and deletions, nonsense single nucleotide

polymorphisms, splice-site alterations) Run MutSigCV on all 268 patients with called mutations to determine significantly mutated genes across the entire cohort, which takes into account gene size, patient-specific mutational rate, mutational spectra, and gene replication rates and times Select genes that are mutated significantly more often in responders or non-responders ($p < 0.05$) and pass MutSigCV significance ($q < 0.1$; FDR).

(390) The identified biomarkers are shown in FIG. 5. Table 3 is a summary of all genes with significantly more nonsynonymous mutations in R (N=98) or NR (N=132).

(391) TABLE-US-00004 TABLE 3 p-value Gene N (R) N (NR) (Fisher's exact) CETP 4 0 0.032 COL3A1 18 10 0.015 COL5A1 22 15 0.029 DDX60 13 5 0.012 **DXH8 10 2 0.005** DLEC1 16 9 0.031 DNAH2 23 14 0.011 FHOD1 8 2 0.020 HK3 7 2 0.040 **KALRN 27 10 0.000093** KIF21B 15 9 0.049 KIF5A 11 5 0.036 **KRAS 11 3 0.0095** MCTP1 12 6 0.045 MGAM 33 26 0.022 **NARS2 6 0 0.0055** ZNF253 10 4 0.047 NF1 25 17 0.016 *NR1H4* 2 11 0.046 PBRM1 10 4 0.047 PKP1 7 2 0.040 **PLCB4 21 11 0.0065** POLR2A 10 4 0.047 PREX2 25 19 0.042 *RB1* 1 10 0.026 ROCK1 8 2 0.020 SAFB2 9 3 0.032 SERPINB3 11 5 0.036 SPATA16 12 6 0.045 STAB1 17 10 0.037 TGM3 11 5 0.036 TSC1 13 6 0.027 TSPAN2 7 2 0.040 ZNF207 6 1 0.044 *Genes in italics (i.e., *NR1H4* and *RB1*) mutated more frequently in NR vs. R. Genes in plain text mutated more frequently in R vs. NR. **Bolded genes (i.e., **DXH8**, **KALRN**, **KRAS**, **NARS2**, and **PLCB4**) have $p < 0.01$.

(392) After limiting analyses to comparing patients with objective tumor response (CR, PR, or SD by RECIST vs. PD by RECIST) in non-melanoma cancer types, it was observed that a striking association exists between mutations in one or more SWI/SNF complex subunits and response to immune checkpoint therapy (Tables 4-5 and FIGS. 6-8).

(393) For example, truncating alterations in PBRM1 and response to immune checkpoint therapy, driven by nonsense, frameshift, and splice site mutations in bladder cancer, lung cancer, and renal cell carcinoma (9/75 responders vs. 0/41 non-responders, $p = 0.026$). Additionally, it was observed that ARID2 truncating mutations enriched in responders in melanoma across multiple clinical cohorts (6/68 responders vs. 2/96 non-responders), as well as in isolated cases in other tumor types (one frameshift deletion in lung cancer PR and one frameshift deletion in one SD RCC).

Interestingly, the two ARID2 alterations occurring in non-responders occurred in patients receiving anti-CTLA4 therapies (rather than anti-PD1 therapies), though one patient with PR to anti-CTLA4 also had an ARID2 splice site mutation. Lastly, it was observed that mutations in SMARCA4 were associated with response in head and neck squamous cell carcinoma (3/6 responders vs. 0/9 non-responders, $p = 0.044$, Fisher's exact test). Thus, alterations in the SWI/SNF pathway were found to be predictive of response to immune checkpoint therapy across cancer types.

(394) TABLE-US-00005 TABLE 4 Complete list of all identified SWI/SNF mutations Entrez_Hugo_Gene_Chromo- Start_End_Variant_Tumor_Sample_Barcode pair_id Symbol Id some position position Classification response MEL-IPI_Pat132-Tumor-SM- MEL-IPI_Pat132-TP-NB-SM- ACTL6A 86 3 179294461 179294461 Missense_ + 5VWJA 5VWJA-SM-5VWHR Mutation MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- ACTL6A 86 3 179298455 179298455 Missense_ + 5VWJG 5VWJG-SM-5VWHX Mutation Pt20 Pt20 ACTL6A 86 3 179294018 179294018 Missense_ - Mutation CR04885 CR04885 ACTL6B 51412 7 100244379 100244379 Missense_ + Mutation Lung-DFCI-11-104-009- Lung-DFCI-11-104-009-TM- ACTL6B 51412 7 100244451 100244451 Missense_ + Tumor-SM-5YS7O NB-SM-5YS70-SM-5YS7P Mutation MEL-IPI_Pat130-Tumor-SM- MEL-IPI_Pat130-TP-NT-SM- ACTL6B 51412 7 100247744 100247744 Silent - 5X2R8 5X2R8-SM-5X2RJ MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- ACTL6B 51412 7 100240903 100240903 Missense_ + 5VWJG 5VWJG-SM-5VWHX Mutation MEL-IPI_Pat139-Tumor-SM- MEL-IPI_Pat139-TP-NB-SM- ACTL6B 51412 7 100244908 100244908 Splice_Site - 5VWJH 5VWJH-SM-5VWHY MEL-IPI_Pat28-Tumor-SM- MEL-IPI_Pat28-TP-NB-SM- ACTL6B 51412 7 100244416 100244416 Silent - 4DK10 4DK10-SM-4NFUU MEL-IPI_Pat38-Tumor-SM- MEL-IPI_Pat38-TP-NT-SM- ACTL6B 51412 7 100244877 100244877 Silent + 53U3Z 53U3Z-SM-53U5L MEL-IPI_Pat38-Tumor-SM-

MEL-IPI_Pat38-TP-SM- ACTL6B 51412 7 100244902 100244902 Silent + 53U3Z 53U3Z-
SM-53U5L PR4035 PR4035 ACTL6B 51412 7 100253064 100253064 Missense_ + Mutation Pt14
Pt14 ACTL6B 51412 7 100253200 100253200 Missense_ - Mutation Pt4 Pt4 ACTL6B 51412 7
100246273 100246273 Missense_ + Mutation RH090935 RH090935 ACTL6B 51412 7
100253478 100253478 Missense_ + Mutation SD1494 SD1494 ACTL6B 51412 7 100253045
100253045 Splice_Site ± SU2C_Lung-SU2C-DFCI- SU2C_Lung-SU2C-DFCI- ACTL6B 51412 7
100252740 100252740 Missense_ - LUAD-1011-Tumor-SM- LUAD-1011-TM-NB-SM- Mutation
AOL75 AOL75-SM-A46NN AL4602 AL4602 ARID1A 8289 1 27023633 27023633 Missense_ ±
Mutation BLADDER- BLADDER- ARID1A 8289 1 27101612 27101612 Frame_Shift_ -
15330_CCPM_0700692- 15330_CCPM_0700692-TM- Del Tumor-SM-AVI11 NB-SM-AVI11-
SM-AVHZM BLCA-IM01-Tumor-SM- BLCA-IM01-TP-NB-SM- ARID1A 8289 1 27057664
27057664 Missense_ - 79XD9 79XD9-SM-7AABJ Mutation BLCA-IM01-Tumor-SM- BLCA-
IM01-TP-NB-SM- ARID1A 8289 1 27058092 27058097 Splice_Site - 79XD9 79XD9-SM-
7AABJ BLCA-IM01-Tumor-SM- BLCA-IM01-TP-NB-SM- ARID1A 8289 1 27057642 27057642
Splice_Site - 79XD9 79XD9-SM-7AABJ HNSCC-287-Tumor-SM- HNSCC-287-TP-NB-SM-
ARID1A 8289 1 27100352 27100355 Frame_Shift_ - AXGEI AXGEI-SM-ADP7M Del LO3793
LO3793 ARID1A 8289 1 27105688 27105688 Nonsense ± Mutation LUAD-BS-13-F33496-
LUAD-BS-13-F33496-TP- ARID1A 8289 1 27056342 27056343 Frame_Shift_ ± Tumor-SM-
9J2XU NB-SM-9J2XU-SM-9HBZX Del Lung-DFCI-11-104-009- Lung-DFCI-11-104-009-IM-
ARID1A 8289 1 27106621 27106621 Nonsense + Tumor-SM-5YS7O NB-SM-5YS7O-SM-5YS7P
Mutation MEL-IPI_Pat7-Tumor-SM- MEL-IPI_Pat07-IP-NB-SM- ARID1A 8289 1 27105918
27105918 Silent + 4DK13 4DK13-SM-4NFU9 MEL-IPI_Pat11-Tumor-SM- MEL-IPI_Pat11-TP-
NB-SM- ARID1A 8289 1 27101642 27101642 Missense_ - 4DK17 4DK17-SM-4NFUD Mutation
MEL-IPI_Pat110-Tumor-SM- MEL-IPI_Pat110-TP-NT-SM- ARID1A 8289 1 27106693 27106693
Missense_ - 4CU6X 4CU6X-SM-4MGPO Mutation MEL-IPI_Pat133-Tumor-SM- MEL-
IPI_Pat133-TP-NB-SM- ARID1A 8289 1 27089607 27089607 Missense_ - 5VWJB 5VWJB-SM-
5VWHS Mutation MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- ARID1A 8289 1
27023487 27023487 Missense_ + 5VWJG 5VWJG-SM-5VWHX Mutation MEL-IPI_Pat138-
Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- ARID1A 8289 1 27106461 27106461 Silent + 5VWJG
5VWJG-SM-5VWHX MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- ARID1A 8289
1 27058070 27058070 Missense_ + 5VWJG 5VWJG-SM-5VWHX Mutation MEL-IPI_Pat159-
Tumor-SM- MEL-IPI_Pat159-TP-NB-SM- ARID1A 8289 1 27087452 27087452 Missense_ -
5VWK2 5VWK2-SM-5VWIJ Mutation MEL-IPI_Pat163-Tumor-SM- MEL-IPI_Pat163-TP-NB-
SM- ARID1A 8289 1 27107176 27107176 Missense_ - 5VWK6 5VWK6-SM-5VWIN Mutation
MEL-IPI_Pat37-Tumor-SM- MEL-IPI_Pat37-TP-NB-SM- ARID1A 8289 1 27101712 27101712
Splice_Site - 53U3Y 53U3Y-SM-4NFV4 MEL-IPI_Pat37-Tumor-SM- MEL-IPI_Pat37-TP-NB-
SM- ARID1A 8289 1 27105837 27105837 Silent - 53U3Y 53U3Y-SM-4NFV4 MEL-IPI_Pat38-
Tumor-SM- MEL-IPI_Pat38-TP-NT-SM- ARID1A 8289 1 27087393 27087393 Missense_ +
53U3Z 53U3Z-SM-53U5L Mutation MEL-IPI_Pat39-Tumor-SM- MEL-IPI_Pat39-TP-NB-SM-
ARID1A 8289 1 27057822 27057822 Silent + 4DK1Z 4DK1Z-SM-4NFV6 MEL-IPI_Pat62-
Tumor-SM- MEL-IPI_Pat62-TP-NB-SM- ARID1A 8289 1 27087494 27087494 Missense_ -
4DK2N 4DK2N-SM-4NFVT Mutation MEL-IPI_Pat64-Tumor-SM- MEL-IPI_Pat64-TP-NB-SM-
ARID1A 8289 1 27099940 27099940 Missense_ - 4DK2P 4DK2P-SM-4NFVV Mutation MEL-
IPI_Pat85-Tumor-SM- MEL-IPI_Pat85-TP-NB-SM- ARID1A 8289 1 27058033 27058033
Nonsense - 53U2Y 53U2Y-SM-4NFWH Mutation PR11217 PR11217 ARID1A 8289 1 27093053
27093053 Missense_ + Mutation PR4077 PR4077 ARID1A 8289 1 27089494 27089494 Missense_
+ Mutation Pt15 Pt15 ARID1A 8289 1 27101090 27101090 Intron + SU2C_Lung-SU2C-DFCI-
SU2C_Lung-SU2C-DFCI- ARID1A 8289 1 27101525 27101525 Frame_Shift_ + LUAD-1017-
Tumor-SM- LUAD-1017-TM-NB-SM- Del AOL99 AOL99-SM-A46NT SU2C_Lung-SU2C-
DFCI- SU2C_Lung-SU2C-DFCI- ARID1A 8289 1 27106878 27106878 Frame_Shift_ + LUAD-

1017-Tumor-SM- LUAD-1017-TM-NB-SM- Del AOL99 AOL99-SM-A46NT Y2087 Y2087
 ARID1A 8289 1 27023360 27023360 Missense_ ± Mutation ZA6965 ZA6965 ARID1A 8289 1
 27023690 27023690 Missense_ + Mutation BLCA-IM01-Tumor-SM- BLCA-IM01-TP-NB-SM-
 ARID1B 57492 6 157100005 157100005 Silent - 79XD9 79XD9-SM-7AABJ HNSCC-239-
 Tumor-SM- HNSCC-239-TP-NB-SM- ARID1B 57492 6 157100005 157100005 Silent ± AXGCS
 AXGCS-SM-ADP7K HNSCC-243-Tumor-SM- HNSCC-243-TP-NB-SM- ARID1B 57492 6
 157100005 157100005 Silent - CLFNS CLFNS-SM-AV34T LUAD-BS-11-R21845- LUAD-BS-
 11-R21845-TP- ARID1B 57492 6 157100005 157100005 Silent + Tumor-SM-9J2YH NT-SM-
 9J2YH-SM-9J2YI LUAD-BS-13-X14864- LUAD-BS-13-X14864-TP- ARID1B 57492 6
 157528372 157528372 Missense_ ± Tumor-SM-9J2XQ NB-SM-9J2XQ-SM-9HBZU Mutation
 LUAD-BS-14-G65174- LUAD-BS-14-G65174-TP- ARID1B 57492 6 157100005 157100005
 Silent - Tumor-SM-9J2YF NT-SM-9J2YF-SM-9J2YG MEL-682321-Tumor-SM- MEL-682321-
 TP-NB-SM- ARID1B 57492 6 157099512 157099512 Missense_ + CN21G CN21G-SM-CJP7S
 Mutation MEL-682321-Tumor-SM- MEL-682321-TP-NB-SM- ARID1B 57492 6 157099511
 157099511 Missense_ + CN21G CN21G-SM-CJP7S Mutation MEL-IPI_Pat132-Tumor-SM-
 MEL-IPI_Pat132-TP-NB-SM- ARID1B 57492 6 157517305 157517305 Missense_ + 5VWJA
 5VWJA-SM-5VWHR Mutation MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM-
 ARID1B 57492 6 157522597 157522597 Silent + 5VWJG 5VWJG-SM-5VWHX MEL-
 IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- ARID1B 57492 6 157527479 157527479
 Missense_ + 5VWJG 5VWJG-SM-5VWHX Mutation MEL-IPI_Pat138-Tumor-SM- MEL-
 IPI_Pat138-TP-NB-SM- ARID1B 57492 6 157502265 157502265 Silent + 5VWJG 5VWJG-SM-
 5VWHX MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- ARID1B 57492 6
 157511325 157511325 Silent + 5VWJG 5VWJG-SM-5VWHX MEL-IPI_Pat139-Tumor-SM-
 MEL-IPI_Pat139-TP-NB-SM- ARID1B 57492 6 157527627 157527627 Silent - 5VWJH 5VWJH-
 SM-5VWHY MEL-IPI_Pat139-Tumor-SM- MEL-IPI_Pat139-TP-NB-SM- ARID1B 57492 6
 157505558 157505558 Missense_ - 5VWJH 5VWJH-SM-5VWHY Mutation MEL-IPI_Pat174-
 Tumor-SM- MEL-IPI_Pat174-TP-NB-SM- ARID1B 57492 6 157528667 157528667 Missense_ +
 5VOB4 5VOB4-SM-5VWIY Mutation MEL-IPI_Pat21-Tumor-SM- MEL-IPI_Pat21-TP-NT-SM-
 ARID1B 57492 6 157522222 157522222 Missense_ + 4DK1H 4DK1H-SM-53U5G Mutation
 MEL-IPI_Pat39-Tumor-SM- MEL-IPI_Pat39-TP-NB-SM- ARID1B 57492 6 157511303
 157511303 Missense_ + 4DK1Z 4DK1Z-SM-4NFW6 Mutation MEL-IPI_Pat74-Tumor-SM- MEL-
 IPI_Pat74-TP-NB-SM- ARID1B 57492 6 157100377 157100377 Silent - 4DK2Z 4DK2Z-SM-
 4NFW6 MEL-IPI_Pat74-Tumor-SM- MEL-IPI_Pat74-TP-NB-SM- ARID1B 57492 6 157100376
 157100376 Missense_ - 4DK2Z 4DK2Z-SM-4NFW6 Mutation PR11217 PR11217 ARID1B
 57492 6 157522344 157522344 Missense_ + Mutation PR4092 PR4092 ARID1B 57492 6
 157521990 157521990 Missense_ + Mutation Pt8 Pt8 ARID1B 57492 6 157222594 157222594
 Missense_ + Mutation SA9755 SA9755 ARID1B 57492 6 157522508 157522508 Missense_ +
 Mutation SD1494 SD1494 ARID1B 57492 6 157522095 157522095 Missense_ ± Mutation
 SU2C_Lung-SU2C-DFCI- SU2C_Lung-SU2C-DFCI- ARID1B 57492 6 157100005 157100005
 Silent - LUAD-1006-Tumor-SM- LUAD-1006-TP-NB-SM- AOL5E AOL5E-SM-A46NI
 SU2C_Lung-SU2C-DFCI- SU2C_Lung-SU2C-DFCI- ARID1B 57492 6 157222621 157222621
 Missense_ - LUAD-1011-Tumor-SM- LUAD-1011-TM-NB-SM- Mutation AOL75 AOL75-SM-
 A46NN Y2087 Y2087 ARID1B 57492 6 157099481 157099481 Missense_ ± Mutation Case1-
 BaselineTumor Case1-TP-NB-Zaretsky ARID2 196528 12 46287234 46287234 Missense_ +
 Mutation Case3-BaselineTumor Case3-TP-NB-Zaretsky ARID2 196528 12 46243857 46243857
 Nonsense_ + Mutation HNSCC-323-Tumor-SM- HNSCC-323-TP-NB-SM- ARID2 196528 12
 46240638 46240638 Splice_Site + CK9WS CK9WS-SM-AV34N LSD6819 LSD6819 ARID2
 196528 12 46243857 46243857 Nonsense_ + Mutation LUAD-BS-08-013532- LUAD-BS-08-
 013532-TP-NT- ARID2 196528 12 46245525 46245525 Frame_Shift_ + Tumor-SM-9J2Y1 SM-
 9J2Y1-SM-9J2Y2 Del LUAD-BS-13-J60666- LUAD-BS-13-J60666-TP-NB- ARID2 196528 12

46246071 46246071 Missense_ + Tumor-SM-9J2YL SM-9JBZYL-SM-9HBJZYL Mutation MEL-
650366-Tumor-SM- MEL-650366-TP-NB-SM- ARID2 196528 12 46245857 46245857 Silent -
CN221 CN221-SM-CJP7U MEL-IPI_Pat100-Tumor-SM- MEL-IPI_Pat100-TP-NT-SM- ARID2
196528 12 46230641 46230641 Missense_ - 53U2D 53U2D-SM-53U4M Mutation MEL-
IPI_Pat100-Tumor-SM- MEL-IPI_Pat100-TP-NT-SM- ARID2 196528 12 46242701 46242701
Nonsense_ - 53U2D 53U2D-SM-53U4M Mutation MEL-IPI_Pat103-Tumor-SM- MEL-
IPI_Pat103-TP-NT-SM- ARID2 196528 12 46243514 46243514 Missense_ + 4CU6Q 4CU6Q-
SM-53U4P Mutation MEL-IPI_Pat109-Tumor-SM- MEL-IPI_Pat109-TP-NT-SM- ARID2 196528
12 46243825 46243825 Missense_ - 4CU6W 4CU6W-SM-4MGPN Mutation MEL-IPI_Pat109-
Tumor-SM- MEL-IPI_Pat109-TP-NT-SM- ARID2 196528 12 46243824 46243824 Missense_ -
4CU6W 4CU6W-SM-4MGPN Mutation MEL-IPI_Pat115-Tumor-SM- MEL-IPI_Pat115-TP-NT-
SM- ARID2 196528 12 46211474 46211474 Missense_ - 5X2QS 5X2QS-SM-5X2RA Mutation
MEL-IPI_Pat117-Tumor-SM- MEL-IPI_Pat117-TP-NT-SM- ARID2 196528 12 46244997
46244997 Nonsense_ + 5X2QU 5X2QU-SM-5X2RC Mutation MEL-IPI_Pat117-Tumor-SM-
MEL-IPI_Pat117-TP-NT-SM- ARID2 196528 12 46245843 46245843 Nonsense_ + 5X2QU
5X2QU-SM-5X2RC Mutation MEL-IPI_Pat132-Tumor-SM- MEL-IPI_Pat132-TP-NB-SM-
ARID2 196528 12 46243530 46243530 Missense_ + 5VWJA 5VWJA-SM-5VWHR Mutation
MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- ARID2 196528 12 46242749
46242749 Missense_ + 5VWJG 5VWJG-SM-5VWHX Mutation MEL-IPI_Pat139-Tumor-SM-
MEL-IPI_Pat139-TP-NB-SM- ARID2 196528 12 46205217 46205217 Missense_ - 5VWJH
5VWJH-SM-5VWHY Mutation MEL-IPI_Pat151-Tumor-SM- MEL-IPI_Pat151-TP-NB-SM-
ARID2 196528 12 46287240 46287240 Missense_ - 5VWJT 5VWJT-SM-5VWIB Mutation MEL-
IPI_Pat159-Tumor-SM- MEL-IPI_Pat159-TP-NB-SM- ARID2 196528 12 46244889 46244889
Nonsense_ - 5VWK2 5VWK2-SM-5VWIJ Mutation MEL-IPI_Pat174-Tumor-SM- MEL-
IPI_Pat174-TP-NB-SM- ARID2 196528 12 46215271 46215271 Splice_Site + 5VOB4 5VOB4-
SM-5VWIY MEL-IPI_Pat58-Tumor-SM- MEL-IPI_Pat58-TP-NB-SM- ARID2 196528 12
46245648 46245648 Missense_ - 4DK2J 4DK2J-SM-4NFVP Mutation MEL-IPI_Pat58-Tumor-
SM- MEL-IPI_Pat58-TP-NB-SM- ARID2 196528 12 46240672 46240672 Missense_ - 4DK2J
4DK2J-SM-4NFVP Mutation MEL-IPI_Pat58-Tumor-SM- MEL-IPI_Pat58-TP-NB-SM- ARID2
196528 12 46287428 46287428 Missense_ - 4DK2J 4DK2J-SM-4NFVP Mutation MEL-
IPI_Pat66-Tumor-SM- MEL-IPL_Pat66-TP-NB-SM- ARID2 196528 12 46230691 46230691
Missense_ + 4DK2R 4DK2R-SM-4NFVX Mutation PR11217 PR11217 ARID2 196528 12
46233249 46233249 Nonsense_ + Mutation PR11217 PR11217 ARID2 196528 12 46245639
46245639 Nonsense_ + Mutation PR4077 PR4077 ARID2 196528 12 46243857 46243857
Nonsense_ + Mutation PR4092 PR4092 ARID2 196528 12 46242619 46242619 Splice_Site + Pt1
Pt1 ARID2 196528 12 46123846 46123846 Missense_ - Mutation Pt31 Pt31 ARID2 196528 12
46243467 46243467 Missense_ - Mutation Pt37 Pt37 ARID2 196528 12 46211600 46211600
Frame_Shift_ + Del SU2C_Lung-SU2C-DFCI- SU2C_Lung-SU2C-DFCI- ARID2 196528 12
46244393 46244393 Silent ± LUAD-1016-Tumor-SM- LUAD-1016-TM-NB-SM- AOL8W
AOL8W-SM-A46NS WA7899 WA7899 ARID2 196528 12 46244529 46244529 Missense_ -
Mutation DM123062 DM123062 BRD7 29117 16 50384049 50384049 Missense_ - Mutation
Lung-DFCI-11-104-009- Lung-DFCI-11-104-009-TM- BRD7 29117 16 50388348 50388348
Missense_ + Tumor-SM-5YS7O NB-SM-5YS7O-SM-5YS7P Mutation MEL-IPI_Pat03-Tumor-
SM- MEL-IPI_Pat03-TP-NB-SM- BRD7 29117 16 50357497 50357497 Splice_Site - 4DJZY
4DJZY-SM-4NFU5 MEL-IPI_Pat110-Tumor-SM- MEL-IPI_Pat110-TP-NT-SM- BRD7 29117 16
50368748 50368748 Missense_ - 4CU6X 4CU6X-SM-4MGPO Mutation MEL-IPI_Pat03-Tumor-
SM- MEL-IPI_Pat03-TP-NB-SM- DPF1 8193 19 38706825 38706825 Missense_ - 4DJZY
4DJZY-SM-4NFU5 Mutation MEL-IPI_Pat132-Tumor-SM- MEL-IPI_Pat132-TP-NB-SM- DPF1
8193 19 38709622 38709622 Silent + 5VWJA 5VWJA-SM-5VWHR MEL-IPI_Pat134-Tumor-
SM- MEL-IPI_Pat134-TP-NB-SM- DPF1 8193 19 38709646 38709646 Silent - 7A151 7A151-

SM-5VWHT MEL-IPI_Pat138-Tumor-SM- TP-NB-SM- DPF1 8193 19
38702995 38702995 Silent + 5VWJG 5VWJG-SM-5VWHX MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- DPF1 8193 19 38704352 38704352 Missense_ + 5VWJG 5VWJG-SM-5VWHX Mutation MEL-IPI_Pat74-Tumor-SM- MEL-IPI_Pat74-TP-NB-SM- DPF1 8193 19
38712998 38712998 Splice_Site - 4DK2Z 4DK2Z-SM-4NFW6 Pt27 Pt27 DPF1 8193 19
38709621 38709621 Missense_ + Mutation LUAD-BS-12-R10269- LUAD-BS-12-R10269-TP- DPF2 5977 11 65108997 65108997 Silent + Tumor-SM-9J2XO NB-SM-9J2XO-SM-9HBZT MEL-IPI_Pat132-Tumor-SM- MEL-IPI_Pat132-TP-NB-SM- DPF2 5977 11 65107914 65107914 Missense_ + 5VWJA 5VWJA-SM-5VWHR Mutation MEL-IPI_Pat159-Tumor-SM- MEL-IPI_Pat159-TP-NB-SM- DPF2 5977 11 65113741 65113741 Intron - 5VWK2 5VWK2-SM-5VWIJ MEL-IPI_Pat32-Tumor-SM- MEL-IPI_Pat32-TP-NT-SM- DPF2 5977 11 65108462 65108462 Silent - 53U3T 53U3T-SM-53U67 MEL-IPI_Pat38-Tumor-SM- MEL-IPI_Pat38-TP-NT-SM- DPF2 5977 11 65123565 65123565 IGR + 53U3Z 53U3Z-SM-53U5L MEL-IPI_Pat58-Tumor-SM- MEL-IPI_Pat58-TP-NB-SM- DPF2 5977 11 65113530 65113530 Intron - 4DK2J 4DK2J-SM-4NFVP MEL-IPI_Pat74-Tumor-SM- MEL-IPI_Pat74-TP-NB-SM- DPF2 5977 11 65113812 65113812 Intron - 4DK2Z 4DK2Z-SM-4NFW6 Pt13 Pt13 DPF2 5977 11 65111304 65111304 Intron + Pt14 Pt14 DPF2 5977 11 65109007 65109007 Missense_ - Mutation SU2C_Lung-SU2C-DFCI- SU2C_Lung-SU2C-DFCI- DPF2 5977 11 65113251 65113251 Intron - LUAD-1011-Tumor-SM- LUAD-1011-TM-NB-SM- AOL75 AOL75-SM-A46NN DFCI_MM_2-Tumor-SM- DFCI_MM_2-TP-NB-SM- DPF3 8110 14 73137945 73137945 Intron + BZRJA BZRJA-SM-BZRJD DFCI_MM_2-Tumor-SM- DFCI_MM_2-TP-NB-SM- DPF3 8110 14 73238507 73238507 Missense_ + BZRJA BZRJA-SM-BZRJD Mutation FR9547 FR9547 DPF3 8110 14 73140993 73140993 Missense_ + Mutation Lung-DFCI-11-104-009- Lung-DFCI-11-104-009-TM- DPF3 8110 14 73137964 73137964 Intron + Tumor-SM-5YS7O NB-SM-5YS7O-SM-5YS7P MEL-IPI_Pat110-Tumor-SM- MEL-IPI_Pat110-TP-NT-SM- DPF3 8110 14 73137905 73137905 Intron - 4CU6X 4CU6X-SM-4MGPO MEL-IPI_Pat110-Tumor-SM- MEL-IPI_Pat110-TP-NT-SM- DPF3 8110 14 73137904 73137904 Intron - 4CU6X 4CU6X-SM-4MGPO MEL-IPI_Pat139-Tumor-SM- MEL-IPI_Pat139-TP-NB-SM- DPF3 8110 14 73138006 73138006 Intron - 5VWJH 5VWJH-SM-5VWHY MEL-IPI_Pat139-Tumor-SM- MEL-IPI_Pat139-TP-NB-SM- DPF3 8110 14 73138005 73138005 Intron - 5VWJH 5VWJH-SM-5VWHY MEL-IPI_Pat16-Tumor-SM- MEL-IPI_Pat16-TP-NT-SM- DPF3 8110 14 73220050 73220050 Missense_ - 53U3E 53U3E-SM-53U5B Mutation MEL-IPI_Pat21-Tumor-SM- MEL-IPI_Pat21-TP-NT-SM- DPF3 8110 14 73190370 73190370 Missense_ + 4DK1H 4DK1H-SM-53U5G Mutation SA9755 SA9755 DPF3 8110 14 73190391 73190391 Missense_ + Mutation BLCA-IM07-Tumor-SM- BLCA-IM07-TP-NB-SM- PBRM1 55193 3 52621431 52621431 Missense_ + 79XDD 79XDD-SM-7AABP Mutation CR04885 CR04885 PBRM1 55193 3 52597336 52597336 Missense_ + Mutation M4945 M4945 PBRM1 55193 3 52696148 52696148 Splice_Site + MA7027 MA7027 PBRM1 55193 3 52598231 52598231 Missense_ - Mutation MEL-IPI_Pat103-Tumor-SM- MEL-IPI_Pat103-TP-NT-SM- PBRM1 55193 3 52620592 52620592 Missense_ + 4CU6Q 4CU6Q-SM-53U4P Mutation MEL-IPI_Pat103-Tumor-SM- MEL-IPI_Pat103-TP-NT-SM- PBRM1 55193 3 52620593 52620593 Missense_ + 4CU6Q 4CU6Q-SM-53U4P Mutation MEL-IPI_Pat118-Tumor-SM- MEL-IPI_Pat118-TP-NT-SM- PBRM1 55193 3 52692325 52692325 Missense_ - 5X2QV 5X2QV-SM-5X2RD Mutation MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- PBRM1 55193 3 52595959 52595959 Missense_ + 5VWJG 5VWJG-SM-5VWHX Mutation MEL-IPI_Pat151-Tumor-SM- MEL-IPI_Pat151-TP-NB-SM- PBRM1 55193 3 52643530 52643530 Missense_ - 5VWJT 5VWJT-SM-5VWIB Mutation MEL-IPI_Pat38-Tumor-SM- MEL-IPI_Pat38-TP-NT-SM- PBRM1 55193 3 52643692 52643692 Missense_ + 53U3Z 53U3Z-SM-53U5L Mutation MEL-IPI_Pat70-Tumor-SM- MEL-IPI_Pat70-TP-NB-SM- PBRM1 55193 3 52584527 52584527 Missense_ - 4DK2V 4DK2V-SM-4NFW2 Mutation MEL-IPI_Pat79-Tumor-SM- MEL-IPI_Pat79-TP-NB-SM- PBRM1 55193 3 52621315 52621315 Intron + 53U2S 53U2S-SM-4NFWB MEL-

IPI_Pat88-Tumor-SM- MEL-IPI_Pat88-TP-NT-SM- PBRM1 55193 3 52668815 52668815 Silent +
 4DK3E 4DK3E-SM-53U4C MEL-IPI_Pat88-Tumor-SM- MEL-IPI_Pat88-TP-NT-SM- PBRM1
 55193 3 52668765 52668765 Missense_ + 4DK3E 4DK3E-SM-53U4C Mutation PR4035 PR4035
 PBRM1 55193 3 52643768 52643768 Nonsense_ + Mutation Pt13 Pt13 PBRM1 55193 3
 52643768 52643768 Nonsense_ + Mutation SU2C_Lung-SU2C-DFCI- SU2C_Lung-SU2C-DFCI-
 PBRM1 55193 3 52651406 52651406 Nonsense_ + LUAD-1017-Tumor-SM- LUAD-1017-TM-
 NB-SM- Mutation AOL99 AOL99-SM-A46NT MEL-IPI_Pat110-Tumor-SM- MEL-IPI_Pat110-
 TP-NT-SM- PHF10 55274 6 170112483 170112483 Splice_Site - 4CU6X 4CU6X-SM-4MGPO
 MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- PHF10 55274 6 170116103
 170116103 Missense_ + 5VWJG 5VWJG-SM-5VWHX Mutation MEL-IPI_Pat58-Tumor-SM-
 MEL-IPI_Pat58-TP-NB-SM- PHF10 55274 6 170117919 170117919 Splice_Site - 4DK2J 4DK2J-
 SM-4NFVP Pt1 Pt1 PHF10 55274 6 170112579 170112579 Missense_ - Mutation Pt2 Pt2 PHF10
 55274 6 170116131 170116131 Missense_ + Mutation SD1494 SD1494 PHF10 55274 6
 170117924 170117924 Missense_ ± Mutation BLCA-IM10-Tumor-SM- BLCA-IM10-TP-NB-SM-
 SMARCA2 6595 9 2181573 2181573 Missense_ + 79XDG 79XDG-SM-9QSPX Mutation BLCA-
 IM11-Tumor-SM- BLCA-IM11-TP-NT-SM- SMARCA2 6595 9 2033008 2033008 Missense_ +
 79XDH 79XDH-SM-79XDI Mutation LSD0167 LSD0167 SMARCA2 6595 9 2161819 2161819
 Missense_ + Mutation MEL-IPI_Pat132-Tumor-SM- MEL-IPI_Pat132-TP-NB-SM- SMARCA2
 6595 9 2039901 2039901 Splice_Site + 5VWJA 5VWJA-SM-5VWHR MEL-IPI_Pat132-Tumor-
 SM- MEL-IPI_Pat132-TP-NB-SM- SMARCA2 6595 9 2186134 2186134 Silent + 5VWJA
 5VWJA-SM-5VWHR MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- SMARCA2
 6595 9 2056722 2056722 Silent + 5VWJG 5VWJG-SM-5VWHX MEL-IPI_Pat15-Tumor-SM-
 MEL-IPI_Pat15-TP-NB-SM- SMARCA2 6595 9 2070473 2070473 Splice_Site - 4DK1B 4DK1B-
 SM-4NFUH MEL-IPI_Pat21-Tumor-SM- MEL-IPI_Pat21-TP-NT-SM- SMARCA2 6595 9
 2161845 2161845 Missense_ + 4DK1H 4DK1H-SM-53U5G Mutation MEL-IPI_Pat38-Tumor-
 SM- MEL-IPI_Pat38-TP-NT-SM- SMARCA2 6595 9 2039623 2039623 Silent + 53U3Z 53U3Z-
 SM-53U5L MEL-IPI_Pat90-Tumor-SM- MEL-IPI_Pat90-TP-NB-SM- SMARCA2 6595 9
 2186142 2186142 Missense_ + 4DK3G 4DK3G-SM-4NFWM Mutation PR4092 PR4092
 SMARCA2 6595 9 2161836 2161836 Missense_ + Mutation Pt31 Pt31 SMARCA2 6595 9
 2104046 2104046 Missense_ - Mutation RH090935 RH090935 SMARCA2 6595 9 2123911
 2123911 Missense_ + Mutation SD2056 SD2056 SMARCA2 6595 9 2077654 2077654 Missense_
 + Mutation Y2087 Y2087 SMARCA2 6595 9 2047355 2047355 Missense_ ± Mutation BLCA-
 IM07-Tumor-SM- BLCA-IM07-TP-NB-SM- SMARCA4 6597 19 11134267 11134267 Missense_
 + 79XDD 79XDD-SM-7AABP Mutation BLCA-IM09-Tumor-SM- BLCA-IM09-TP-NB-SM-
 SMARCA4 6597 19 11097617 11097617 Missense_ + 79XDF 79XDF-SM-7AABN Mutation
 FR9547 FR9547 SMARCA4 6597 19 11136975 11136975 Splice_Site + HNSCC-186-Tumor-SM-
 HNSCC-186-TP-NB-SM- SMARCA4 6597 19 11144853 11144853 3'UTR + AXGDN AXGDN-
 SM-ADP7L HNSCC-186-Tumor-SM- HNSCC-186-TP-NB-SM- SMARCA4 6597 19 11144853
 11144853 3'UTR + AXGDN AXGDN-SM-ADP7L HNSCC-258-Tumor-SM- HNSCC-258-TP-
 NB-SM- SMARCA4 6597 19 11170556 11170556 Missense_ + AXGAI AXGAI-SM-ADP7G
 Mutation HNSCC-323-Tumor-SM- HNSCC-323-TP-NB-SM- SMARCA4 6597 19 11096069
 11096069 Nonsense_ + CK9WS CK9WS-SM-AV34N Mutation LUAD-BS-13-J60666- LUAD-
 BS-13-J60666-TP-NB- SMARCA4 6597 19 11132428 11132428 Missense_ + Tumor-SM-9J2YL
 SM-9J2YL-SM-9HBZW Mutation LUAD-BS-14-G65174- LUAD-BS-14-G65174-TP-
 SMARCA4 6597 19 11145805 11145805 3'UTR - Tumor-SM-9J2YF NT-SM-9J2YF-SM-9J2YG
 M4945 M4945 SMARCA4 6597 19 11144072 11144072 IGR + MA7027 MA7027 SMARCA4
 6597 19 11134207 11134207 Missense_ - Mutation MEL-IPI_Pat08-Tumor-SM- MEL-IPI_Pat08-
 TP-NB-SM- SMARCA4 6597 19 11121151 11121151 Missense_ - 4DK14 4DK14-SM-4NFUA
 Mutation MEL-IPI_Pat110-Tumor-SM- MEL-IPI_Pat110-TP-NT-SM- SMARCA4 6597 19
 11096986 11096986 Silent - 4CU6X 4CU6X-SM-4MGPO MEL-IPI_Pat110-Tumor-SM- MEL-

IPI_Pat110-TP-NT-SM- SMARCA4 6597 19 11134230 11134230 Missense_ - 4CU6X 4CU6X-SM-4MGPO Mutation MEL-IPI_Pat132-Tumor-SM- MEL-IPI_Pat132-TP-NB-SM- SMARCA4 6597 19 11144106 11144106 IGR + 5VWJA 5VWJA-SM-5VWHR MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- SMARCA4 6597 19 11144028 11144028 IGR + 5VWJG 5VWJG-SM-5VWHX MEL-IPI_Pat139-Tumor-SM- MEL-IPI_Pat139-TP-NB-SM- SMARCA4 6597 19 11136986 11136986 Missense_ - 5VWJH 5VWJH-SM-5VWHY Mutation MEL-IPI_Pat16-Tumor-SM- MEL-IPI_Pat16-TP-NT-SM- SMARCA4 6597 19 11141561 11141561 IGR - 53U3E 53U3E-SM-53U5B MEL-IPI_Pat19-Tumor-SM- MEL-IPI_Pat19-TP-NB-SM- SMARCA4 6597 19 11144856 11144856 3'UTR - 4DK1F 4DK1F-SM-4NFUL MEL-IPI_Pat28-Tumor-SM- MEL-IPI_Pat28-TP-NB-SM- SMARCA4 6597 19 11137018 11137018 Nonsense_ - 4DK1O 4DK1O-SM-4NFUU Mutation MEL-IPI_Pat49-Tumor-SM- MEL-IPI_Pat49-TP-NT-SM- SMARCA4 6597 19 11121110 11121110 Missense_ ± 4DK2A 4DK2A-SM-53U5W Mutation MEL-IPI_Pat52-Tumor-SM- MEL-IPI_Pat52-TP-NT-SM- SMARCA4 6597 19 11097614 11097614 Missense_ - 4DK2D 4DK2D-SM-53U5Z Mutation MEL-IPI_Pat54-Tumor-SM- MEL-IPI_Pat54-TP-NB-SM- SMARCA4 6597 19 11100047 11100047 Silent - 4DK2F 4DK2F-SM-4NFVL MEL-IPI_Pat58-Tumor-SM- MEL-IPI_Pat58-TP-NB-SM- SMARCA4 6597 19 11098595 11098595 Silent - 4DK2J 4DK2J-SM-4NFVP Pt31 Pt31 SMARCA4 6597 19 11170804 11170804 Nonsense_ - Mutation SA9755 SA9755 SMARCA4 6597 19 11123685 11123685 Missense_ + Mutation SD1494 SD1494 SMARCA4 6597 19 11118614 11118614 Missense_ ± Mutation HNSCC-181-Tumor-SM- HNSCC-181-TP-NB-SM- SMARCB1 6598 22 24129440 24129440 Silent - CK9X1 CK9X1-SM-AV34P MEL-IPI_Pat110-Tumor-SM- MEL-IPI_Pat110-TP-NT-SM- SMARCB1 6598 22 24143148 24143148 Intron - 4CU6X 4CU6X-SM-4MGPO MEL-IPI_Pat110-Tumor-SM- MEL-IPI_Pat110-TP-NT-SM- SMARCB1 6598 22 24143149 24143149 Intron - 4CU6X 4CU6X-SM-4MGPO MEL-IPI_Pat130-Tumor-SM- MEL-IPI_Pat130-TP-NT-SM- SMARCB1 6598 22 24143281 24143281 Intron - 5X2R8 5X2R8-SM-5X2RJ MEL-IPI_Pat62-Tumor-SM- MEL-IPI_Pat62-TP-NB-SM- SMARCB1 6598 22 24145537 24145537 Silent - 4DK2N 4DK2N-SM-4NFVT Pt2 Pt2 SMARCB1 6598 22 24133958 24133958 Missense_ + Mutation Lung-DFCI-11-104-009- Lung-DFCI-11-104-009-TM- SMARCC1 6599 3 47823230 47823230 Missense_ + Tumor-SM-5YS7O NB-SM-5YS7O-SM-5YS7P Mutation MEL-IPI_Pat03-Tumor-SM- MEL-IPI_Pat03-TP-NB-SM- SMARCC1 6599 3 47680267 47680267 Missense_ - 4DJZY 4DJZY-SM-4NFU5 Mutation MEL-IPI_Pat03-Tumor-SM- MEL-IPI_Pat03-TP-NB-SM- SMARCC1 6599 3 47787455 47787455 Missense_ - 4DJZY 4DJZY-SM-4NFU5 Mutation MEL-IPI_Pat08-Tumor-SM- MEL-IPI_Pat08-TP-NB-SM- SMARCC1 6599 3 47777539 47777539 Silent - 4DK14 4DK14-SM-4NFUA MEL-IPI_Pat110-Tumor-SM- MEL-IPI_Pat110-TP-NT-SM- SMARCC1 6599 3 47632172 47632172 Silent - 4CU6X 4CU6X-SM-4MGPO MEL-IPI_Pat151-Tumor-SM- MEL-IPI_Pat151-TP-NB-SM- SMARCC1 6599 3 47787430 47787430 Silent - 5VWJT 5VWJT-SM-5VWIB MEL-IPI_Pat28-Tumor-SM- MEL-IPI_Pat28-TP-NB-SM- SMARCC1 6599 3 47742863 47742863 Missense_ - 4DK1O 4DK1O-SM-4NFUU Mutation MEL-IPI_Pat58-Tumor-SM- MEL-IPI_Pat58-TP-NB-SM- SMARCC1 6599 3 47651680 47651680 Silent - 4DK2J 4DK2J-SM-4NFVP MEL-IPI_Pat58-Tumor-SM- MEL-IPI_Pat58-TP-NB-SM- SMARCC1 6599 3 47770567 47770567 Silent - 4DK2J 4DK2J-SM-4NFVP Pt4 Pt4 SMARCC1 6599 3 47629788 47629788 Missense_ + Mutation CR04885 CR04885 SMARCC2 6601 12 56565627 56565627 Missense_ + Mutation MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- SMARCC2 6601 12 56558459 56558459 Missense_ + 5VWJG 5VWJG-SM-5VWHX Mutation MEL-IPI_Pat139-Tumor-SM- MEL-IPI_Pat139-TP-NB-SM- SMARCC2 6601 12 56558475 56558475 Missense_ - 5VWJH 5VWJH-SM-5VWHY Mutation MEL-IPI_Pat27-Tumor-SM- MEL-IPI_Pat27-TP-NB-SM- SMARCC2 6601 12 56578857 56578857 Missense_ - 4DK1N 4DK1N-SM-4NFUT Mutation MEL-IPI_Pat32-Tumor-SM- MEL-IPI_Pat32-TP-NT-SM- SMARCC2 6601 12 56572223 56572223 Silent - 53U3T 53U3T-SM-53U67 MEL-IPI_Pat58-Tumor-SM- MEL-IPI_Pat58-TP-NB-SM- SMARCC2 6601 12 56563668 56563668 Missense_ - 4DK2J 4DK2J-SM-

4NFVP Mutation MEL-IPI_Pat64-Tumor-SM- MEL-IPI_Pat64-TP-NB-SM- SMARCC2 6601 12 56563227 56563230 Intron – 4DK2P 4DK2P-SM-4NFVV MEL-IPI_Pat71-Tumor-SM- MEL-IPI_Pat71-TP-NB-SM- SMARCC2 6601 12 56566475 56566475 Missense_ – 4DK2W 4DK2W-SM-4NFW3 Mutation MEL-IPI_Pat77-Tumor-SM- MEL-IPI_Pat77-TP-NT-SM- SMARCC2 6601 12 56575309 56575309 Missense_ + 4DK33 4DK33-SM-53U63 Mutation MEL-IPI_Pat77-Tumor-SM- MEL-IPI_Pat77-TP-NT-SM- SMARCC2 6601 12 56575308 56575308 Missense_ + 4DK33 4DK33-SM-53U63 Mutation MEL-IPI_Pat62-Tumor-SM- MEL-IPI_Pat62-TP-NB-SM- SMARCD1 6602 12 50480624 50480624 Missense_ – 4DK2N 4DK2N-SM-4NFVT Mutation Pt37 Pt37 SMARCD1 6602 12 50484135 50484135 Nonsense_ + Mutation LUAD-BS-13-J60666-LUAD-BS-13-J60666-TP-NB- SMARCD2 6603 17 61912836 61912836 Missense_ + Tumor-SM-9J2YL SM-9J2YL-SM-9HBZW Mutation MEL-IPI_Pat119-Tumor-SM- MEL-IPI_Pat119-TP-NT-SM- SMARCD2 6603 17 61914856 61914856 Nonsense_ – 7459N 7459N-SM-7459Q Mutation MEL-IPI_Pat119-Tumor-SM- MEL-IPI_Pat119-TP-NT-SM- SMARCD2 6603 17 61914857 61914857 Silent – 7459N 7459N-SM-7459Q MEL-IPI_Pat151-Tumor-SM- MEL-IPI_Pat151-TP-NB-SM- SMARCD2 6603 17 61912922 61912922 Silent – 5VWJT 5VWJT-SM-5VWIB MEL-IPI_Pat21-Tumor-SM- MEL-IPI_Pat21-TP-NT-SM- SMARCD2 6603 17 61911039 61911039 Missense_ + 4DK1H 4DK1H-SM-53U5G Mutation MEL-IPI_Pat38-Tumor-SM- MEL-IPI_Pat38-TP-NT-SM- SMARCD2 6603 17 61914827 61914827 Silent + 53U3Z 53U3Z-SM-53U5L Case3-BaselineTumor Case3-TP-NB-Zaretsky SMARCD3 6604 7 150939235 150939235 Silent + MEL-IPI_Pat11-Tumor-SM- MEL-IPI_Pat11-TP-NB-SM- SMARCD3 6604 7 150939045 150939045 Missense_ – 4DK17 4DK17-SM-4NFUD Mutation BLADDER- BLADDER- SMARCE1 6605 17 38793628 38793632 Intron + 15330_CCPM_0700694- 15330_CCPM_0700694-TM- Tumor-SM-AVI16 NB-SM-AVI16-SM-AVHZK MEL-IPI_Pat123-Tumor-SM- MEL-IPI_Pat123-TP-NB-SM- SMARCE1 6605 17 38787856 38787856 Silent + 5X2R1 5X2R1-SM-5VWHL MEL-IPI_Pat132-Tumor-SM- MEL-IPI_Pat132-TP-NB-SM- SMARCE1 6605 17 38788513 38788513 Silent + 5VWJA 5VWJA-SM-5VWHR MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- SMARCE1 6605 17 38787103 38787103 Missense_ + 5VWJG 5VWJG-SM-5VWHX Mutation MEL-IPI_Pat138-Tumor-SM- MEL-IPI_Pat138-TP-NB-SM- SMARCE1 6605 17 38785098 38785098 Missense_ + 5VWJG 5VWJG-SM-5VWHX Mutation MEL-IPI_Pat70-Tumor-SM- MEL-IPI_Pat70-TP-NB-SM- SMARCE1 6605 17 38792665 38792665 Silent – 4DK2V 4DK2V-SM-4NFW2

(395) All samples in Table 4 were from broad.mit.edu with NCBI-build no. of 37 and can be further identified based on the following information: ##Oncotator v1.2.7.0 Flat File Reference hg19|GENCODE v19|UnProt_AAxfom 2011_09|ClinVar 12.03.20|ESP 6500SI-V2|ORegAnno UCSC Track|dbSNP build 134|CCLE_By_GP 09292010|COSMIC v62_291112|1000Genome phase1|UniProt_AA 2011_09|dbNSFP v2.4|ESP 6500SI-V2|COSMIC_FusionGenes v62_291112|gencode_xref_refseq metadata v19|CCLE_By_Gene 09292010|ACHILLES_Lineage_Results 110303|CGC full_2012-03-15|UniProt 2011_09|HumanDNARepairGenes 20110905|HGNC Nov2013|COSMIC_Tissue 291112|Familial_Cancer_Genes 20110905|TUMORScape 20100104|Ensembl ICGC MUCOPA|TCGAScape 110405|MutSig Published Results 20110905.

(396) For responses, “+” represents having clinical benefit; “±” represents having intermediate benefit; and “–” represents having no clinical benefit.

(397) TABLE-US-00006 TABLE 5 Hugo_Symbol n_cb_truncating n_ncb_truncating n_truncating n_cb_nonsyn n_ncb_nonsyn n_nonsyn ARID1A 3 6 11 9 14 27 SMARCE1 0 0 0 2 0 2 ARID1B 0 0 0 11 3 17 SMARCA4 2 2 4 7 7 16 PBRM1 4 0 4 11 4 15 SMARCA2 1 1 2 9 2 12 ARID2 12 2 14 18 14 32 SMARCD3 0 0 0 0 1 1 ACTL6B 0 1 2 6 3 10 SMARCC2 0 0 0 4 4 8 DPF3 0 0 0 4 1 5 BRD7 0 1 1 1 3 4 SMARCB1 0 0 0 1 0 1 DPF2 0 0 0 1 1 2 SMARCD2 0 1 1 2 1 3 SMARCC1 0 0 0 2 3 5 DPF10 1 1 2 2 4 PHF10 0 2 2 2 3 6 ACTL6A 0 0 0 2 1 3 SMARCD1 1 0 1 1 1 2

(398) All samples in Table 5 were taken from 98 patients with clinical benefit and 132 patients with

no clinical benefit from immune checkpoint therapy “n_cb_truncating” refers to the total number of patients with truncating mutation in a given gene with clinical benefit from immune checkpoint therapy; “n_ncb_truncating” refers to the total number of patients with truncating mutation in a given gene with no clinical benefit from immune checkpoint therapy; “n_truncating” refers to the total number of truncating mutations in a given gene in the cohort (includes patients with intermediate clinical benefit); “n_cb_nonsyn” refers to the total number of patients with nonsynonymous mutation in a given gene with clinical benefit from immune checkpoint therapy; “n_ncb_nonsyn” refers to the total number of patients with nonsynonymous mutation in a given gene with no clinical benefit from immune checkpoint therapy; and “n_nonsyn” refers to the total number of nonsynonymous mutations in a given gene in the cohort (includes patients with intermediate clinical benefit).

(399) A summary of SWI/SNF complex is illustrated in FIG. 9. ARID2 and PBRM1 are two representative genes in the SWI/SNF complex, which were found in this study as relevant to sensitivity to immunotherapies such as those antagonizing immune checkpoints. SMARCA2 (also known as BRM) and ARID1B did not pass cohort-wide MutSig significance, but were mutated significantly more often in responders vs. non-responders (FIG. 10).

(400) Alterations in PBRM1 are a common driver in clear-cell renal cell carcinoma (up to 40%), where it has a tumor suppressor function, but are rarer in other cancer types. This cohort contained 14 patients (8.3%) with nonsynonymous mutations in PBRM1 and 4 patients (1.5%, 2 with melanoma and 2 with non-small-cell lung cancer) with truncating alterations.

(401) Similarly, ARID2 is a common driver mutation in hepatocellular carcinoma and melanoma. This cohort contained 26 patients (15.4%) with nonsynonymous mutations in ARID2 and 12 (7.1%, 10 with melanoma, 1 with head and neck squamous cell carcinoma, and 1 with non-small-cell lung cancer) with truncating alterations. Truncating (but not nonsynonymous) mutations in ARID2 were significantly associated with clinical benefit vs. no clinical benefit after controlling for nonsynonymous mutational load ($p=0.0051$; logistic regression) (FIG. 11). Nonsynonymous alterations in PBRM1 were marginally associated with clinical benefit ($p=0.058$) after controlling for mutational burden, while truncating mutations were not ($p=0.98$), perhaps due to the relative rarity of these events.

(402) KDM6A encodes an enzyme called lysine-specific demethylase 6A that functions as a histone demethylase (FIG. 12). Truncating alterations in KDM6A were seen in 8 patients (4.8%, 5 with bladder cancer and 3 with melanoma) in this cohort. Truncating (but not nonsynonymous) alterations in KDM6A were marginally associated with clinical benefit ($p=0.089$; logistic regression) after controlling for mutational burden.

(403) Immune checkpoint therapies can yield durable responses and long-lasting survival benefit across many cancer types, and checkpoint therapies have been approved for use in metastatic melanoma, non-small cell lung cancer, bladder cancer, and renal cell carcinoma, including as a first-line therapy for lung cancer. While past studies have highlighted mutational load, neoantigen presentation, transcriptomic signatures, microbiome features, and immune cell infiltration as correlated with response to immune checkpoint therapies in melanoma, non-small-cell lung cancer, and bladder cancer, the results described herein indicate that nonsynonymous alterations in the SWI/SNF chromatin remodeling complex has predictive value for patient response to immune checkpoint therapies. Moreover, other biomarkers described herein, such as additional chromatin modifying genes like KDM6A and EGFR (resistance) biomarkers, were identified. In particular, EGFR showed a strong trend with intrinsic resistance to immune checkpoint therapy in lung cancer (FIG. 6). In addition, as described further in Example 4 below, cancers with hotspot mutations in EGFR are significantly less likely to respond to immune checkpoint therapies.

(404) Thus, these results are believed to have wide-ranging implications for patient stratification for immune checkpoint therapies and those treated with other therapies, such as EGFR signaling inhibitors. Additionally, this finding drew from the largest set of clinically annotated cancer types

yet collected (>200 pre-treatment patient tumors) across both well-studied and more poorly understood cancer types, lending great statistical power to detect associations. Finally, these results provide biomarkers, drug design, and combination treatment strategies across cancer types.

Example 3: Meta-Analysis of Genomic Predictors of Response to Immune Checkpoint Therapy in Metastatic Melanoma

(405) Since immune checkpoint therapies only benefit a subset of patients with metastatic melanoma and the ability to predict clinical outcomes is limited, a meta-analysis of genomic predictors of outcomes to anti-PD1 blockade and anti-CTLA4 blockade in melanoma combining 220 sequenced tumors from 3 published cohorts was conducted in order to validate existing hypotheses regarding response to immune checkpoint therapies and discover new relationships with greater power.

(406) Nonsynonymous mutational burden was significantly higher in clinical benefit (CB) vs. no clinical benefit (NCB) using all 3 response metrics, though the significance was less pronounced when using PFS alone ($p < 0.01$ vs. $p < 0.0001$; Wilcoxon rank sum), partially due to 3 patients with high mutational burden who experienced PR lasting <6 months, potentially representing early acquired rather than intrinsic resistance. In order to assess the impact of mutational processes contributing to overall mutational burden, a non-negative matrix factorization framework was used to infer mutational activity in tumors from 6 signatures previously seen in melanoma: aging (S1), T>C substitutions (S5), UV (S7), mismatch repair (S6), alkylating agents (S11), and T>G substitutions (S17). Across all samples, the proportion of mutations in S7 or S11 was positively correlated with mutational burden (Spearman's $\rho = 0.66$), while S5 and S1 were anti-correlated ($\rho = -0.62$). Additionally, in a multivariate logistic model, S7 and S11 activity were independent predictors of clinical benefit adjusting for mutational load ($p < 0.05$), with the sum of S7 and S11 activity being a strong predictor ($p < 0.001$). Of the patients with low mutational burden (<median) with CB, 79% had >1/2 of mutations in S7 or S11, compared to only 51% of NCB ($p < 0.01$; Pearson's chi-squared). Neoantigen burden was strongly correlated with mutational burden, and did not improve ability to predict CB. In examining mutations in specific genes, >500 genes were mutated significantly more frequently in CB or NCB ($p < 0.05$, Fisher's exact). Restricting analysis to recurrently mutated genes in cancer and correcting for patient mutational burden by permutation, nonsynonymous mutations in ACSL3 and MET and truncating alterations in ARID2 were significantly enriched in CB.

(407) In this meta-analysis of 220 patients, harmonized clinical and whole exome analysis confirmed that mutational burden correlates with CB from anti-PD1 and anti-CTLA4 therapy, with mutational signatures and alterations in specific genes potentially providing additional predictive power.

Example 4: SU2C Cohort Study for Lung Cancer Immunotherapy

(408) Analyses were also performed for a cohort of patients receiving lung cancer immunotherapy. For these patients with metastatic lung cancer treated with anti-PD1/PD-L1 therapies at the Dana-Farber Cancer Institute, whole exome sequencing was performed from the clinically annotated pre-treatment biopsies, including: 36 “pairs” of samples (pre-treatment tumor+matched germline normal tissue) and 3 “trios” of samples (LUAD-1020: 4 pre-treatment tumors (1 primary+3 metastases); LUAD-1007: 2 pre-treatment tumors; and LUAD-1011: 2 pre-treatment tumors). The baseline clinical characteristics and prior therapies is summarized in Table 6 below.

(409) TABLE-US-00007 TABLE 6 Characteristic Patients (N = 39) Age (years) - Median (range) 60 (32-83) Age >75 - No. 3 (7.7) Male sex - No. (%) 15 (38.5) Smoking status - No. (%) Current 11 (28.2) Former 17 (43.6) Never 11 (28.2) No. of prior systemic regimens - No. (%) 0 3 (7.7) 1-2 19 (48.7) 3-4 16 (41.0) 5-6 1 (2.6)

(410) The resulting Kaplan-Meier analysis is compared for baseline clinical variables as predictors of PFS for SU2C cohort (N=39) (FIG. 13A-13D). The corresponding quality control processes are summarized in FIG. 14. As for clinical stratification, patients were divided into three groups

according to their response to immunotherapy. The definition of “clinical benefit,” as for the first group of patients, includes CR or PR by RECIST or SD with PFS>12 months. The definition of “no clinical benefit” includes PD by RECIST with PFS<3 months. The definition of “stable disease” (intermediate clinical benefit) includes SD with PFS<12 months or PD with PFS>3 months. A summary of different responses of 39 SU2C lung cancer patients to immunotherapy is shown in FIG. 15. Their mutational burden and response to immune checkpoint therapies is also compared (FIG. 16, N=31). Another cohort previously reported by Rizvi et al. (2015), supra was similarly analyzed (FIG. 17 and FIG. 18), showing that pre-treatment tumor mutational load was a strong predictor of response to immune checkpoint therapy in anti-PD1/PD-L1-treated lung cancer (FIG. 18). The current cohort of lung cancer patients were tested for any mutations to genes commonly mutated in lung cancers. As shown in FIG. 19, NF1 alterations were more frequent in responders (3/6 clinical benefit, 3/13 stable disease, 0/12 NCB). EGFR hotspot alterations were seen more frequently in nonresponders. KRAS hotspot alterations were observed more frequently in responders (1/6 clinical benefit, 4/13 SD, 1/12 NCB). The following provides additional genetic observations: SU2C-1006: splice site mutation in MET; missense mutation in LTBP1; SU2C-1066: 3 missense mutations in LEPR; SU2C-1068: 2 missense mutations in LEPR; SU2C-1067: missense mutations in STAG2 and SRCAP and an observed EGFR hotspot was L85. Sample SU2C-1066 may be excluded since its purity=0.36. A summary of significantly mutated genes in these patients is shown in FIG. 20. Patients with hotspot mutations in EGFR uniformly did not respond to immune checkpoint therapy (FIG. 21). SAFB2 indels were likely caused by sequencing artifacts (FIG. 22).

INCORPORATION BY REFERENCE

(411) All publications, patents, and patent applications mentioned herein are hereby incorporated by reference in their entirety as if each individual publication, patent or patent application was specifically and individually indicated to be incorporated by reference. In case of conflict, the present application, including any definitions herein, will control.

(412) Also incorporated by reference in their entirety are any polynucleotide and polypeptide sequences which reference an accession number correlating to an entry in a public database, such as those maintained by The Institute for Genomic Research (TIGR) on the world wide web and/or the National Center for Biotechnology Information (NCBI) on the world wide web.

EQUIVALENTS

(413) Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the present invention described herein. Such equivalents are intended to be encompassed by the following claims.

Claims

1. A method of treating melanoma, non-small cell lung cancer (NSCLC), or head and neck squamous cell carcinoma (HNSCC) in a subject likely to be responsive to immune checkpoint therapy against PD-1 or CTLA-4, the method comprising: i) selecting the subject, the subject having been identified according to: a) determining the copy number of ARID2 in a sample from the subject having cancer selected from the group consisting of melanoma, non-small cell lung cancer (NSCLC), and head and neck squamous cell carcinoma (HNSCC), wherein the sample comprises nucleic acid molecules from the subject's cancer; and b) comparing said copy number to that of a control sample, wherein an increased copy number of ARID2 encoding a loss of function mutation in the subject sample, relative to the control sample identifies the cancer as being more likely to be responsive to the immune checkpoint therapy against PD-1 or CTLA-4; and ii) administering the immune checkpoint therapy against PD-1 or CTLA-4 to the selected subject.
2. The method of claim 1, wherein the control sample is determined from a non-cancerous sample from either the subject or a member of the same species to which the subject belongs.

3. The method of claim 2, wherein the control sample is a non-cancerous sample from the subject obtained from an earlier point in time than the subject sample.
 4. The method of claim 3, wherein the control sample is obtained before the subject has received immune checkpoint therapy and the subject sample is obtained after the subject has received immune checkpoint therapy.
 5. The method of claim 2, wherein the control sample does not comprise cells.
 6. The method of claim 2, wherein the control sample comprises cells.
 7. The method of claim 6, wherein the cells are cancer cells known to be non-responsive to the immune checkpoint therapy.
 8. The method of claim 1, wherein the subject sample and/or the control sample has not been contacted with a renal cell cancer treatment or an inhibitor of an immune checkpoint.
 9. The method of claim 1, wherein the subject has not been administered a renal cell cancer treatment or an inhibitor of an immune checkpoint.
 10. The method of claim 1, wherein the subject sample is selected from the group consisting of serum, whole blood, plasma, urine, cells, cell lines, and biopsies.
 11. The method of claim 1, further comprising recommending, prescribing, or administering at least one additional anti-cancer therapeutic agent.
 12. The method of claim 1, wherein the immune checkpoint therapy comprises at least one antibody selected from the group consisting of anti-PD-1 antibodies, anti-CTLA-4 antibodies, and combinations thereof.
 13. The method of claim 12, wherein the immune checkpoint therapy comprises an anti-PD-1 antibody.
 14. The method of claim 12, wherein the immune checkpoint therapy comprises an anti-CTLA4 antibody.
 15. The method of claim 1, wherein the likelihood of the cancer in the subject being responsive to immune checkpoint therapy is indicated by at least one of cellular proliferation, tumor burden, m-stage, metastasis, progressive disease, clinical benefit rate, survival until mortality, pathological complete response, semi-quantitative measures of pathologic response, clinical complete remission, clinical partial remission, clinical stable disease, recurrence-free survival, metastasis free survival, disease free survival, circulating tumor cell decrease, circulating marker response, and RECIST criteria.
 16. The method of claim 1, wherein the subject is a mammal.
 17. The method of claim 16, wherein mammal is an animal model of the cancer.
 18. The method of claim 16, wherein the mammal is a human.
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